ACE MANUFACTURING SYSTEMS LTD

INDEX - MACHINE HISTORY

AMS — In Pursuit of Excellence

MACHINE NUMBER : M30 - 209

SL NO	DESCRIPTION	FORMAT NUMBER	ENCLOSED
1	BASE BUILDING	F-QA-IPR-BBA- 01	✓
2	BALL SCREW ASSEMBLY	F-QA-IPR-BSA- 01	✓
3	SPINDLE ASSEMBLY	F-QA-IPR-SPL- 01	✓
4	GEAR BOX ASSEMBLY	F-QA-IPR-GBA- 01	—
5	AXIS CALIBRATION REPORT	F-QA-FA-01	✓
6	BALL BAR REPORT	F-QA-FA-02	✓
7	GEOMETRICAL TEST REPORT	DS - DSN-PROD-TC 01~ 06	✓
8	PRACTICAL TEST REPORT	DS - DSN-PROD-TC-07	✓
9	SPINDLE BALANCING REPORT	F-QA-FA-04	✓
10	HYDRAULIC ASSEMBLY REPORT	F-QA-IPR-HYD-01	— N/A —
11	FINAL ASSEMBLY REPORT-MECHANICAL	F-QA-FA-05	✓
12	FINAL ASSEMBLY REPORT- ELECTRICAL	F-QA-FA-06	✓
13	CHECK LIST -MECHANICAL	F-QA-FA-07	✓
14	CHECK LIST - ELECTRICAL	F-QA-FA-08	✓
15	MACHINE HEALTH CHECK POINTS	F-QA-FA-09	✓
16	MACHINE HISTORY-MECHANICAL	F-QA-FA-10	✓
17	SPINDLE STABILISATION REPORT	F-QA-FA-11	✓
18	MACHINE SEQUENCE RUNNING REPORT	F-QA-FA-12	✓
19	MACHINE HISTORY-ELECTRICAL	F-QA-FA-14	✓
20	WORK ORDER	F-TSG-15	✓
21	MINUTES OF METEING COPY	F-TSG-17	— N/A —
22	DESPATCH DETAILS	F-TSG-18	✓

Date: 20/12/09 CHECKED BY APPROVED BY

Packed

ACE MANUFACTURING SYSTEMS LTD.

MACHINE DISPATCH CHECKLIST

CUSTOMER'S NAME : Mahaveer die Casters Pvt Ltd., Delhi. Dispatch 22/12/09

MODEL : MCV-450 MACHINE NUMBER : M30-209 DATE: 18/12/09 TIME 4-00 PM

S.NO.	DESCRIPTION	YES / NO / NA	S.NO	DESCRIPTION	YES / NO / NA
1	PACKING LIST AVAILABLE	Yes	21	CHIP BIN MESH CLEANED & PUT BACK	Yes
2	HYD / PNE COUNTER BALANCE CYLINDER REMOVED & PACKED	NA	22	CHIP TROLLEY CLEANED	NA
3	Z AXIS MOTOR COVER REMOVED & PACKED	NO	23	WANNER PUMP & COVER CLEANED & PUT BACK	NA
4	SIDE DOOR HANDLES REMOVED & PACKED	NO	24	CHILLER UNIT CLEANED	NA
5	BASE GUARD REMOVED & PACKED	Yes	25	OIL SKIMMER BIN CLEANED	NA
6	COOLANT GUN TIED PROPERLY	NA	26	MACHINE OIL BIN CLEANED	Yes
7	AIR GUN TIED PROPERLY	NA	27	MACHINE PLATFORM CLEANED	NA
8	LEVELLING SCREWS AND NUTS REMOVED & PACKED	NO	28	TPM POINTS STICKERS FIXED PROPERLY	NA
9	LEVELLING BLOCKS : HT - 100 / (180) / 220 / 300 & QTY - 4 / 6 / 7 / (8) / 11 / 20	Yes	29	FIXTURE / S PACKED PROPERLY	NA
10	AUTOMATIC TOOL CHANGER (ATC)	NO	30	INDEX TABLE PACKED PROPERLY	NA
11	APC ASSEMBLY PACKED PROPERLY	NA	31	INDEX TABLE COVER PACKED PROPERLY	NA
12	PALLET REMOVED & PACKED	NA	32	ROTARY TABLE PACKED PROPERLLY	NA
13	APC LEVELLING BLOCK SCREWS - 1 SET	NA	33	TAIL STOCK PACKED PROPERLY	NA
14	POWER PACK STAND	NA	34	TOOLINGS PACKED PROPERLY	NA
15	LUB TANK OIL DRAINED (RETAIN 2 LITER)	Yes	35	CUSTOMER COMPONENT PACKED PROPERLLY	NA
16	VOLTAGE STABILISER PACKED	NA	36	PRACTICAL TEST PIECE PACKED PROPERLY	NA
17	STD SPARES & MACHINE MANUALS AVAILABLE	Yes	37	FUMIGATION CERTIFICATE AVAILABLE	NA
18	COOLANT TANK MESH CLEANED & PUT BACK	Yes	38	DESSICANT BAGS HUNG	NA
19	COOLANT TANK ASSEMBLLY PACKED PROPERLY	Yes	39	AIR REMOVED FROM COVER WITH VACCUM CLEANER	NA
20	CHIP BIN CLEANED & PUT BACK	Yes	40		

REMARKS / SHORTAGES IF ANY : — Shortage —

1. Pull stud BT-40 (CMAS-403) — 20 NO Shortage

ASSEMBLY	STORES	FIXTURE	TRYOUT	ELECTRICAL	DISPATCH	AUTHORISED SIGNATORY
	RS Bhandari	—	—		tayana	24/12/09

F-PAC-03-R0

ACE MANUFACTURING SYSTEMS LTD.

DISPATCH CHECK POINTS

AMS

CUSTOMER'S NAME : Maharees die casters Pvt ltd. Delhi

MACHINE NUMBER : M30-209 MCV-450

DATE : 18/12/09

S.NO	CHECK POINTS	STATUS - YES / NO	REMARKS
1	PACKING CASE IS IN GOOD CONDITION.	Yes	
2	CHIP CONVEYOR IS FREE FROM CHIPS & UNWANTED MATERIALS.	NO	
3	PAINT SHOULD NOT BE PEELED OFF IN ANY PART OF THE MACHINE.	NO	
4	ALL POCKETS IN ATC ARE EMPTY & CLEANED	Yes	
5	NO TOOL IN SPINDLE & CLEANED	Yes	
6	NO LOOSE PARTS KEPT ANYWHERE IN THE MACHINE.	Yes	
7	ENSURE RUST PREVENTIVE OIL IS APPLIED TO ALL MACHINED SURFACE.	Yes	
8	BOLT & NUT ARE FITTED AT ALL RELAVENT PLACES ON PACKING CASE.	Yes	
9	SPARE BOX KEPT INSIDE THE PACKING CASE.	Yes	
10	GEOMETRICAL, AXIS CALIBRATION, CUTTING TEST REPORTS ARE SENT WITH THE MACHINE (inside the cabinet)	Yes	
11	ELECTRICAL CABINET CLEANED.	Yes	
12	ENSURE MACHINE IS CLEANED & NO DUST ON ANY WHERE ON MACHINE.	NO	
13	SUPRABHA COVER IS SEALED PROPERLY.	Yes	
14	ENSURE PACKING BOX SHOULD HAVE MINIMUM 3" GAP ALL OVER	Yes	
15	COOLANT WATER FROM ALL PIPES REMOVED BY AIR JET.	Yes	
16	NO RUST OBSERVED ANYWHERE IN THE MACHINE & RUST PREVENTIVE SPRAYED TO ALL FASTENERS & BLACKEDNED PARTS	Yes	
17	THERE IS NO TOUCHUP PAINT DEPOSITED ON MATCHING PARTS.	NO	
18	MACHINE LIFTING PROCEDURE CHART FIXED ON THE PACKING CASE	Yes	
19	CABINET KEY SENT WITH MACHINE TIED TO THE AC UNIT EYE BOLT	Yes	
20	PANEL COOLER FILTER IS CLEANED	Yes	
21	PAOWERPACK SUCTION & RCF ARE CLEANED	—	
22			
23			
24			
25			

REMARKS / SHORTAGES : — Shortage —

1. Pull Stud @ BT-40 (MAS-403) — 20 NO Shortage.

ATTENDED BY

DISPATCH - INCHARGE

ACE MANUFACTURING SYSTEMS LTD

PACKING CASE IDENTIFICATION LIST

AMS

MACHINE NUMBER : M30-209 MCV-450 DATE : 18/12/09

SL.NO	CASE - 1 CONTENTS	QUANTITY
1	Machine assembly	01NO
2	Std. Spears and Machine Manuals	01BOX
3	SPim-12 Lubricating oil	01Liter
4	Leveling blocres 180 MM	08NO
5		
6		
7		
8		
9		
10		
11		
12		

CASE - 2 CONTENTS

1	Coolant tank assembly	01NO
2	coolant Pump	03NO
3	Mesh	08NO
4	chip bin	01NO
5	chip bin Mesh (inside the coolant tank)	02NO
6	chuter Plate	01NO
7	base guard	01NO
8	oil bin	01NO
9		01NO
10		
11		
12		

CASE- 3 CONTENTS

1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		

CHECKED BY : Jayaroma. M.D

ACE MANUFACTURING SYSTEMS LTD.,

AMS
In Pursuit of Excellence

PACKING LIST ANNEXTURE-II

209

M/C. No. M30-209

Customer:M/s, MAHAVEER DIE CASTERS (P) LTD DELHI

Sl.No.	DESCRIPTION	QTY.
1	O'Ring Dia 19.1X1.6	2
2	O' Ring Dia 8.1X1.6	6
3	O' RingDia 11.1X1.6	4
4	O'Ring Dia 57.6X2.4	6
5	O'Ring Dia 24.5x3.0	6
6	O'Ring Dia 51.6x2.4	2
7	Cord Of O'Ring Dia 3 X 2 Mtrs	1
8	Spanner 10 x 11	1
9	Spanner 12 x 13	1
10	Spanner 18 x 19	1
11	Spanner 30 x 32	1
12	Spanner 36 x 41	1
13	Allen key set Metric.	1
14	Extension Allen Key 8MM	1
15	Panel key	1
16	~~Split Collet Set, Stud Push & Spring~~ STEEL BALL ϕ8.0M	28 NO
17	Fuse Set (Fanuc Make)	1
18	Glass Fuse Set (Kept in Electrical Cabinet)	1
19	IML-Process Oil - 12 (Kept in Machine) **Servo Spin-12**	1
20	Pull Stud BT-40 (**DIN** / MAS-403)	20 NO short
21	Fanuc Series **Oi Mate Model C / Oi-Model C** OPERATOR`S MANUAL	1
22	Fanuc Series Oi-Model C / Oi Mate Model C MAINTENANCE MANUAL	1
23	Fanuc AC SERVO & SPINDLE MOTOR **Bis series** MAINTENANCE MANUAL	1
24	Operating Instructions And Maintainance Manual	1
25	Spare Parts Manual	1
26	Air Cooler Manual (ACE Make - IC100)	1
27	Electrical Document Manual & Back up CD	1 + 1
28	Fanuc system garranty Card	1 no ~~28~~ 24
29		
30		
31		
32		
33		
34		

Sign...... **DATE.** 18/12/09

F-STR-26-R0

Document Control Stamp

ACE MANUFACTURING SYSTEMS LTD.

DISPATCH ADVICE

AMS
In Pursuit of Excellence

Date :22/12/2009

Sl. No.	Description Of Dispatch details	
1	Customer Name	M/S. MAHAVIR DIE CASTERS PVT. LTD.,
2	Telephone No.	0129 – 2443736 / 4068556 / 57
3	Fax No.	0129 – 2443735
4	E-Mail	mdc@oswalelectricals.com
5	Customers ECC No.	AABCM1069RXM002
6	Customers Sales Tax No.& Date	CST: FBD/CST/1316253 Dt.04.05.98 TIN: 06251316253
7	Contact Person	Mr. T.K.Mukherjee (Mob : 099588-47191)
8	Machine Model	CNC VERTICAL MACHINING CENTER MODEL MCV-450
9	Machine No.	M30-209
10	Place of Delivery	Works: Plot No.153, Sector-24, Faridabad (Haryana), INDIA.
11	Purchase Order No. &Date	PO-001533 Dated 06.11.2009
12	P.O Value (Excluding Taxes & Duties)	Rs.20,45,000/-
13	Transportation	Door delivery through any reliable carriers and freight to-pay basis.
14	Insurance	To be arranged & borne by us.Dispatch details to be faxed to the customer.
15	Scheduled dispatch	22.12.2009
16	Payment	Refer payment collection detail sheet.
17	Special Instructions	Keep the machine ready for dispatch. Contact HOD before dispatch Shortage list to be made while preparing machine Invoice All pending points to be attended before machine dispatch.

Copy: Purchase/Accounts/Dispatch/TSG/Stores.

21/12/09

ACE MANUFACTURING SYSTEMS LTD.

PACKING LIST

AMS
An Pursuit of Excellence

CUSTOMER NAME: M/S.MAHAVEER DIE CASTERS PVT LTD., DELHI	MACHINE NO : M30-209
REFERENCE: Your Purchase Order No. PO-001533 Dated 06.11.2009	

Sl No.	Description
1.0	**CNC Vertical Machining Center Model "MCV-450"**
	Spindle nose taper 7/24 No.40
	Spindle drive (Fanuc) 15 Min./Cont. Rating - 11/7.5 KW
	AC Servo axes drive (Fanuc)
	20 Station tool magazine
	Twin Arm (Random) type High Speed Auto tool changer
	Tool shank BT-40
	LM Guideways for all the three axes
	CNC System Fanuc 0IM with 7.2" Monochrome LCD
	Ball screw stretching for X & Y Axes
	Panel cooler for electronics
	Counter balance for Z-Axis
	Fail safe brake for Z-Axis
	Standard work light
	Coolant tank and chip tray
	Three tier indication lamp
	Spindle load meter
	Flexible coupling for X, Y & Z Axes
	Manual pulse generator
	Rigid tapping
	Total machine enclosure
	Chip disposal to the rear side of the machine through two chutes on the side
	One set of manuals
	Spindle speed of 80 - 8000 rpm (Direct drive)
	Tool life management
	Air drier unit (Small)
	Spindle with MAS-403 pull stud clamping system
2.0	Flood coolant system around the spindle
3.0	Chip flushing system on the base
4.0	4th axis provision on the machine (only interface wiring)
5.0	MAS-403 pull studs (20 Nos.)
6.0	1 set of glass fuses(inside the cabinet)
7.0	SERVO SPIN - 12 lubricating oil (1 liter)
8.0	Electrical documentation manual & back up CD
9.0	Following reports to be sent along with the machine.
	1. Geometrical test
	2. NASA test
	3. Laser calibration test
	4. Heavy cut & Rigid tapping test

inside the cabinet

21/12/09

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

Machine Sequence Running Test Report

M/c No : M30/209 | Customer : Mahavir die off.

Preconditions for Machine before starting Machine Running

1. Machine should be fitted with all accessories as per Work Order ✓
2. Machine should be cleared from all Electrical & Mechanical Remarks ✓
3. Machine should be run with Sufficient Coolant. ✓
4. ATC should be full tool holders ✓
5. No Bypass of any Safety switches and cabinet sealed. ✓

Date	Time	No of Hours Running	Date	Time	No of Hours Running	Remarks / Alarm Observed
16/12/09	2:30	1 ✓			17 ✓	
		2 ✓			18 ✓	
		3 ✓			19 ✓	
		4 ✓			20 ✓	
		5 ✓			21 ✓	
		6 ✓			22 ✓	
		7 ✓			23 ✓	
		8 ✓			24 ✓	
		9 ✓			25 ✓	
		10 ✓			26- ✓	
		11 ✓			27 ✓	
		12 ✓			28 ✓	
		13 ✓			29	M/c running completed.
		14 ✓			30	
		15 ✓			31	
		16 ✓			32	

Nature of Remarks Related to	1.Hydraulic leakage	2.Coolant leakage	3.Electrical alarm	4.ATC	5.Spindle	6.Axes	7.Others

Note : Incase machine Running is stopped due to any reason, Please write the above nature of remarks Sl.No.against no of Hours and Restart the Machine Running with initial hours.

Incharge-Mechanical

Incharge- QA

ACE MANUFACTURING SYSTEMS LTD.

MACHINE INSPECTION REPORT -MECHANICAL

AMS — In Pursuit of Excellence

M/c No: M30-209 Assly Technician : Dhananjaya DATE 15/12/09

Customer : Mahaveer die. Casting

S.No	Remarks Observed	Status	Sign
✓	Oil leakage from gear box near proxy mtg area	✓	✓
✓	X axis lub hose to be changed	-BB-	✓
✓	front RH door movement is very free		
	Torque mtr cover to be fixid.	-S-	✓
✓	Z axis Tc cover noise to be checked at 100% rapid.	-BB-	✓
✓	Coolant gun mtr to be removed & Dummy plate to be fixid		✓
✓	ATC assy report & Practical test		✓
1	M/c No. Plate to be fixed		
2	Wrong MPCB Setting (R2)		
3	Wrong Value in Position Switch		

QUALITY ASSURANCE

ASSEMBLY INCHARGE

ACE MANUFACTURING SYSTEMS LTD

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SL- 08647 **SPINDLE STABLISATION REPORT**

MACHINE NUMBER: M30 - 209	MAXIMUM RPM: 8000	DRIVE: ID / DD / IDCTS / DDCTS

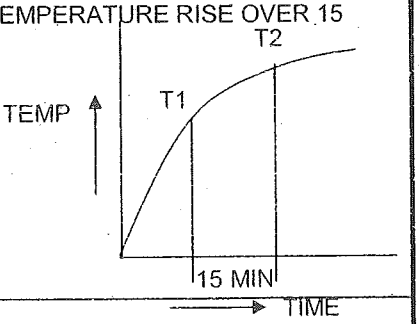
SPINDLE RPM	TIME	ROOM TEMPERATURE	SPINDLE TEMPERATURE
500	8:00 AM	22°C	—
1000	—		
2000	8.10 AM		
3000	—		
4000	8.20 AM		
5000	—		
6000	8.30 AM		
7000	8.40 AM		
8000	8:50 AM	22°C	31°C
	9:40 AM	23.8	35°C
	10:20 AM	25°C	37°C
	10:00 AM	25.8	38°C
	11:20	26°C	39°C
	11:30	26°C	39°C
	11:45	26°C	39°C
	12:00	26°C	39°C
		39-26° = 13°C	

1. THE STABALISATION POINT IS THE ONE AFTER WHICH THE RATE OF TEMPERATURE RISE OVER 15 MINUTES IS LESS THAN 3%.

= [(T2 – T1) / (T2 –Ta)] X 100

= ≤5%

(T1 = PREVIOUS TEMP, T2 = TEMPRATURE AFTER 15 MIN & Ta = AMBIENT TEMP)



2. RUN SPINDLE AFTER STABALISATION FOR 3 HOURS.

15/12/09 Date:	Checked By:

ACE MANUFACTURING SYSTEMS LTD.

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M30-209

GEOMETRICAL TEST CHART

1. SPIRIT LEVEL READING |0.040 / mt|:

Along X-Axis			Along Y-Axis		
1 →	0	← 1/4	0	0	1/4 ↑
1 →	0	← 1/4	0	0	1/4 ↑
1 →	0	← 1/4	1/2 ↑	0	0

2. TABLE TOP READINGS | 0.020 / 500 mm|:

Along X-Axis		
0	-6	0
+6	-4	0
+8	-4	0

Along Y-Axis		
0	3	7
0	2	2
0	0	0

3. T SLOT READING |0.020/500 mm|:

0	+6

PALLET - II

N.A

PALLET - II

Along X-Axis	Along Y-Axis
N.A	N.A

4. COLUMN READINGS :
|0.020 / 300 mm|

FRONT	SIDE
+2	+14
0	0

5. MANDREL READINGS:
| 0.020 / 300 mm|

Front	Side
0	0
-5	-10

RUNOUT:

a. TOP : | 0.010mm|
1. 0.002

b. BOTTOM : | 0.020mm|
1. 0.010
2. 0.010
3. 0.009
4. 0.009

6. TURN AROUND READING:
|0.020 / 300 mm Swing Dia|

-11, -10, -12, -5, -9, -2, -7, 0

7. SQUARENESS READING | X & Y|:
|0.015 / 300 mm|

0, -1, 0, +4, +6

8. PALLET HEIGHT VARIATION:
|0.03 mm|

Pallet-1 — N.A —

Pallet-2 — N.A —

9. PALLET REPEATABILITY:
|0.020mm|

Pallet 1 Pallet 2

	X	Y	X	Y
1				
2				
3	N.A			
4				
5				

DATE : 15/12/09

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MACHINE HEALTH CHECK POINTS

I	**AXIS**	
1	Ensure plugs are fitted for L M Guide ways.	✓
2	Ensure effective wiping of saddle cover wipers.	Tic covers
3	Check the tightness of coupling and is set to 17/35Nm (M6 / M8) Torque.	—
4	Ensure the lube timer is set 12 min for HG and 20 min for LM Guide ways.	✓
5	Ensure leveling bolts are fixed with locknut.	✓
6	Check the Table tenon slot 14H8 / 18H8 (Tolerance 0 to +0.027 mm) 20H8	✓
7	Ensure butt reference function.	✓
8	Check the all axes reference arrows are matched.	✓
9	The Machine/APC/Double Hand Emergency Push Button to be checked	✓
10	Ensure all screws marking neatly in Yellow paint marker	✓
11	Ensure cover fixed to Index table, rotary joint & tool breakage sensor	N/A
12	Ensure free moment of sliding door with out play and noise	✓
II	**SPINDLE**	
1	Ensure the belt tension using belt tension meter.	A.A
2	Ensure the proper supply of air to spindle at oriented condition.	✓
3	Ensure the proper supply of air to spindle Air Purge system	N/A
4	Check the tightness of push stud.	✓
5	Ensure grub screws are fitted to spindle flange.	✓
6	Check spindle orientation position and note down the value. 4031	266
7	Ensure spindle balancing	✓
8	Ensure the gap between draw bar bush & ejection screw / roller (with out tool - 0.8 to 1 mm / with tool 2.8 to 3 mm).	✓
III	**ATC**	
1	Ensure ATC / APC / Counter balance bracket pinning.	✓
2	Check oil in ATC gear box.	✓
3	Check the Tool not in spindle proximity message.	✓
4	Check the Tool Number display in panel board as same as in spindle	✓
5	Check tool change position limit switch input & band width.	✓
	CHECKED BY V. Gurudy	

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MACHINE HEALTH CHECK POINTS

AMS — In Pursuit of Excellence

IV	COOLANT	
1	Ensure no coolant leakage around the machine & lubrication oil drain pipe	
2	Ensure proper working of float switch.	
3	Check the function of ~~M7, M30, M48~~, M52 & Coolant gun.	
N/A	Ensure drum wash in conveyors (Optional)	
N/A	Check conveyor overload message & Direction (Optional)	
N/A	Check oil in conveyor gear box. (Optional)	
N/A	Ensure proper working of screw conveyor..	
N/A	Ensure CTS flow switch setting (Optional)	
N/A	Ensure suction strainer & Check Valve in CTS tank & pressure gauge working (Optional)	
N/A	Ensure CTS pin locator hole has plugged (Optional)	
N/A	Ensure gap between twin arm plunger CTS pin locator screw (Optional)	
N/A	Ensure bag filter clogging. (Optional)	
N/A	Ensure High pressure air blast after CTS switched off (Optional)	
V	APC	
N/A	Check pallet initialization in APC Machine	
N/A	Ensure effective supply of air jet to pallet rest pad cleaning.	
N/A	Ensure pallet change takes place smoothly with load without Heavy noise	
VI	HYD,PNEU & LUBRICATION	
1	Ensure air pressure switch is working properly.	
2	Ensure proper setting of oil in FRL unit. & Oil Level low message	
3	Check the FRL drain hose fixed.	
N/A	Ensure identification name plates are fixed in power pack hoses.	
N/A	Check the Hydraulic oil level low message & Pressure gauge function	
6	Ensure Ball valve fixed to FRL Unit inlet	
N/A	Ensure no oil leakage in hydraulic valves & fittings	
8	Ensure lubrication pressure switch is set to 15 bar.	
9	Ensure pneumatic quick change coupling fixed to 4th axis provision	
VII	GENERAL	
N/A	Check all the points of MOM are attended.	
2	Ensure coolant tank capacity engraved or marked in name plate	
N/A	Check Hydraulic circuit diagram pasted in power pack and specification is as per circuit diagram	
4	Ensure axis calibration & ball bar parameter is same on the system and report.	
5	Ensure Ball screw pitch in the machine and report is same.	
6	Ensure spindle SL NO and spindle parameters are same in the spindle report & machine.	

V.G.L-dy

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CHECK LIST - MECHANLCAL

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I	Check the Z-axis motor current during tool de clamp (Turret type ATC only) : Permissible BT 40 - 130 to 150 %, BT 30 -110 to 150 %	
A	840 Feed rate in Jog mode	— N.A —
B	Auto cycle	
II	Z- axis drop during power off (Permissible - + ve direction 0.005 , - ve direction 0.010 mm , Difference between system & dial is with in 0.020 mm)	

		A	B
A	+ve direction (Towards table)		↑
B	-ve direction	-250.000 ↓	-250.000
		-249.998	-249.998
	System:	-0.002	-0.002
	Dial:	-0.001	-0.004

III AXIS 2373 - 300

% of axis current drawn (Permissible as per Data sheet DS -QA -02)

A	Axis	25 % Rapid Rate	50% Rapid Rate	100% Rapid Rate	600 mm Feed rate
	X	17%	20%	49%	13%
	Y	16%	20%	48%	14%
	Z Up	58%	60%	70%	57%
	Z Down	38%	36%	32%	40%
	B- Forw				
	B - Rev				

B % of Axis holding current (Permissible as per Data sheet DS -QA -02)

	Axis	100% Rapid Rate		2000 mm Feed Rate	
		Motor end	Break end	Motor end	Break end
	X	11%	14%	11%	12%
	Y	12%	13%	13%	13%
	Z	55%	40%	55%	40%

C Over travel against Axis dead stopper in mm (Permissible as per Drawing)

	Axis	Break end	Motor End
	X	24.50 mm	22.49 mm
	Y	18.61 mm	13.00 mm
	Z	15.85 mm	17.40 mm

IV SPINDLE

A	Tool ejection	Permissible in mm	Auto	Manual
	✓ BT – 40	0.7 to 0.8 / 0.9 to 1.1	1 mm	1 mm
	BT – 30	0.5 to 0.6	—	—
	HSK A50	0.3 to 0.4 mm	—	—
	HSK A63	0.4 to 0.5 mm	—	—
B	Tool clamping force	Permissible as per data sheet	720 Kgf	

Checked By

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CHECK LIST - MECHANICAL

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C	Axial play in the spindle (Permissible - with in 0.005 mm)		
1	Tool de-clamping	0 ——▷+0.075——▷ 0	Difference = mm
2	100% Load by wood	0 ——▷-0.045——▷	0.002
D	% of current drawn at max RPM of the spindle : (Per missible " Direct Drive - B3 < 15 %, B6 & above < 10%" In Direct Drive B3 < 22 %, B6 & above < 15%")		5 %
E	Spindle noise at max RPM [Permissible - 80 dB(A)]		72 dB (A)
V	Turret type ATC		
A	Check the spring lever compression (mm)		
	Z – axis below zero position (BT 40 - 2.0 to 2.5 mm , BT 30 - 1.5 to 2 mm)		
	After the magazine buts against the dead stopper (BT 40 - 4.5 to 5 mm , BT 30 - 3 to 3.5 mm)		
	Difference – (BT 40 - 2.5 to 3 mm , BT 30 - 1.5 to 2 mm)		
B	Tool picking position in mm -Turret Type ATC (BT 40 - 60 to 80 mm , BT 30 - 50 to 65 mm)		
VI	A.P.C		
A	Pressure setting for the pallet transfer cylinder in kg/cm^2 (As per circuit diagram)	— N/A —	
B	Pressure setting for the pallet clamp-de clamp in kg/cm^2 (As per circuit diagram)	— N/A —	
VII	Cycle Time in seconds (Refer Work order)		
1	Check the chip to chip time with Z axis only	3.70 Sec	
2	Check the tool to tool time	2.20 Sec	
3	Check the chip to chip time with all axes	5.30 Sec	
4	Check the pallet changing time	— N/A —	— NA —

V.G L d

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MACHINE HISTORY

1	ATC DETAILS :	4020V / 4024V / 4030V / 4020H / 4024H		
1.1	ATC (Turret Type / Twin Arm)	Sl.No	08374	
1.2	Tool Capacity	No's	6 / 12 / 16 / 20 / 24 / 30 / 40 /	
1.3	ATC Magazine Encoder	Make	Model	Sl. No
		Pragati	ATC12 / ATC16	
1.4	ATC Torque Motor	TM 50 / TM 80 / TM 100 / TM 125 / Servo ATC		
2	HEAD STOCK DETAILS:	2 NO		
2.1	Spindle SL NO:	08647		
2.2	Spindle Drive	Direct Drive / Indirect Drive		
2.3	Pully Ratio	1:1 / 1:1.25 / 1:1.5 /1:1.75 / 1:2 /		
3	BELT DETAILS : NOT APPLICABLE TO DIRECT DRIVE :			
		GT BELT	COUPLING	
3.1	Model			
3.2	Make	Unitta / Power Grip		
4	COOLANT SYSTEM DETAILS :			
	Tank Capacity (LTRS)	200 / 300 / 400 / 500		
	Make : Rajmane Industries	Model	Type	
4.1	Flood Coolant Pump [Crown / Flexinox / Disc Nozzle]	RKM 206 / RV 160 / 270	Centrifugal Pump	
4.2	Chip Pushing Coolant Pump	RKM 206 / RV 160 / 270		
4.3	Base Flushing Coolant Pump	RKM 206 / RV 160 / 270		
4.4	Coolant Gun & Drum Wash Pump	RKM 206 / RV 160 / 270		
4.5	Oil Skimmer - Make / Model	N.A		
4.6	High Pressure Coolant Pump	Make	Grundfos/Wanner	
		Model		
4.7	Pressure Relief Valve	Make	Polyhydron	
		Model	DPRH-10T-100-04	
4.8	Rotary Joint	Make	Deublin / RIX	
		Model		
5	FLOAT SWITCH :			
5.1	Make	Filpro Elecdtronics		
6	CHIP CONVEYOR DETAILS :			
6.1	Type of Conveyor :- Scraper along X or Y axis / Slat cum Scraper along X or Y axis.			
6.2	Model			
6.3	Make	Universal Conveyor		
6.4	Sl.No	N/A		

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MACHINE HISTORY

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7	GEAR BOX DETAILS :			
7.1	Model		N.A	
7.2	Make	Universal Conveyor		
8	FILTER SPECIFICATIONS :			
8.1	Model	8 /15 ; Max Working Pressure 50 to 100 bar		
8.2	Make	Universal Conveyor		
8.3	Sl.No			
9	COIL / SCREW CONVEYOR DETAILS : Qty : 1 / 2			
9.1	Model			
9.2	Make		Universal Conveyor	
10	HYDRAULIC SYSTEM :			
10.1	Product No		QC. No :	
10.2	Tank Capacity (LTRS)		32 / 40 / 63 / 70	
10.3	Maximum Working Pressure			
10.4	Make		Pascal / Hyd Tech / Rexorth / Yuken /Jai Chamundi	
11	PRESSURE SWITCHES :			
11.1	Model		1PS 30-50 / 30-100 / 10 - 30 /	
11.2	Make		Polyhydron	
11.3	Quantity			
12	ACCUMULATOR :			
12.1	Make	FOX / BOSCH	FOX / BOSCH	
12.2	Set Oil Pressure	Tool de clamp :	Counter balance:	
12.3	Gas Pressure	Tool de clamp :	Counter balance:	
12.4	Volume in Lts	1.4	4 or 10	
13	HYDRAULIC CYLINDERS :			
	Make	Model	Function	Qty
13.1	Island Hydraulics	Φ40 X 22 X [125/152/180/1150]	Pallet F / R	
14	HYDRAULICS VALVES :			
	Type	Make	Model	Qty
14.1	Directional Valve (APC)	Yuken	DSG-01-3C2-D24-N1-50	
14.2	Reducing Modular Valve	Yuken	MRP-01-B30-H12	
14.3	Pilot operated check Valve	Yuken	MPW-01-2-30	
14.4	Throttle & Check Modular Valve	Yuken	MSW-01-X-30	
14.5	Break modular valve	Yuken	MBR-01-C-30	
14.6	Directional Valve	Rexroth	4WE6T 62/E G24 N9K4	
14.7	Reducing Modular Valve	Rexroth	ZDR6 DP 2-32/75YM	
14.8	Pilot operated check Valve	Rexroth	Z2S6-1-64	
14.9	Throttle & Check Modular Valve	Rexroth	Z2FS6-2-43/20QV	
	Directional Valve	Rexroth	4 WE10D 33/C G24 N9K4	
14.1	(Tool De clamp)		SINGLE / DOUBLE	
	Directional Valve	Yuken		
14.11	(Tool De clamp)			

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MACHINE HISTORY

15	HYDRAULIC PUMP :	Make	Model
		YUKEN / REXROTH	
	MOTOR : N/A	Make	Model
		Rotomotive	90L-4-1400 / 90S-4-1390
16	LUBRICATION UNIT DETAILS	Make	Pressure Setting
		CENLUB / DROPCO	15 Kg/ cm2
		CENLUB / DROPCO	

17	FRL UNIT :			
	Make	Model	Operating Range	Grade of Oil
	Shavo / Festo / Norgan	Midi	0-10 Kg/cm2	Servo Way Spin 12
18	DRYER :	Make		Specification
		Trident		VXD - 2

19	PNEUMATIC VALVES :			
	Application	Specs	Make	Qty
19.1	Spindle Air Jet	MFH 3-1/8	Festo	01
19.2	Spindle Air Purge	MFH 3-1/8	Festo	
19.3	ATC Pocket up & Down	JMFH 5- 1/8	Festo	01
19.4	Manual Tool De Clamp	MFH 5-1/8 / JMFH 5-1/2	Festo	01
19.5	APC Auto Door	JMFH 5-1/8	Festo	
19.6	CTS Air Jet	MFH 3-1/8	Festo	
19.7	Front Auto Door	JMFH 5-1/8	Festo	
		MFH 3-1/8	Festo	
19.8	Pallet Rest Pad Cleaning	MFH 3-1/8	Festo	
19.9	Fixture		Festo	

20	PRESSURE SWITCHES :	
20.1	Operating Range	2 to 15 Kg /cm2
20.2	Make	SUCO
20.3	Quantity	1 + 1
21	Tool de clamp system:	Mechanical / Pneumatic / Hydraulic

	Function	Make	Model	Qty
	Manual Tool De Clamp - Pneumatic	Festo	DNU80-80-PPV / DNU-100-60-PPV	1
		Rotex	SCN D -100 / 60 SCN D 80 / 80	1
	Manual Tool De Clamp - Hydraulic	Island Hydraulic	ϕ40 X 20 X 65	1

22	PNEUMATIC CYLINDERS USED :			
	Function	Make	Model	Qty
	Front Auto Door	Festo		
	APC Auto Door	Festo		

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MACHINE HISTORY

23	TURRET / INDEXING TABLE (OPTIONAL)			
	Make	Model	Sl.No	No of Station
	Pragathi / AMS			8 /12 /
24	ROTARY TABLE	Make	Model	Sl.No
		NIKKEN / UCAM		
25	TAIL STOCK	Make	Model	Sl.No
26	PNEUMATIC COUNTER BALANCING :			
26.1	Make		SMC / Rotex / Festo	
26.2	Cylinder Diameter :		80 / 100 / 125	
26.3	Air Reservoir used		Yes / No	
26.4	HYDRAULIC COUNTER BALANCING			
26.5	Make		Island Hydraulic	
26.6	Model		Φ50 X 32 X 570	

27	SPECIAL INFORMATIONS / REMARKS
1	Pneumatic tool de clamping system
2	
3	

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MACHINE STATUS

MACHINE NUMBER: VM30-209 | CUSTOMER: MAHAVEER

SL. No.	DESCRIPTION OF WORK	STATUS	DATE & SIGN	REMARKS
1	STROKE SETTING	Done	2/12/09 Louis2	
2	GEOMETRICAL TEST	Done	3/12/09 Louis2	
3	SPINDLE BALANCING	Done	11/12/09 Krishna	
4	HEAVY CUT & RIGID TAPPING	DONE	11/12/09 Bhavath. P.R	
5	AXES LASER CALIBRATION	Done	12/12/09 Shankar	~~Squareness~~
6	3 AXIS BALL BAR	Done	12/12/09 Shankar	Squareness is 13 micron in X2 & Y2 plane for 300mm
7	GUARDING	Done	9/12/09 Louis2	
8	ATC RUNNING	Done	9/12/09 Louis2	
9	APC RUNNING	NA	NA	
10	INDEX TABLE RUNNING	NA	NA	
11	COOLANT TANK ASSY/CHIP CONVEYOR ASSEMBLY	Done	10/12/09 Louis2	
12	COOLANT RUNNING	Done	11/12/09 Louis2	
13	NASA TEST	Done	11/12/09	
14	ANY OTHER REQUIREMENTS AS PER CUSTOMER NEEDS			
15	MACHINE RUNNING WITH ALL FUNCTIONS	Done	14/12/09 Louis2	
16	INTERNAL CHECK LIST	~~Louis2~~ done	10/12/09 Louis2	
17	QUALITY CHECKING	Done	14/12	

SIGN & DATE	PREPARED BY	CHECKED BY	DOCUMENT NUMBER
As Louis2		~~DHANANJAY A.C.~~	F-PDN-MEC-FA-04-R0

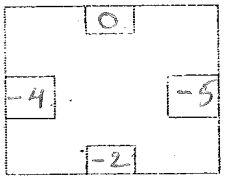
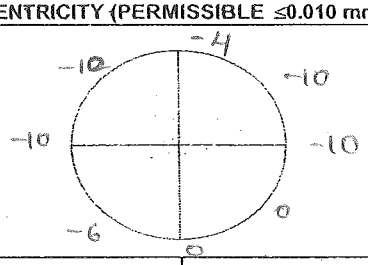
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FINAL ASSEMBLY - INSPECTION REPORT

MACHINE NUMBER: M30-209 | CUSTOMER : MAHAVEER

I. MOTOR MOUNTING BRACKET FACE OUT AND CONCENTRICITY

	FACE OUT (PERMISSIBLE ≤0.015 mm)	CONCENTRICITY (PERMISSIBLE ≤0.010 mm)
		

S.No.	DESCRIPTION	PERMISSIBLE	ACTUAL
II	DIRECT DRIVE		
1	SPINDLE CONNECTOR RUNOUT	0.01 MM	
2	MOTOR CURRENT AT MAX. RPM	B3 < 15%, B6 & ABOVE <10%	11%
3	CTS ROD GUIDE RUNOUT (FOR DDCTS)	0.005 MM	NA
4	SPINDLE MOTOR SPECIFICATION	AS PER WORK ORDER	B12 8000i
III	INDIRECT DRIVE		
1	MOTOR PULLEY RUNOUT	0.050MM	
2	SPINDLE PULLEY RUNOUT	0.050MM	
3	BELT TENSION	AS PER DATA SHEET	
4	MOTOR CURRENT AT MAX. RPM	B3 < 20%, B6 & ABOVE <15%	
5	ROTARY JOINT RUNOUT (FOR IDCTS)	0.040MM	
6	SPINDLE CLAMPING FORCE	AS PER DATA SHEET	720 KGF
7	SPINDLE MOTOR SPECIFICATION	AS PER WORK ORDER	
8	PULLEY RATIO	AS PER WORK ORDER	

NA

IV	AXIS DEAD STOPPER VALUE / BUTT REFERENCING VALUE (AS PER DRAWING)		
	AXIS	MOTOR END	BREAK END
1	X	22.49 mm	24.5 mm
2	Y	13 mm	18.61 mm
3	Z	17.4 mm	15.85 mm

V	% of axis current drawn (Permissible as per Data sheet DS -QA -02)		
	AXIS	100 % rapid rate	
	X	50 %	
	Y	48 %	
	Z	ME - 70 %	BE - 30 %
VI	ACCUMULATOR OIL PRESSURE	AS PER CIRCUIT DIAGRAM	
	ACCUMULATOR GAS PRESSURE	AS PER CIRCUIT DIAGRAM	

NA

NAME & SIGN	DNANANJAY A.C.		1...2
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S.No.	DESCRIPTION	PERMISSIBLE	ACTUAL
IV	ATC ALIGNMENT (TURRET TYPE)		
1	ATC DISC ALIGNMENT	0.050 MM	
2	TOOL EJECTION	AS PER DATA SHEET	
3	SWING LEVER GAP	AS PER DATA SHEET	NA
4	SIDE CAM ALIGNMENT	0.020 MM	
5	GAP BETWEEN DRAW BAR BUSH AND ROLLER / SCREW	1 ± 0.1MM	
V	ATC ALIGNMENT (TWIN ARM)		
1	X - AXIS ALIGNMENT	0.1 MM	0.05 ok
2	Y - AXIS ALIGNMENT	0.1 MM	0.1 ok
3	Z - AXIS ALIGNMENT	0.050 MM	
4	TWIN ARM ALIGNMENT	< 0.1MM	0.1 ok
5	GAP BETWEEN DRAW BAR BUSH & ROLLER	1 ± 0.1	1.1 ok
6	TOOL EJECTION	AS PER DATA SHEET	manual ok (motor Pneumatic)
7	GAP BETWEEN DEFLECTION PADS	0.02 TO 0.050MM	— NA —
VI	APC ALIGNMENT - LINEAR TYPE		
1	HEIGHT DIFFERENCE OF REST PADS	0.010MM	
2	PALLET CHANGE TIME	AS PER WORK ORDER	
VII	APC ALIGNMENT- ROTARY TYPE		NA
1	CLEARANCE BETWEEN ARM & PALLET	0.1 MM	
2	PALLET SAGGING	1 MM	
3	PALLET CHANGE TIME	AS PER WORK ORDER	
NAME & SIGN	DHANANJAY A.C.		2..3
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FINAL ASSEMBLY INSPECTION REPORT

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S.No.	CHECK POINTS	REMARKS
A	**AXIS**	
1	Ensure geometrical accuracies as per test chart (Enclose Test Chart)	
2	Ensure butt reference function.	✓
3	Check the all axes reference arrows are matched & Direction name plate fixed.	✓
4	Ensure all screws marking neatly in Yellow paint marker	✓
5	Ensure proper setting of Limit switches.	✓
B	**SPINDLE**	
1	Ensure the proper supply of air to spindle at oriented condition.	✓
2	Ensure the proper supply of air to spindle Air Purge system	NA
3	Ensure proper setting of proximities.	✓
4	Ensure smooth runing of spindle with out heavy noise.	✓
5	Ensure dummy plate fixed for CTS slot in Non CTS machine.	NA
C	**ATC**	
1	Ensure ATC / APC / Counter balance bracket pinning.	NA
2	Check oil in ATC gear box.	✓
3	Check the Tool not in spindle proximity message.	
4	Ensure smooth function of Tool change function without heavy noise.	✓
5	Ensure ATC bracket & Magazine Dowelling.	✓
D	**COOLANT**	
1	Ensure no coolant leakage around the machine & lubrication oil drain pipe	✓
2	Ensure proper working of float switch.	✓
3	Ensure all the hoses, fittings tightened and routed properly.	✓
4	Ensure CTS flow switch setting (Optional)	NA
5	Ensure suction strainer & Check Valve in CTS tank & pressure gauge working (Optional)	NA
6	Ensure CTS pin locator hole has plugged (Optional)	NA
7	Ensure gap b/w twin arm plunger CTS pin locator screw (Optional)	NA
8	Ensure bag filter clogging & Pressure switch setting. (Optional)	NA
9	Ensure High pressure air blast after CTS switched off (Optional)	NA
10	Ensure drum wash in conveyors (Optional)	NA
11	Check conveyor overload message & Direction (Optional)	NA
12	Ensure splash gauard fixed to conveyor.	NA
13	Check the function of M7, M30, M48, M52 & Coolant gun.	✓
NAME & SIGN	DHANANJAY A.C — CHECKED BY / APPROVED BY	3...4 — PAGE NO

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FINAL ASSEMBLY INSPECTION REPORT

	CHECK POINTS	
E	APC	
1	Ensure proper setting of proximities.	
2	Ensure effective supply of air jet to pallet rest pad cleaning.	
3	Ensure smooth function of pallet change function without heavy noise with weight on pallets.	NA
4	Ensure APC base Dowelling.	
F	HYD,PNEU & LUBRICATION	
1	Ensure pneumatic main line pressure switch is working properly with message.	L
2	Ensure Spindle air jet pressure switch is working properly with message.	L
3	Ensure Spindle air purge pressure switch is working properly with message.	NA
4	Ensure proper setting of oil in FRL unit. & Oil Level low message	NA
5	Ensure proper routing of FRL drain hose.	L
6	Ensure identification name plates are fixed in power pack hoses.	NA
7	Ensure Ball valve fixed to FRL Unit inlet	L
8	Ensure no oil leakage in hydraulic valves & fittings	NA
9	Ensure pneumatic counter balance bracket Dowelling.	NA
G	GAURDING ASSEMBLY	
1	Ensure all internal gaurds are assembled.	L
2	Ensure free moment of sliding door with out play and noise	L
3	Ensure Nylock nuts are fixed to door rollers.	L
4	Ensure ATC door LM rail plugs fixed.	NA
5	Ensure Proper tuning of Front auto door / ATC auto door cylinder.	NA
	ANY OTHER POINTS	
1	Torque motor Cover	Locxt

NAME & SIGN	DHANANJAY A.C.		4...
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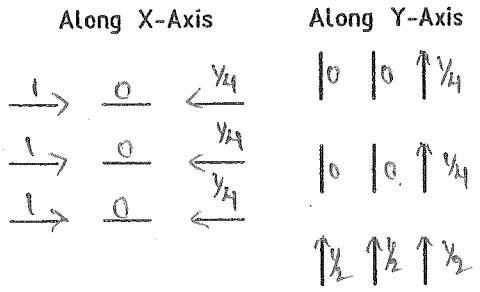
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GEOMETRICAL TEST CHART - VMC

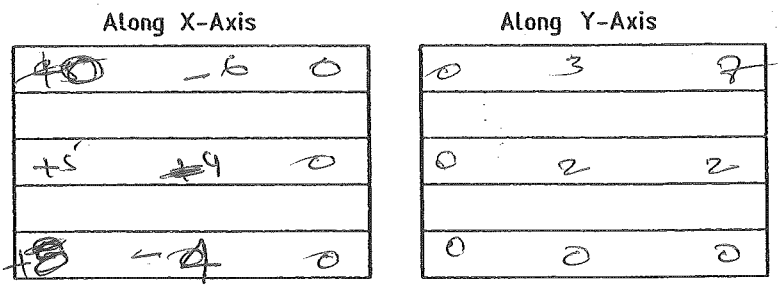
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PERMISSIBLE TOLERANCE 50% OF TEST CHART VALUE

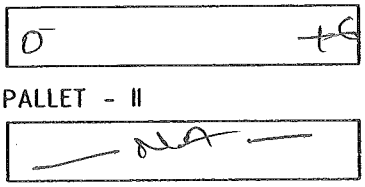
1. SPIRIT LEVEL READING:



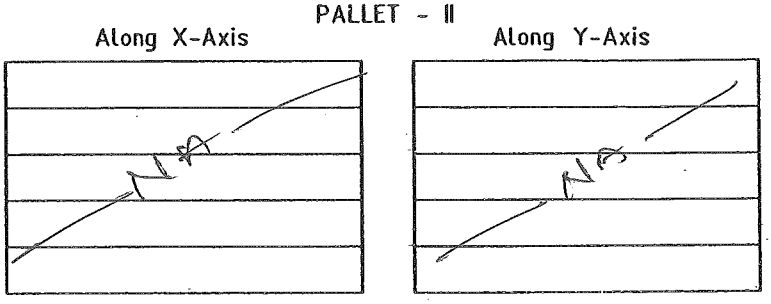
2. TABLE TOP READINGS:



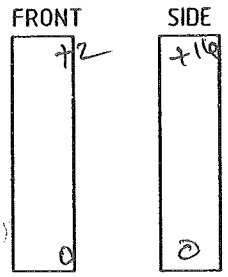
3. T SLOT READING:



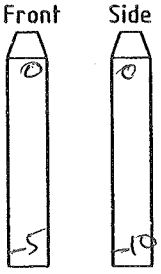
PALLET - II



4. COLUMN READINGS :



5. MANDREL READINGS:

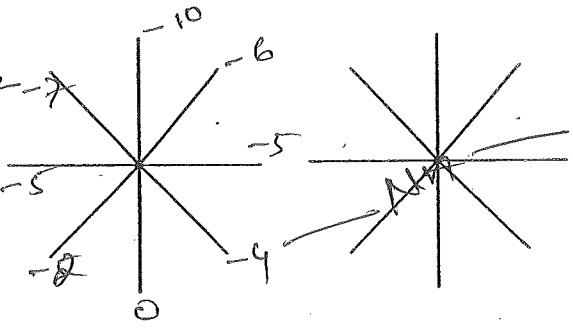


6. TURN AROUND READING:

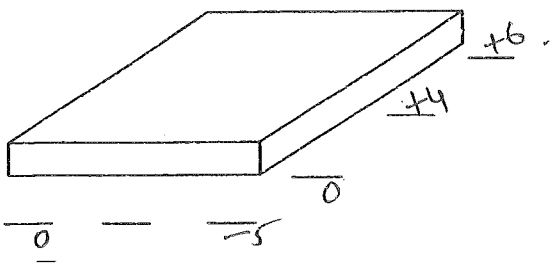
RUNOUT:

a. TOP : 0.002
   1.

b. BOTTOM :
   1. 0.010
   2. 0.010
   3. 0.009
   4. 0.009



7. SQUARENESS READING (X&Y):



8. PALLET HEIGHT VARIATION:

Pallet-1: NA

Pallet-2 : NA

9. PALLET REPEATABILITY:

	Pallet 1		Pallet 2	
	X	Y	X	Y
1				
2.				
3.				
4.				
5.				

NA

SIGN & DATE	DLaneyjeep		5....	F-QA-FA-05-R0
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WORK ORDER

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Poornatica/WB

Customer	:	M/s. Mahavir Die Casters Pvt. Ltd., Faridabad
Machine	:	CNC Vertical Machining Center Model MCV-450 **(Qty. - 2 Nos.)**
P.O. No.	:	PO-001491 03.11.2009
O.A. No.	:	OA-M30-209-09-00 21.11.2009
W.O. No.	:	WO-M30-209-09-00 21.11.2009

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WORK ORDER

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01.	Work Order No. & Date	:	WO-M30-209-09-00 21.11.2009
02.	Customer Name	:	M/s. Mahavir Die Casters Pvt. Ltd.,
03.	Address	:	Plot No.153, Sector-24, Faridabad, Haryana.
04.	Phone No.	:	0129 - 2443736 / 4068556 / 57
05.	Fax No.	:	0129 - 2443735
06.	E-Mail	:	mdc@oswalelectricals.com
07.	Excise Control Code No.	:	AABCM 1069 RXM002
08.	Sales Tax No. & Date	:	CST: FBD/CST/1316253 Dated 04.05.98 LST: FBD/HGST/1316253 Dated 04.05.98 TIN: 06251316253
09.	Contact Person	:	Mr. T.K. Mukherjee (Mobile: 099588-47191)
10.	Our Quotation No. & Date	:	Discussion had with Mr. Ravi Rana of MMT-Gurgaon
11.	Our Tooled-up Proposal No. & Date	:	Not applicable
12.	Customer P.O. No. & Date	:	PO-001491 Dated 03.11.2009
13.	Machine	:	CNC Vertical Machining Center Model MCV-450 (Qty. - 2 Nos.)
14.	Colour of Machine	:	Blue and light gray combination (Blue - Munsell No. 4PB4/10 Light Gray - Munsell No. 2.5y7/0.5)
15.	Scope of supply, Technical Specs & CNC features	:	As per Anneuxre-I
16.	Machine Inspection	:	Geometrical accuracy's and machine acceptance as per test chart & NASA Test.
17.	Component trials	:	Not applicable
18.	Price	:	Rs. 20,45,000/- for each machine. **Total Rs. 40,90,000/- for TWO machines.** **(Rupees Forty lakhs ninety thousand only)**
19.	Delivery	:	Machines to be kept ready for dispatch as follows: 1st Machine by 23.11.2009 2nd Machine by 27.11.2009
20.	Payment terms	:	40% advance along with the order and balance payment along with 100% of taxes & duties to be paid against Proforma Invoice before machine dispatch.

21.	Duties & Taxes	:	CED @ 8.24% & CST @ 2% against Form-C. However, Taxes and Duties prevailing at the time of despatch will be applicable.
22.	Payment method	:	RBI has started discouraging payment through cheque for high value clearances. In order to hasten up the receipt of payment, we request you to remit the payment through RTGS to our below mentioned account before machine dispatch: Our Bank Name & Address: Canara Bank No.83, Venkatadri Complex Richmond Road, Bangalore – 560 058 Nature of Account: Open Cash Credit Account No.: 1927261000252 RTGS/IFSC Code No.: CNRB0001927 MICR Code: 560015088
23.	Exchange variation	:	Not applicable
24.	Packing charges	:	Inclusive.
25.	Transportation	:	**Door delivery through any reliable carriers and freight to-paybasis.**
26.	Insurance	:	**To be arranged by us.**
27.	Commissioning	:	Free of cost by Engineers from our Sales & Service Associates M/s.Micromatic Machine Tools Pvt. Ltd.,
28.	Warranty	:	a) For Machine: 15 Months from the date of supply of the machine or 12 months from the date of installation at the customer end whichever is earlier.
		:	b) For Fanuc System & Drives: 27 months from the date of supply of machine OR 24 months from the date of installation at the customer end whichever is earlier.
29.	Training	:	Three persons for One week at our works. Training includes. a) Programming and Operation - 1 Person b) Electrical / Electronic / Mechanical trouble shooting and maintenance - 2 Persons

For **Ace Manufacturing Systems Ltd.,**

(P. Ramadas)
Managing Director

(T.S. Umesh)
Sr. Vice President

ANNEXURE - I
SPECIFICATION OF CNC VERTICAL MACHINING CENTER
MODEL MCV-450 (Qty. 2 Nos.)

Sl. No.	Description	Unit	MCV-450
1	**TABLE**		
1.1	TABLE SIZE	mm	1000 x 450
1.2	T-SLOT DIM.	N x W x P	4 x 20 x 100
1.3	MAX. LOAD ON TABLE	Kgf	400
1.4	DIST. FROM FLOOR TO TABLE TOP	mm	925
2	**MACHINE CAPACITY**		
2.1	X - AXIS TRAVEL	mm	800
2.2	Y - AXIS TRAVEL	mm	450
2.3	Z - AXIS TRAVEL	mm	500
2.4	DIST. FROM SPINDLE FACE TO TABLE TOP	mm	175 - 675
2.5	DIST. FROM SPINDLE CENTER TO COLUMN GUIDEWAYS	mm	510
2.6	COOLANT SUPPLY - FLOOD COOLANT AROUND THE SPINDLE	lpm	**50**
3	**SPINDLE**		
3.1	SPINDLE NOSE TAPER		7 / 24 No.40
3.2	SPINDLE SPEED - DIRECT DRIVE	rpm	80 - 8000
3.3	SPINDLE BEARING DIAMETER	mm	70
3.4	TOOL CLAMPING FORCE	Kgf	575 (720 Kgf)
4	**FEED**		
4.1	RAPID TRAVERSE X, Y & Z AXES	m / min.	30 / 30 / 30
4.2	CUTTING FEED	mm / min.	1 - 10000
4.3	BALL SCREW DIA FOR X, Y & Z AXES	mm	40 / 40 / 40
4.4	GUIDEWAYS FOR X, Y & Z AXES		Linear Motion
5	**AUTOMATIC TOOL CHANGER**		
5.1	NUMBER OF TOOLS	Nos.	20
5.2	MAX. TOOL DIA	mm	80
5.3	MAX. TOOL DIA. WITH ADJACENT POCKETS EMPTY	mm	155
5.4	TOOL SELECTION		Bi - Directional
5.5	TOOL LENGTH MAX.	mm	250
5.6	TOOL WEIGHT MAX.	Kg.	8
5.7	CHIP TO CHIP TIME	Sec.	4 / 5.3
5.8	TOOL CHANGE SYSTEM		Twin Arm
5.9	TOOL SHANK TYPE		BT - 40
5.10	PULL STUD TYPE		**MAS-403**
6	**ACCURACY WITH ROTARY ENCODER**		
6.1	POSITIONING FOR X, Y & Z AXES AS PER JIS B6330	mm	0.010
6.2	REPEATABILITY FOR X, Y & Z AXES AS PER JIS B6330	mm	± 0.003
7	**MOTORS**		
7.1	SPINDLE POWER 15 MIN./CONT. (FANUC)	kW	11 / 7.5
7.2	FEED MOTOR X, Y & Z AXES (FANUC)	Nm	11 / 11 / 20

ANNEXURE - I
SPECIFICATION OF CNC VERTICAL MACHINING CENTER
MODEL MCV-450 (Qty. 2 Nos.)

Sl. No.	Description	Unit	MCV-450
8	**INSTALLATION DATA - STANDARD MACHINE**		
8.1	MACHINE WIDTH	mm	2200
8.2	MACHINE DEPTH	mm	3200
8.3	MACHINE HEIGHT	mm	3250
8.4	MACHINE WEIGHT - NET	Kgf	4400
8.5	POWER SUPPLY FOR BASIC MACHINE	Kva	25
8.6	CNC CONTROL (WITH 7.2 " MONOCHROME LCD)		FANUC 0IM
8.7	PNEUMATIC PRESSURE REQUIRED	Kgf / Cm2	5
8.8	PNEUMATIC SUPPLY REQUIRED AT ATMOSPHERIC PR.	Lpm	300
9	**TANK CAPACITY - STANDARD MACHINE**		
9.1	COOLANT TANK	litr	300
9.2	LUBRICATION TANK (USEFUL QUANTITY)	litr	5 (4)
10	**MACHINE FEATURES / ACCESSORIES - STANDARD**		
10.1	PANEL COOLER FOR ELECTRICAL CABINET		
10.2	FAIL SAFE BRAKE FOR Z - AXIS		
10.3	STANDARD WORK LIGHT		
10.4	COOLANT TANK & CHIP TRAY		
10.5	THREE TIER INDICATION LAMP TO INDICATE CYCLE RUNNING READY FOR NEXT CYCLE & MACHINE IN ALARM STATUS		
10.6	FLEXIBLE COUPLING FOR X, Y & Z AXES		
10.7	MANUAL PULSE GENERATOR		
10.8	RIGID TAPPING		
10.9	TOTAL MACHINE ENCLOSURE		
10.10	CHIP DISPOSAL TO THE REAR SIDE OF THE MACHINE THROUGH TWO CHUTES ON THE SIDES		
10.11	MANUALS - ONE SET		
10.12	BALL SCREW STRETCHING FOR X & Y AXES		
10.13	TOOL LIFE MANAGEMENT		
10.14	AIR DRIER UNIT (SMALL)		
11	**STANDARD CNC FEATURES - FANUC 0IM**		
11.1	FOUR CONTROLLED AXES		
11.2	SIMULTANEOUS CONTROLLABLE AXES - 4 AXES		
11.3	FEED RATE / RAPID OVERRIDE FACILITY		
11.4	CIRCULAR INTERPOLATION - ANY TWO OUT OF FOUR AXES		
11.5	JOG MODE		
11.6	BACKLASH COMPENSATION		
11.7	PITCH ERROR COMPENSATION		
11.8	MANUAL DATA INPUT		
11.9	DRY RUN		
11.10	SINGLE BLOCK		
11.11	PROGRAM PROTECT KEY		
11.12	SELF DIAGNOSTIC FUNCTION WITH SYSTEM & MACHINE ALARM STATUS DISPLAY		

ANNEXURE - I
SPECIFICATION OF CNC VERTICAL MACHINING CENTER
MODEL MCV-450 (Qty. 2 Nos.)

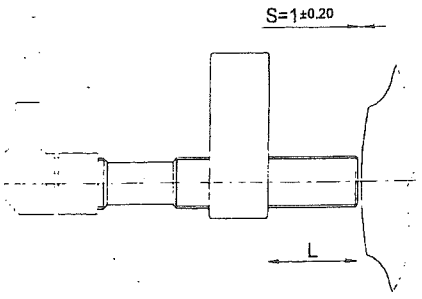
Sl. No.	Description	Unit	MCV-450
11.13	EMERGENCY STOP		
11.14	MIRROR IMAGE (X & Y C-ORDINATES)		
11.15	TOOL OFFSET MEMORY - 400 NOS.		
11.16	TOOL LENGTH COMPENSATION		
11.17	PART PROGRAM MEMORY 320 MTS (256K) BATTERY BACKUP		
11.18	CUTTER DIAMETER COMPENSATION		
11.19	CANNED CYCLES FOR DRILLING, TAPPING AND BORING		
11.20	PROGRAMMABLE DATA INPUT (G10)		
11.21	HELICAL INTERPOLATION		
11.22	RIGID TAPPING		
11.23	CLOCK FUNCTION		
11.24	LOOK AHEAD CONTROL		
11.25	READER PUNCHER INTERFACE (RS 232)		
11.26	WORK CO-ORDINATE SYSTEM (G53 TO G59)		
11.27	ANGLE CHAMFERING / CORNER RADIUS		
11.28	CUSTOM MACRO-A OR MACRO-B		
11.29	CO-ORDINATE SYSTEM ROTATION		
11.30	NUMBER OF REGISTERED PROGRAMS TOTAL 200		
11.31	BACK GROUND EDITING		
11.32	EXTENDED PART PROGRAM EDITING		
11.33	RUN HOUR PARTS COUNT DISPLAY		
11.34	MANUAL PULSE GENERATOR		
11.35	ALARM HISTORY & OPERATION HISTORY		
12	**MACHINE ACCESSORIES - ADDITIONALLY ORDERED BY M/s MAHAVEER DIE CASTINGS PVT. LTD., FARIDABAD.**		
12.1	SPINDLE WITH MAS-403 PULL STUD CLAMPING SYSTEM		
12.2	FLOOD COOLANT SYSTEM AROUND THE SPINDLE		
12.3	CHIP FLUSHING SYSTEM ON THE BASE		
12.4	4TH AXIS PROVISION ON THE MACHINE (ONLY NECESSARY INTERFACE WIRING)		

PRAGATI	AUTOMATIC TOOL CHANGER TEST REPORT	
SL NO: 9374	CUSTOMER I.D: S01-011-78	CUSTOMER ~~AMS~~ AMS
PRODUCT I.D.: ATC-4020V-026	PRODUCT DESCRIPTION: ATC 4020/V DIN W/O/M R-225	
DESPATCH DATE: 23/10/09		

TEST RESULTS:

TESTING HOURS	SPECIFIED: 6 Hrs.	ACTUAL: 6 HOURS

PARAMETERS	SPECIFIED	ACTUAL		
		DURING ASSLY	AFTER 6 Hrs	DURING DESPATCH
ARM STROKE	114 -115mm	114.60	114.70	114.70
ARM AXIAL PLAY(IN PARKING)	0.20-1.50mm	1.10	1.20	1.20
ARM RADIAL PLAY @R250 IN PARKING	0.20-1.0mm	0.25	0.35	0.35
ARM RADIAL PLAY @R250 WHILE MOVING OUT	0.20-1.0mm	0.20	0.30	0.30
ARM RADIAL PLAY @R250 WHILE MOVING IN	0.20-1.0mm	0.25	0.30	0.30
ARM PARLLALITY	0-0.40mm	0.05	0.05	0.05
ARM PARK SIGNAL	Visual Check	ok	ok	ok
TOOL DECLAMP SIGNAL	Visual Check	ok	ok	ok
TOOL CLAMP SIGNAL	Visual Check	ok	ok	ok
BACKLASH OF WHEEL @R300	0.05-0.40mm	0.25	0.35	0.35
AXIAL PLAY OF WHEEL	0.01-0.10mm	0.08		

SENSING DISTANCE	PROXIMITY	SENSING GAP(S)	SENSOR LENGTH(L)
	PS1	1.00	22.90
	PS2	1.00	23.00
	PS3	1.00	22.80
	PS4	1.00	30.00
	PS5	1.00	13.70
	PS6	1.00	13.20
	PS7	1.10	20.00

DESCRIPTION	NAME	SIGNATURE	DATE
ASSEMBLED BY	Narasimha		21/10/09
CHECKED BY AFTER 6Hrs	Narasimha		22/10/09
INSPECTED BY	Ayyappa.e		23/10/09
APPROVED BY	DICBER		23-10-09

得意先名:
CUSTOMER: FANUC INDIA PRIVATE LIMITED
CONTRACT NO.: YD20140- 000 DATE OF MANUFACTURE:2008-07
MODEL NO.: A02B-0309-B520 DATE OF TEST :2008-07-22
SERIAL NO.: E087B2807 CHECKED BY :M. Fujimaki
SETTING OF INPUT: APPROVED BY :MIYASAKA

CNC INDENTIFICATION NUMBER

3C7B5D01
FAE11E06



5E06C389
7C92C4FB





EDITION OF PRINTED BOARD & ROM

N A M E	EDIT	N A M E	EDIT	N A M E	EDIT
A20B-8200-0393	04C	A02B-0299-J571#656V	08		
		A02B-0310-H501#D4C1	11		
		A02B-0309-H580#408A	05		
		A02B-0281-J541#60VI	04		
		A02B-0309-J536#8A02	03		
		A02B-0309-H500#60M7	04		
		A02B-0309-H590#90B5	06		

LIST OF CARD , DIMM & BACK BOARD

A20B-8101-0430	1		
A20B-8200-0360	1		
A20B-3900-0223	1		

本データシートはファナック工場出荷時のCNC装置の装置情報です
Information contained in this Data Sheet is system configuration at the time of FANUC factory shipment.

NAME	NAME	SPECIFICATION	ORDR
BASIC CONTROL UNIT	ﾍﾞｰｼｯｸ ﾕﾆｯﾄ	A02B-0309-B520	1
AXIS CONTROL CARD	ｼﾞｸ ｾｲｷﾞｮ ｶｰﾄﾞ	A02B-0309-H032	1
FROM/SARM MEMORY	FROM/SRAM ﾒﾓﾘ	A02B-0309-H057	1
POWER SUPPLAY UNIT	ﾃﾞﾝｹﾞﾝ ﾕﾆｯﾄ	A02B-0309-H111	1
DISPLAY UNIT	ﾋｮｳｼﾞ ﾕﾆｯﾄ	A02B-0309-H142#M	1
MAIN BOARD	ﾒｲﾝ ﾎﾞｰﾄﾞ	A02B-0309-H204	1
NAME PLATE (FANUC)	ﾒｲﾊﾞﾝ	A02B-0310-H190	1
DIGITAL SERVO FUNCTION	ﾃﾞｨｼﾞﾀﾙ ｻｰﾎﾞｷﾉｳ	A02B-0309-H590#90B5	1
PACKAGE A	ﾊﾟｯｹｰｼﾞ A	A02B-0310-H532	1
CUSTOM SOFT SIZE FOR MAIN CP 512K BYTE	ｶｽﾀﾑｿﾌﾄ ﾖｳﾘｮｳ 512K (ﾒｲﾝCPU)	A02B-0310-J738#512K	1
MACRO EXECUTER	ﾏｸﾛ ｴｸﾞｾﾞｷｭｰﾀ	A02B-0310-J888	1
I/O UNIT FOR 0i	I/O UNIT	A02B-0309-C001	1
I/O MODULE FOR OP.PANEL	ｿｳｻﾊﾞﾝ I/Oｶｰﾄﾞ	A20B-2002-0521	1
*MANUAL PULSE GENERATOR	*ｼｭﾄﾞｳ ﾊﾟﾙｽ ﾊｯｾｲ	A860-0203-T001	1
DIGITAL SERVO UNIT	ﾃﾞｼﾞﾀﾙｻｰﾎﾞﾕﾆｯﾄ	A06B-6130-H002	2
DISCHARGE RESISTOR	ｶｲｾｲﾃｲｺｳﾕﾆｯﾄ	A06B-6130-H402	2
AC LINE FILTER	AC ﾗｲﾝ ﾌｨﾈﾀｰ	A81L-0001-0083#3C	2
SPINDLE SERVO UNIT	ｽﾋﾟﾝﾄﾞﾙｻｰﾎﾞﾕﾆｯﾄ	A06B-6134-H313#D	1
	BIS4/4000	A06B-0063-B003	2
	BIS12/3000	A06B-0078-B103	1
	BIS12/3000	A06B-0078-B103#0100	1
	BIS22B/3000	A06B-0082-B403	1
	SPB6/10000I	A06B-1445-B103	1

ACE MANUFACTURING SYSTEMS LTD

AMS
In Pursuit of Excellence

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

PRODUCT : VMC

MODEL NUMBER : MCV - 450

MACHINE NUMBER : M30 - 209

CUSTOMER : M/S MHAVIR DIE CASTING PVT LTD

CHECKED BY
CUSTOMER / QUALITY ASSURANCE

CHECKED BY
ASSEMBLY - MECHANICAL

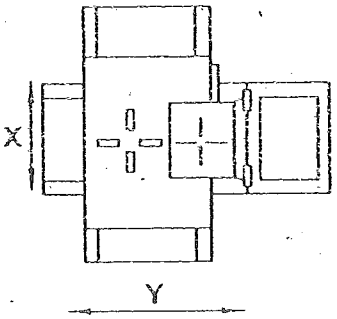
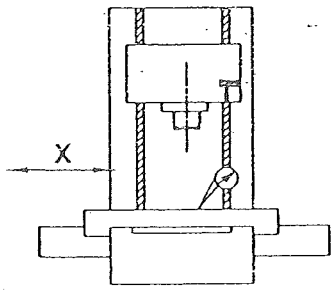
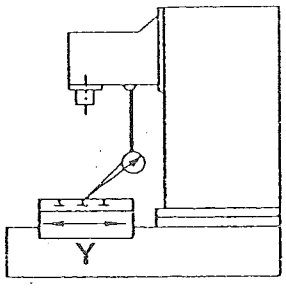
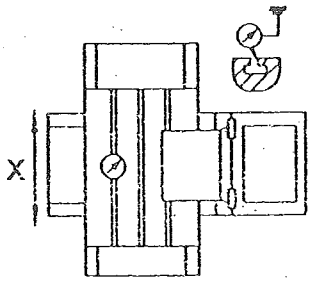
ACE MANUFACTURING SYSTEMS LTD

DESIGN VERIFICATION / VALIDATION PLAN CHART

AMS — In Pursuit of Excellence

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

All measurements in mm

SL. NO.	Description of the test	Figure	Tolerance: Permissible	Tolerance: Actual
G1	LEVELLING OF MACHINE X & Y AXES AT THEIR MIDDLE OF STROKE		0.03 / METER ALONG X & Y AXES	LEVEL ALONG X-AXIS : 0.005 LEVEL ALONG Y-AXIS : 0.005
G2	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO ITS GUIDEWAYS OR X-AXIS MOVEMENT		L ≤ 400 - 0.015 400 ≤ L ≤ 630 - 0.025 630 ≤ L ≤ 1000 - 0.040 1000 ≤ L ≤ 2000 - 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 0.014 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G3	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO Y-AXIS GUIDEWAYS OR Y-AXIS MOVEMENT		L ≤ 400 - 0.015 400 ≤ L ≤ 630 - 0.025 630 ≤ L ≤ 1000 - 0.040 1000 ≤ L ≤ 2000 - 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 0.007 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G4	PARALLELISM OF REFERENCE T-SLOT TO X-AXIS MOVEMENT		L ≤ 400 - 0.015 400 ≤ L ≤ 630 - 0.025 630 ≤ L ≤ 1000 - 0.040 1000 ≤ L ≤ 2000 - 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 0.006 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)

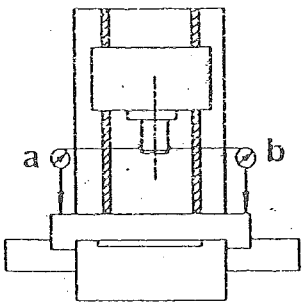
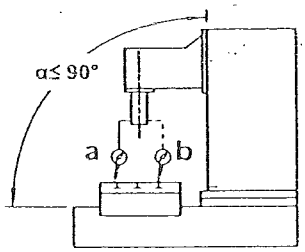
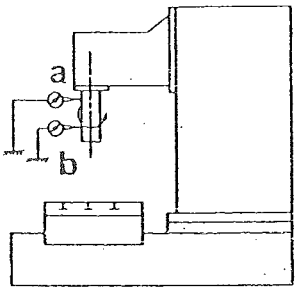
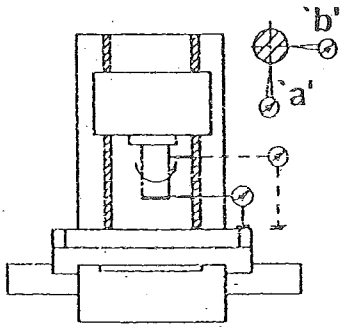
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	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

ACE MANUFACTURING SYSTEMS LTD

DESIGN VERIFICATION / VALIDATION PLAN CHART

AMS — In Pursuit of Excellence

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G5	SQUARENESS OF THE SPINDLE AXIS TO THE TABLE SURFACE IN THE XZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : 0.004 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G6	SQUARENESS OF THE SPINDLE AXIS TO THE TABLE SURFACE IN THE YZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : 0.011 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G7	RUN OUT OF SPINDLE TAPER a) NEAR THE SPINDLE NOSE b) AT A DISTANCE OF 300 FROM THE SPINDLE NOSE		a. 0.01 NEAR THE SPINDLE NOSE b. 0.02 AT A DISTANCE OF 300mm	a. 0.002 b. 0.010
G8	PARALLELISM OF SPINDLE AXIS & Z-AXIS IN a) XZ PLANE b) YZ PLANE TO BE CHECKED AT MEAN POINT OF RUNOUT		a. 0.02 / 300 b. 0.02 / 300	a. 0.005 b. 0.010

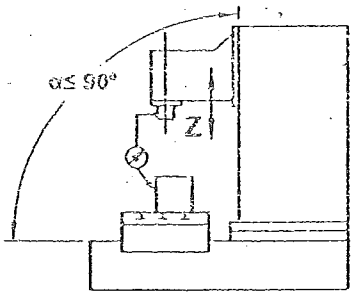
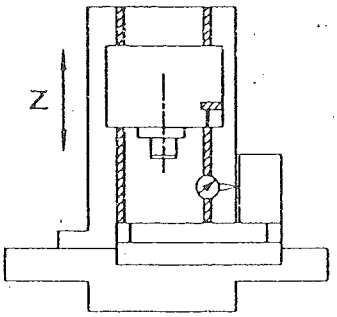
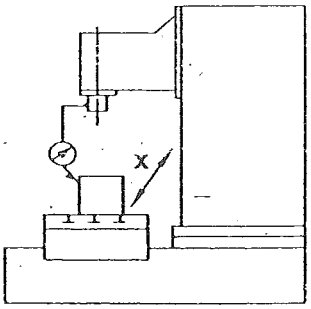
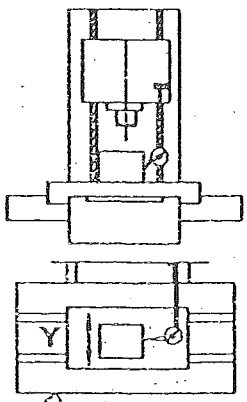
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	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

ACE MANUFACTURING SYSTEMS LTD

DESIGN VERIFICATION / VALIDATION PLAN CHART

AMS — In Pursuit of Excellence

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

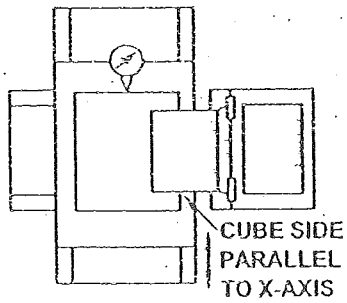
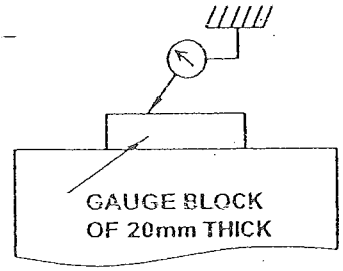
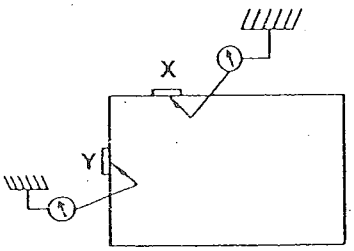
SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G9	SQUARENESS BETWEEN TABLE SURFACE & THE SPINDLE HEAD ALONG THE Z-AXIS IN XZ PLANE		0.015 FOR ANY 300 mm LENGTH	TABLE : 0.002 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G10	SQUARENESS BETWEEN TABLE SURFACE & THE SPINDLE HEAD ALONG THE Z-AXIS IN YZ PLANE		0.015 FOR ANY 300 mm LENGTH	TABLE : 0.014 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G11	STRAIGHTNESS OF X-AXIS MOVEMENT IN XZ - PLANE		0.010 FOR ANY 300 mm LENGTH	0.002
G12	STRAIGHTNESS OF Y-AXIS MOVEMENT IN YZ - PLANE		0.010 FOR ANY 300 mm LENGTH	0.001

SIGN & DATE	GPA	IG	RP	06-11-2009	4...	DS-DSN-TC-PROD-01-R0
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

ACE MANUFACTURING SYSTEMS LTD

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance: Permissible	Actual
G13	SQUARENESS OF Y - AXIS MOVEMENT WITH RESPECT TO X - AXIS MOVEMENT	MEASURE READING ALONG 'Y' 	0.015 FOR ANY 300 mm LENGTH	0.006
G14	VARIATION BETWEEN THE PALLET HEIGHT (FOR APC MACHINE ONLY)		0.030	N.A
G15	PALLET REPEATABILITY IN X & Y AXES WHEN LOADED ON THE MACHINE FROM THE PALLET STATION FOR SAME PALLET (FOR APC MACHINE ONLY)		0.020 IN X & Y AXES DIRECTION	PALLET 1 : N.A PALLET 2 : N.A (FOR APC MACHINE ONLY)

SIGN & DATE	GPA	IG	RP	06-11-2009	5...	DS-DSN-TC-PROD-01-R0
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

M30-209
X'Ass'y

ボールねじ検査成績証
CERTIFICATE

納入先
CUSTOMER M/S ACE MFG SYSTEMS (

形式
TYPE 40TTFB12-1365C3

製番
MFG.No. 51M-0103- 14

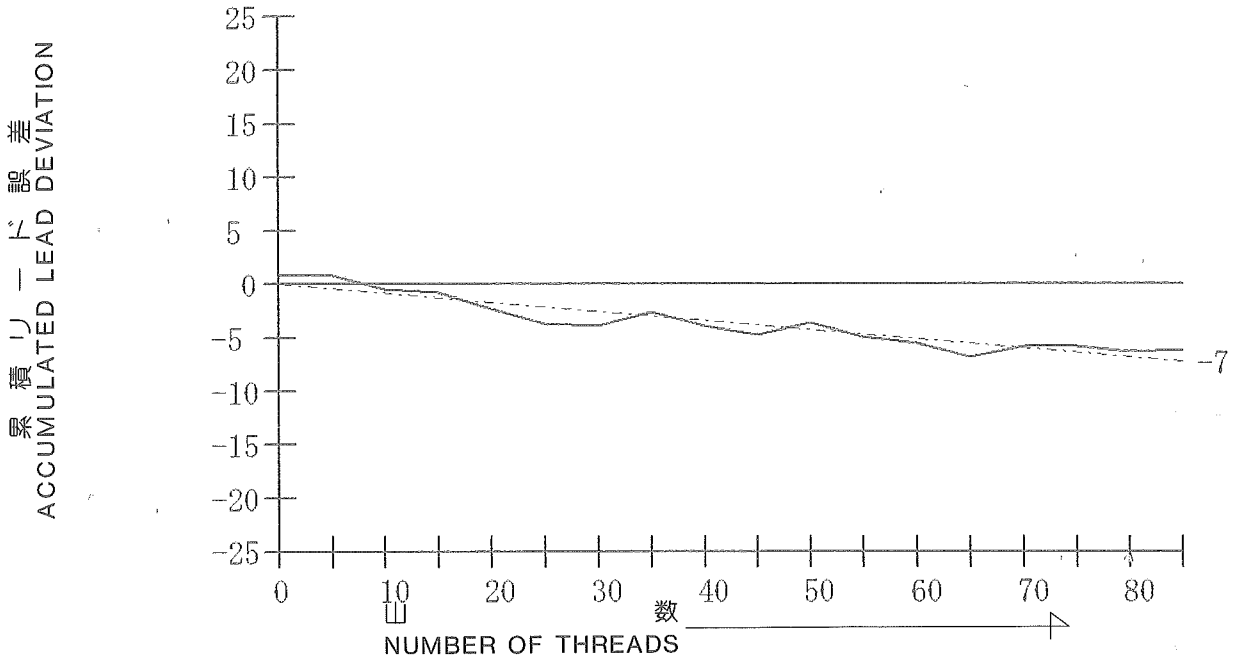
軸番
SHAFT No. H59

備考
REMARKS

X	累積基準リード目標値 TARGET POINT OF ACCUMULATED BASIC LEAD		0.0000	
			測定値 DATA	許容値 TOLERANCE
Y	累積代表リード誤差 ACCUMULATED REFERENCE LEAD DEVIATION		0.0072	0.0240
Z	変動 RELATIVE VARIATION	ねじ部有効長さに対し TOTAL THREAD LENGTH	0.0029	0.0160
Z_{300}		任意の300㎜に対し ANY 300mm LENGTH	0.0029	0.0080

測定器 MEASURING INSTRUMENT	ULMM 3000
測定温度 MEASURING TEMPERATURE	20°C

090902



他の精度項目については全て許容値に合格していることを保証致します。
We certify that all other dimensions are within specifications.

検査印
INSPECTED BY

検査年月日(DATE)

TSUBAKI NAKASHIMA CO.,LTD.

ACE MANUFACTURING SYSTEMS LTD

BALL SCREW ASSEMBLY REPORT

AMS — In Pursuit of Excellence

Machine Number : M30-209	Ball screw Sl No: SIM-0103-14 (1459)	Axis : X

I. Ball screw Make & Specification : T-Subaki (HOTTFB12-1365(3))

II. Ball Scerw Bearing Make & Specification : NSK(25TAC62BSUC10PN7B)

Sl No	Description	Permissible	Motor End	Brake End
1	Ball screw rotational Torque in N-m (Forward / Reverse)	0.9 to 1.8 N-m	0.9N/m	1 N/m
2	Bearing housing Bore diameter	As per drawing	10to8M	10to9H
3	Bearing housing center height difference	0.01 mm	0.01M	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.005M	0.005M

Motor End

Sl No	Description	Part Number	Symbol	Depth / width
1	Flange cover	C24-290-8123	T1	13.847
2	Bearing B1	—	T2	—
3	Bearing B2	NSK	T3	14.935
4	Bearing B3	— " —	T4	14.935
5	Spacer cylindrical B4	—	T5	15.140 ①
6	Flange cover	C24-290-8123	T6	
7	Housing Bearing	C37-0106-8123	T7	72.000

IV. Flange cover thickness "T6" = T7 - (T1+T2+T3+T4+T5)+0.05 0106/02

"T6" = 13.220 mm

Brake End

Sl No	Description	Part Number	Symbol	Depth / width
1	Flange cover	—	T1	—
2	Bearing B1	—	T2	—
3	Bearing B2	— NSK	T3	14.930
4	Bearing B3	— " —	T4	14.930
5	Spacer cylindrical B4	—	T5	—
6	Flange cover	C24-360-8123	T6	—
7	Housing Bearing	C37-0153-8123	T7	43.470

V. Flange cover thickness "T6" = T7 - (T1+T2+T3+T4+T5)+0.05 0153/02

"T6" = 13.680 mm

VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is with in 0.030 mm.	— " —
3	Ensure free rotation of ball screw assembly.	— " —
4	Ensure ball screw run out with in 0.010 mm	— " —
5	Ensure ball Screw report attached to Inspection Report.	— " —
6	Ensure all screw tightness & marking neatly in yellow marker	— " —

Date: 23-11-09	Checked by P.M. Manjunath	Approved by

M30-209
Y'AXIS

ボールねじ検査成績証
CERTIFICATE

納入先
CUSTOMER M/S ACE MFG SYSTEMS (

形式
TYPE 40TTFB12-1060C3

製番
MFG.No. 51L-0313- 14

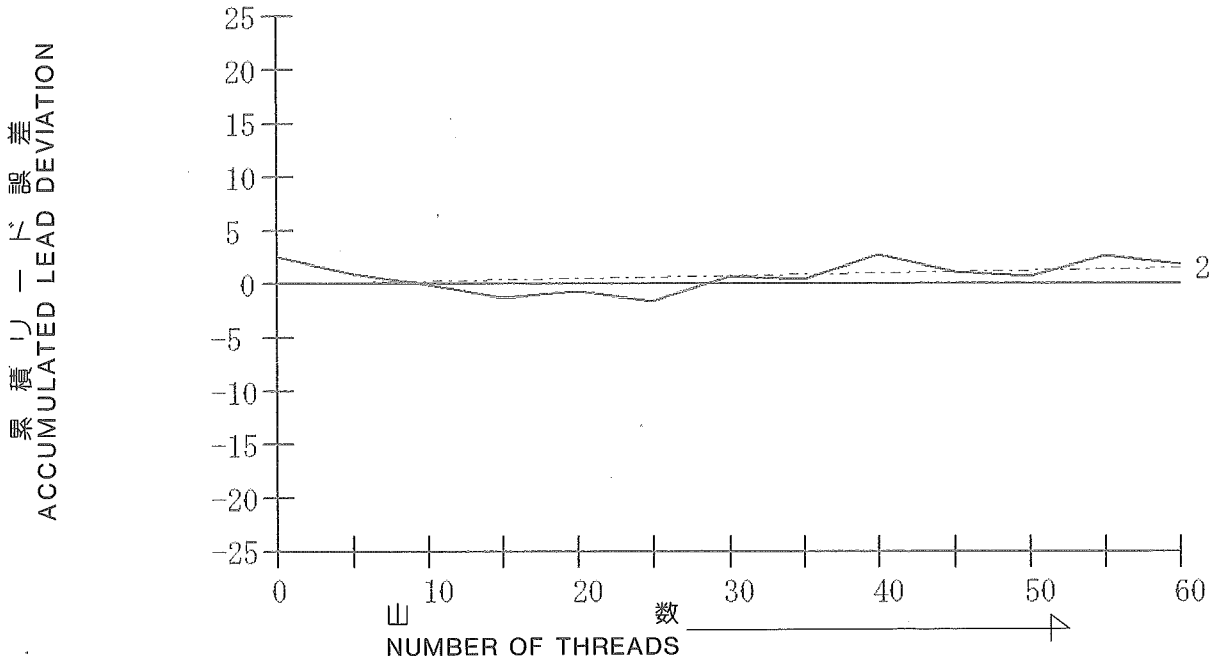
軸番
SHAFT No. H139

備考
REMARKS

X	累積基準リード目標値 TARGET POINT OF ACCUMULATED BASIC LEAD		0.0000	
			測定値 DATA	許容値 TOLERANCE
Y	累積代表リード誤差 ACCUMULATED REFERENCE LEAD DEVIATION		0.0016	0.0180
Z	変動 RELATIVE VARIATION	ねじ部有効長さに対し TOTAL THREAD LENGTH	0.0047	0.0130
Z300		任意の300㎜に対し ANY 300mm LENGTH	0.0047	0.0080

測定器 MEASURING INSTRUMENT	ULMM 3000
測定温度 MEASURING TEMPERATURE	20°C

080827



他の精度項目については全て許容値に合格していることを保証致します。
We certify that all other dimensions are within specifications.

検査印
INSPECTED BY
K.Horiuchi

検査年月日(DATE) AUG. 29. 2008

TSUBAKI NAKASHIMA CO.,LTD.

ACE MANUFACTURING SYSTEMS LTD

BALL SCREW ASSEMBLY REPORT

AMS — In Pursuit of Excellence

Machine Number : M30-209	Ball screw Sl No: SIL-0313-14 H139	Axis : Y

I. Ball screw Make & Specification : Tsubaki (HOTTFB12-1060C3)

II. Ball Scerw Bearing Make & Specification : NSK (25TAC62BSUC10PN7B)

Sl No	Description	Permissible	Motor End	Brake End
1	Ball screw rotational Torque in N-m (Forward / Reverse)	0.9 to 1.8 N-m	0.9 N/m	1 N/m
2	Bearing housing Bore diameter	As per drawing	62.007	72.007
3	Bearing housing center height difference	0.01 mm	0.01 M	
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.005 M	0.005 M

Motor End

Sl No	Description	Part Number	Symbol	Depth / width
1	Flange cover		T1	
2	Bearing B1		T2	
3	Bearing B2	NSK	T3	14.930
4	Bearing B3		T4	14.925
5	Spacer cylindrical B4		T5	
6	Flange cover	C24-275-8123	T6	
7	Housing Bearing	C37-0103-8123	T7	43.470

IV. Flange cover thickness "T6" = T7 - (T1+T2+T3+T4+T5)+0.05 0103) 158

"T6" = mm

Brake End

Sl No	Description	Part Number	Symbol	Depth / width
1	Flange cover		T1	
2	Bearing B1		T2	
3	Bearing B2	NSK	T3	14.940
4	Bearing B3		T4	14.940
5	Spacer cylindrical B4		T5	
6	Flange cover	C24-360-8123	T6	
7	Housing Bearing	C37-0153-8123	T7	43.480

V. Flange cover thickness "T6" = T7 - (T1+T2+T3+T4+T5)+0.05 0153) 158

"T6" = mm

VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is with in 0.030 mm.	
3	Ensure free rotation of ball screw assembly.	
4	Ensure ball screw run out with in 0.010 mm	
5	Ensure ball Screw report attached to Inspection Report.	
6	Ensure all screw tightness & marking neatly in yellow marker	

18-11-09 Date:	Checked by P. M. Manjunath	Approved by

m30-209
Z'Axis

ボールねじ検査成績証
CERTIFICATE

納入先
CUSTOMER M/S ACE MFG SYSTEMS (

形式
TYPE 40TTFC12-1105C3

製番
MFG.No. 51K-0134- 19

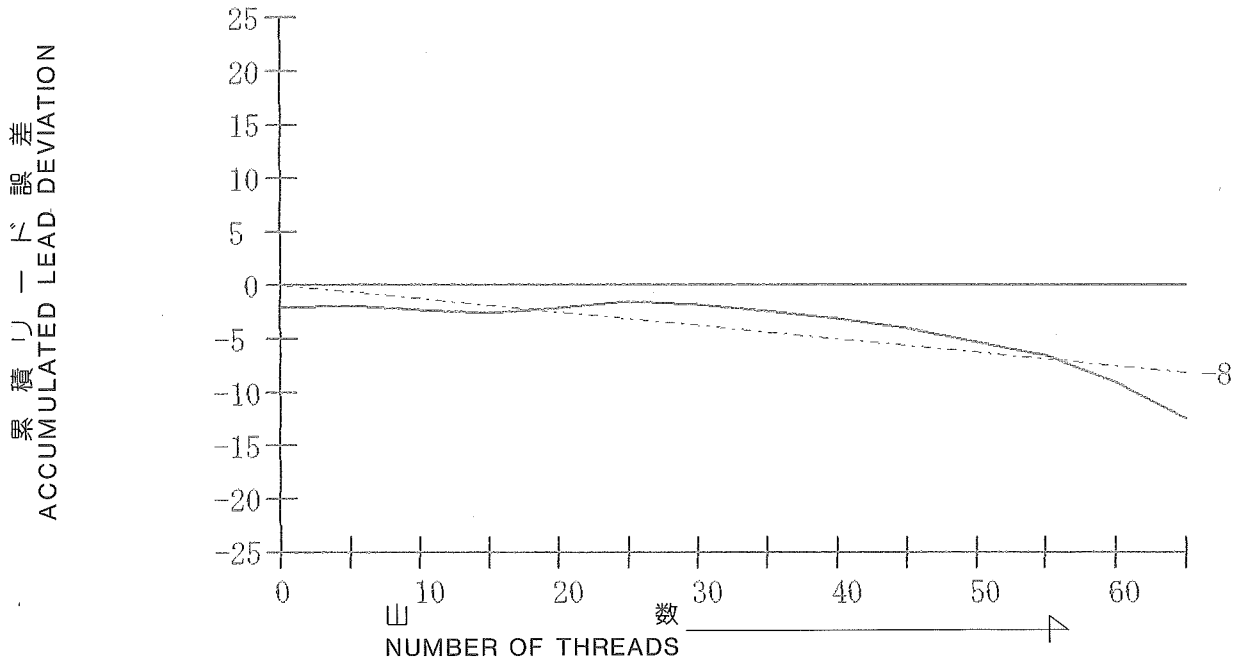
軸番
SHAFT No. F1184

備考
REMARKS

X	累積基準リード目標値 TARGET POINT OF ACCUMULATED BASIC LEAD		0.0000	
			測定値 DATA	許容値 TOLERANCE
Y	累積代表リード誤差 ACCUMULATED REFERENCE LEAD DEVIATION		0.0081	0.0210
Z	変動 RELATIVE VARIATION	ねじ部有効長さに対し TOTAL THREAD LENGTH	0.0063	0.0150
Z_{300}		任意の300㎜に対し ANY 300mm LENGTH	0.0063	0.0080

測定器 MEASURING INSTRUMENT	ULMM 3000
測定温度 MEASURING TEMPERATURE	20°C

070719



他の精度項目については全て許容値に合格していることを保証致します。
We certify that all other dimensions are within specifications.

検査印 INSPECTED BY

検査年月日(DATE) FEB. 24. 2007

TSUBAKI NAKASHIMA CO.,LTD.

ACE MANUFACTURING SYSTEMS LTD

BALL SCREW ASSEMBLY REPORT

AMS — In Pursuit of Excellence

Machine Number : M30-209	Ball screw Sl No: 51K-0134-19 (F1184)	Axis : Z

I. Ball screw Make & Specification : T-SUBAKI (HOTTFC12-1105C3)

II. Ball Scerw Bearing Make & Specification : NSK (25TAC62BSUC10PN7B)

Sl No	Description	Permissible	Motor End	Brake End
1	Ball screw rotational Torque in N-m (Forward / Reverse)	0.9 to 1.8 N-m	1.7 N/m	1.2 N/m
2	Bearing housing Bore diameter	As per drawing	[illegible]	100.010
3	Bearing housing center height difference	0.01 mm	—	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	—	0.005

Motor End

Sl No	Description	Part Number	Symbol	Depth / width
1	Flange cover	C24-291-8123	T1	13.820
2	Bearing B1	—	T2	—
3	Bearing B2	NSK	T3	14.940
4	Bearing B3	—	T4	14.945
5	Spacer cylindrical B4	—	T5	—
6	Flange cover	C24-291-8123	T6	—
7	Housing Bearing	C37-701-8123	T7	57.006

IV. Flange cover thickness "T6" = T7 - (T1+T2+T3+T4+T5)+0.05 70.1/10

"T6" = 13.380 mm

Brake End

Sl No	Description	Part Number	Symbol	Depth / width
1	Flange cover	—	T1	—
2	Bearing B1	—	T2	—
3	Bearing B2	—	T3	—
4	Bearing B3	6305C3	T4	—
5	Spacer cylindrical B4	—	T5	—
6	Flange cover	C24-280.8123	T6	—
7	Housing Bearing	C37-0154-8123	T7	43.490

V. Flange cover thickness "T6" = T7 - (T1+T2+T3+T4+T5)+0.05 0154/04

"T6" = mm

VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is with in 0.030 mm.	—
3	Ensure free rotation of ball screw assembly.	—
4	Ensure ball screw run out with in 0.010 mm	—
5	Ensure ball Screw report attached to Inspection Report.	—
6	Ensure all screw tightness & marking neatly in yellow marker	—

18-11-09 Date:	Checked by P.M.Manju	Approved by BLM

ACE MANUFACTURING SYSTEMS LTD.

STANDARD TEST REPORT

1) HEAVY CUT REPORT :

CUSTOMER : M/s. Mahaveer die casters
M/C NO. : M30-209
TYPE OF MOTOR : β 12/8000 i
PULLEY RATIO : 1:1

BASE SPEED OF MOTOR : 1100 rpm
SPINDLE SPEED RANGE : 80-8000 rpm
MOTOR SPEED RANGE : 80-8000 rpm
MOTOR CAPACITY : 11/7.5Kw

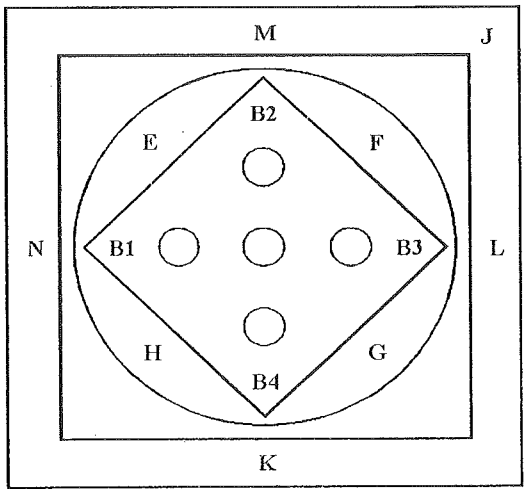
MATERIAL : C.I.(Gr 25)
CUTTER : Ø80 Sq.Sh FMC
WIDTH OF CUT : 55 mm
C.S. (V) : 250 mtr/min.

Spindle speed in rpm	Feed rate mm/ins.	Feed rate mm/rev.	Feed rate mm/min.	Depth Of Cut in mm	M R R cc/min.	'X' Axis % as per CRT	Spl.load % as per CRT	Remarks
1000	0.15	1.05	1050	3.0	173	20	60	SMOOTH CUTTING
1000	0.15	1.05	1050	5.2	300	26	96	SMOOTH CUTTING

RIGID TAPPING : M8×1.25 RIGID TAPPING TEST CONDUCTED, CHECKED WITH GAUGE & FOUND OK.

2) NASA TEST REPORT :

sl. No.	Description of Test	Test Applied to	Permissible deviation	Actual Readings
1	Flatness of Surface	A	0.015	0.005
2	Parallelism of Surfaces	N-L	0.020	0.004
		M-K	0.020	0.006
		H-F	0.030	0.004
		G-E	0.030	0.005
3	Squareness of side faces	K-L	0.020	0.010
		L-M	0.020	0.009
		M-N	0.020	0.007
		N-K	0.020	0.009
		E-F	0.030	0.005
		F-G	0.030	0.008
		G-H	0.030	0.004
		H-E	0.030	0.002
4	Circularity of Machining	J	0.040	0.009
5	Linear pitch in ' X ' axis	B1-B3 (100)	0.025	100.000
	Linear pitch in ' Y ' axis	B2-B4 (100)	0.025	99.998
	Diagonal pitch	C(70.71)	0.035	70.715
6	Bore size variation		0.020	
	Hole No.	Max.	Min.	Ovality
	B1=	37.450	37.445	0.005
	B2=	37.457	37.454	0.003
	B3=	37.451	37.449	0.002
	B4=	37.449	37.445	0.004
7	Strightness	K	0.010	0.003
		L	0.010	0.003
		M	0.010	0.005
		N	0.010	0.003
		E	0.015	0.005
		F	0.015	0.002
		G	0.015	0.003
		H	0.015	0.002
8	Step difference in axis change over		0.010	0.004



SINGLE AXIS CONTROL K=L=M=N = 240 mm

DOUBLE AXIS CONTROL E=F=G=H = 160 mm

CIRCLE " J " = Ø 224.2 mm

Heavy Cut Done By	NASA Test Done By	QC / CMM
PRA	KM	
Date: 9/12/2009	Date: 14/12/2009	Date: 14/12/09

F-TSG-34-R0

ACE MANUFACTURING SYSTEMS LTD

SPINDLE ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

Machine Number : M30-209	Spindle Assembly Number: 08647
Spindle Drive ID / ID -CTS / DD / DD -CTS	Spindle Speed : 8000 RPM
Spindle number CH-276	Cartridge number ST-11
Pull stud type DIN / MAS mas	Draw bar Type ID / ID-CTS / DD / DD-CTS
Bearing Make & Specification : FAG B7011C.T.DUL.SUL	Drawbar serial number 1392

Spindle Nose Taper ISO-30 / 40 / 50 / HSK50 / 63 / 100

SL No.	Description	Permissible	Actual
1	Spindle Bore Seating	Min 70%	70%
2	Locknut Tightening Torque	Refer Data Sheet	115 Nm
3	Spindle Bearing Preload	Refer Data Sheet	-0.010 MM
4	Spindle Runout on taper	0.002 mm	0.002 MM
5	Mandrel Runout near the spindle nose	0.005 mm	0.002 MM
6	Mandrel Runout at 300mm Length @ 4 positions	0.010 mm	1. 3 2. 2 3. 4 4. 3
7	Spindle Axial Play	0.005 mm	
8	Spindle stabilization on Test rig in deg C.	Up to 15° C above the ambient	21°C-32°C
9	Spindle Noise:	< 75 dB(A)	✓
10	Ensure Air coming in taper nose during orientation position.	Refer Data Sheet	
11	Pulley ratio (Not applicable for DD machine)	Refer Work order	—
12	Number of Teeths on spindle pulley (Not applicable for DD machine)	Refer Data Sheet	—
13	Spindle pulley Runout (Not applicable for DD m/c)	0.050 mm	—
14	Spindle connector Runout (Not applicable for ID machine)	0.010 mm	0-010M
15	Disc spring specification (Single / double stack)	Refer Data Sheet	Ø140XØ208X3mm
16	Number of disc springs used	Refer Data Sheet	56 Nos
17	Draw bar Compression Free length -Compression length	Refer Data Sheet	228-210.5=17.5mm
18	Tool clamping force	Refer Data Sheet	7208 —
19	ID-CTS Deublin Runout in Assembly condition (For CTS Spindle) Sl. No.	0.030 mm	✓ —

Date: 30/11/09	mahesh	Checked By	Approved By

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

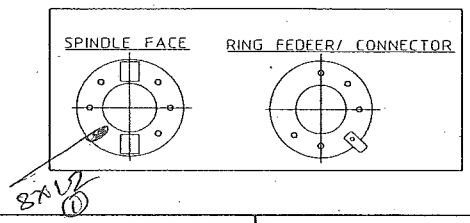
SPINDLE BALANCING REPORT

Machine Number : M30 - 209

Spindle Serial Number : 08647

Spindle Type : 0087d

Spindle Speed : 8000 rpm



Sl No.	Speed in RPM	Before Balancing Along X-axis	After Balancing along X-axis (Maximum -0.8 mm /sec)	After Balancing along Y-axis (Maximum -0.8 mm /sec)
1	500			
2	1000	0·01	0·01	0·01
3	1500	0·07	0·04	0·01
4	2000	0·05	0·06	0·01
5	2500	0·03	0·03	0·05
6	3000	0·14	0·08	0·04
7	3500	0·35	0·20	0·03
8	4000	0·25	0·17	0·09
9	4500	0·15	0·19	0·03
10	5000	0·14	0·13	0·11
11	5500	0·43	0·09	0·13
12	6000	0·52	0·07	0·14
13	6500	0·60	0·42	0·14
14	7000	1·33	0·12	0·13
15	7500	1·80	0·52	0·20
16	8000	1·52	0·47	0·19
17	8500			
18	9000			
19	9500			
20	10000			
21	10500			
22	11000			
23	11500			
24	12000			
25	12500			
26	13000			
27	13500			
28	14000			
29	14500			
30	15000			

Spindle maximum vibration mm / sec : 0·52

Correction in gms : 0.8 gms

Date: 11/12/09

Checked By

APPROVED BY

ACE MANUFACTURING SYSTEMS LTD

TOOL DE CLAMP ASSEMBLY - INSPECTION REPORT

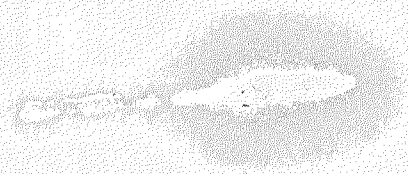
AMS
In Pursuit of Excellence

MACHINE NUMBER : RC-TDL-27-09 M30-209

SL No.	Description	Permissible	Actual
1	De clamp lever roller height variation:	0.050 mm	0.042 mm
2			

Sl No	Check points:	Remark
1	Check bell crank lever / De clamp lever roller rotating freely	✓
2	Check play in the de clamp lever and key.	✓
3	Ensure Retainer spring holds the de clamp lever with tension.	✓
4	Check piston movement in hydraulic de clamp assembly	-NA-
5	Ensure no oil leakage from cylinder & fittings	-NA-
6	Ensure tightness of screws & marking in yellow marker.	✓

5/12/09		
Date:	Checked by	Approved by



RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M30-209-X.RTL

Machine:MCV-450.
Date:13:38 Dec 12 2009
Axis:X.

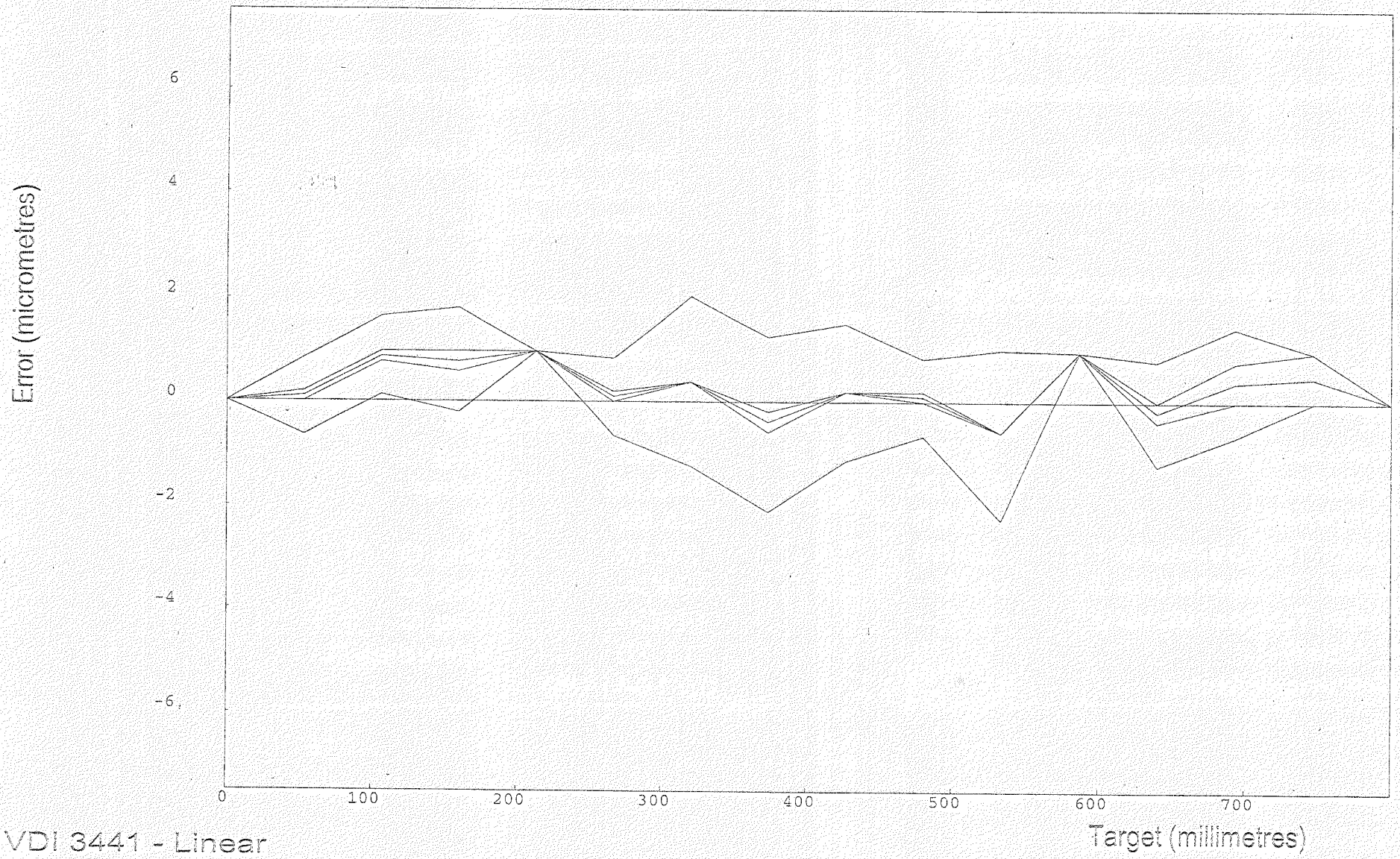
Serial No:M30-209.
By:QAS/SV.
Location:AMS.

TITLE:M/s.MAHAVIR DIE CASTINGS.

Data	Start	End	Units
Time :	13:38 Dec 12 2009	13:48 Dec 12 2009	
Air temperature :	24.80	24.82	deg C
Air pressure :	908.00	908.00	mbar
Rel. humidity :	66.65	66.68	%
Material Temperature 1 :	22.06	22.12	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31641211	0.31641189	

VDI 3441

M/s.MAHAVIR DIE CASTINGS.

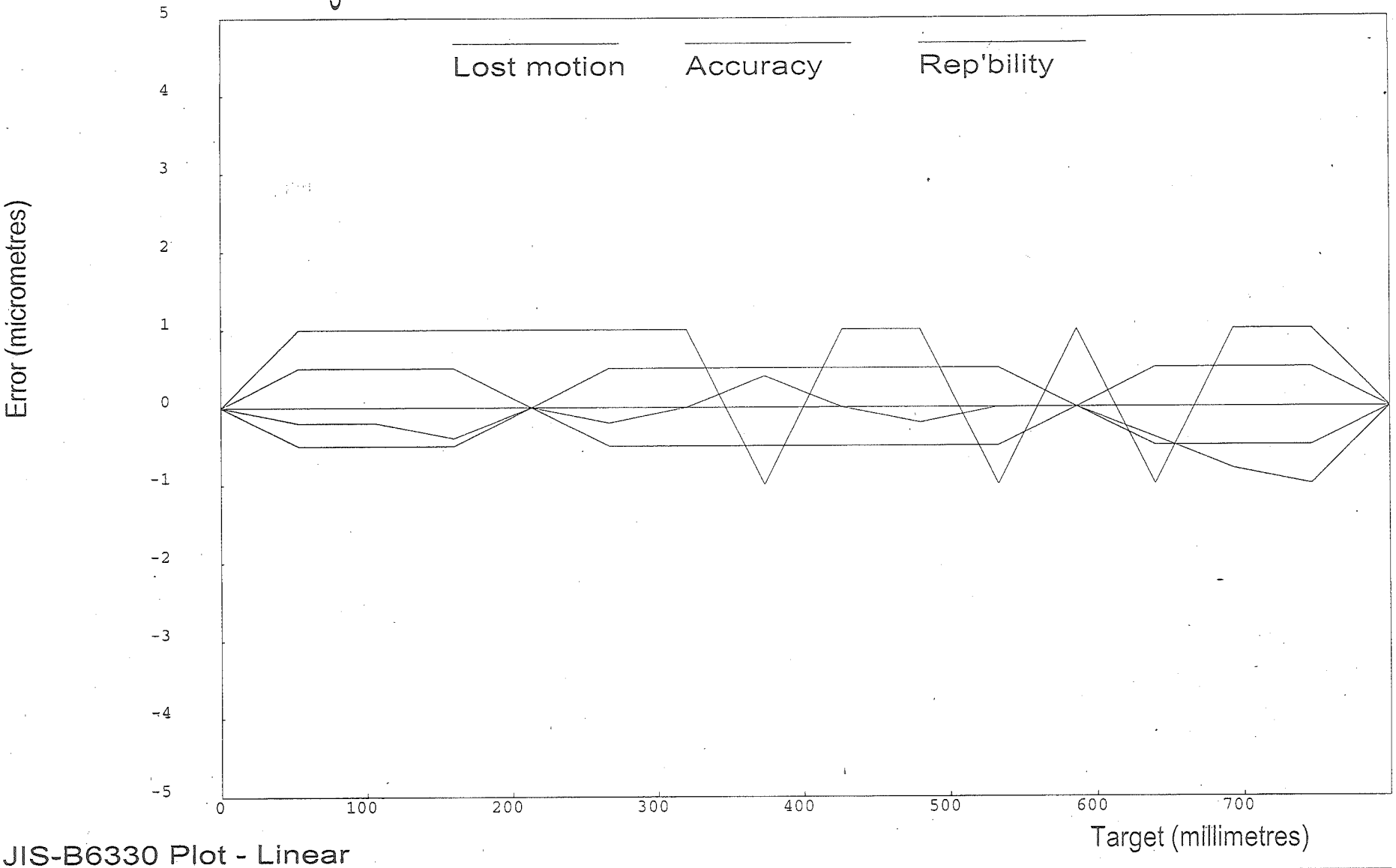


VDI 3441 - Linear

Machine:MCV-450.	Axis:X.	U max : 1.000
Serial No:M30-209.	Location:AMS.	P : 4.286
Date:13:38 Dec 12 2009	U mean : 0.238	Ps max : 3.286
By:QAS/SV.	Ps mean : 1.390	Pa : 1.600

Positioning Error

M/s.MAHAVIR DIE CASTINGS.



JIS-B6330 Plot - Linear

Machine:MCV-450. Serial No:M30-209. Date:13:38 Dec 12 2009 By:QAS/SV.	Axis:X. Location:AMS.	Lost motion: -1.000 Accuracy: 1.000 Rep'bility: 0.500

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-450. Serial No:M30-209.
Date:13:38 Dec 12 2009 By:QAS/SV.
Axis:X. Location:AMS.
TITLE:M/s.MAHAVIR DIE CASTING Filename: M30-209-X.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (53.3330 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 2 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 3 (+) (106.6660 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 0.0
Target 3 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 4 (+) (159.9990 millimetres)
1) 1.0 2) 0.0 3) 1.0 4) 1.0
5) 0.0
Target 4 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 5 (+) (213.3320 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 5 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 6 (+) (266.6650 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 6 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 7 (+) (319.9980 millimetres)
1) 0.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0
Target 7 (-)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 8 (+) (373.3310 millimetres)
1) -1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) 0.0

Target 9 (+) (426.6640 millimetres)
1) 0.0 2) 1.0 3) 0.0 4) 0.0
5) 0.0
Target 9 (-)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 10 (+) (479.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 10 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 11 (+) (533.3300 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) 0.0
5) -1.0
Target 11 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) 0.0

Target 12 (+) (586.6630 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 12 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 13 (+) (639.9960 millimetres)
1) -1.0 2) 0.0 3) -1.0 4) 0.0
5) 0.0
Target 13 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 14 (+) (693.3290 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
arget 14 (-)
1) 1.0 2) 1.0 3) 0.0 4) 1.0
5) 1.0

Target 15 (+) (746.6620 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 15 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 16 (+) (799.9950 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 16 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M30-209-Y.RTL

Machine:MCV-450.
Date:14:48 Dec 12 2009
Axis:Y.

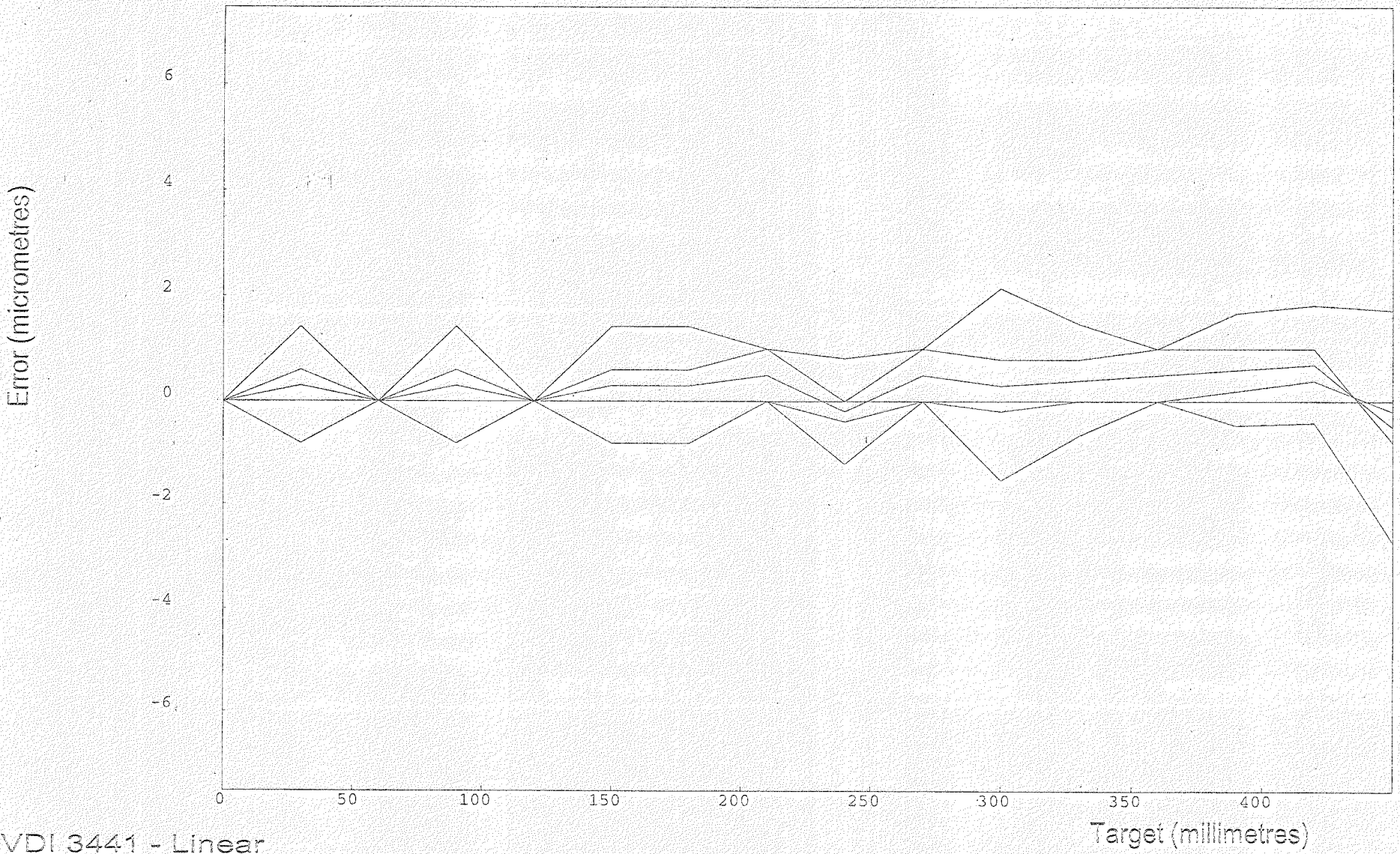
Serial No:M30-209.
By:QAS/SV.
Location:AMS.

TITLE:M/s.MAHAVIR DIE CASTINGS.

Data	Start	End	Units
Time :	14:50 Dec 12 2009	14:59 Dec 12 2009	
Air temperature :	24.49	24.46	deg C
Air pressure :	907.80	907.80	mbar
Rel. humidity :	69.58	69.84	%
Material Temperature 1 :	22.47	22.52	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31641055	0.31641037	

VDI 3441

M/s.MAHAVIR DIE CASTINGS.

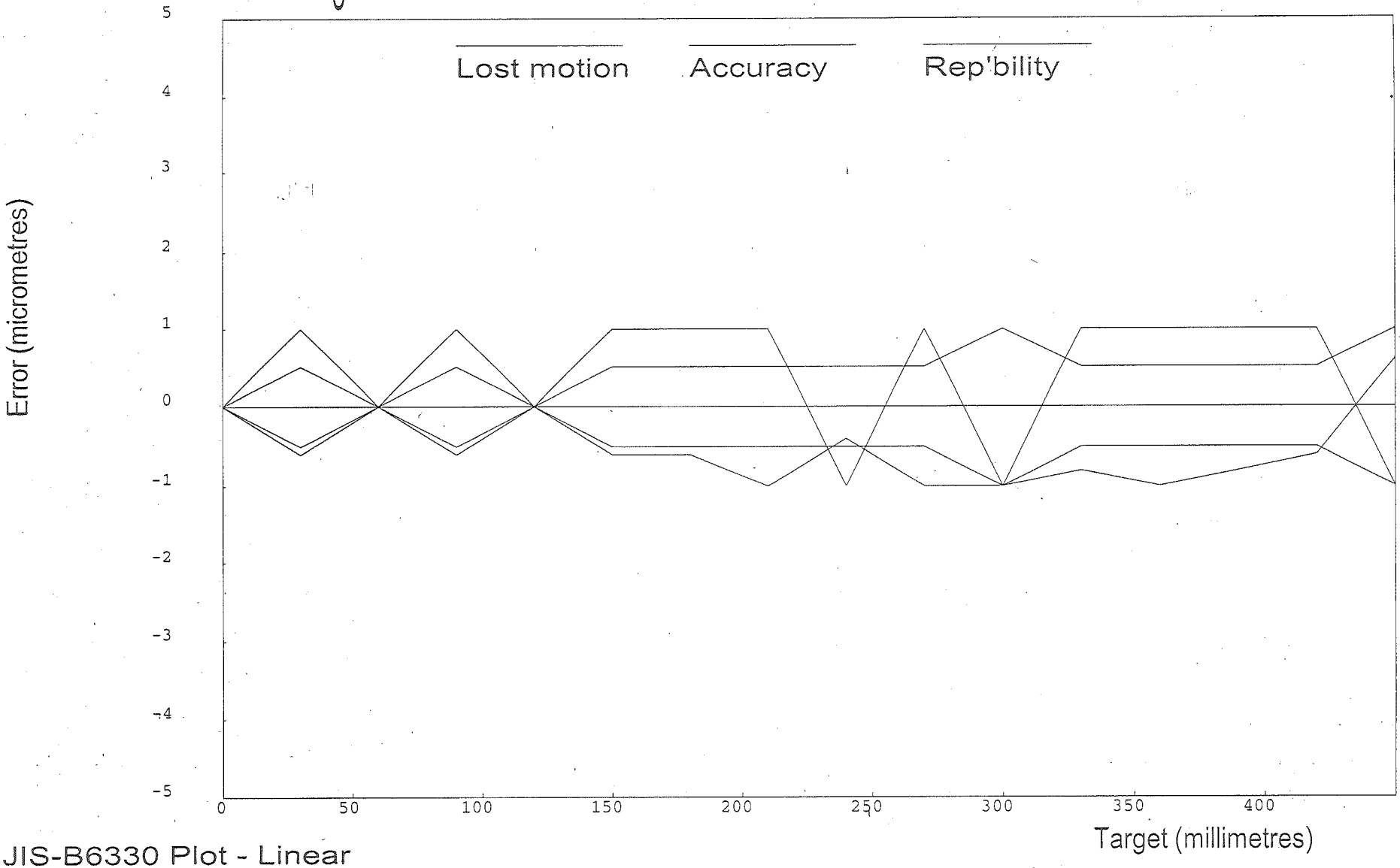


VDI 3441 - Linear

Machine:MCV-450.	Axis:Y.	U max : 1.000
Serial No:M30-209.	Location:AMS.	P : 4.867
Date:14:48 Dec 12 2009	U mean : 0.600	Ps max : 3.852
By:QAS/SV.	Ps mean : 1.192	Pa : 1.200

Positioning Error

M/s.MAHAVIR DIE CASTINGS.



JIS-B6330 Plot - Linear

Machine:MCV-450. Serial No:M30-209. Date:14:48 Dec 12 2009 By:QAS/SV.	Axis:Y. Location:AMS.	Lost motion: -1.000 Accuracy: -1.000 Rep'bility: 1.000

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-450. Serial No:M30-209.
Date:14:48 Dec 12 2009 By:QAS/SV.
Axis:Y. Location:AMS.
TITLE:M/s.MAHAVIR DIE CASTING Filename: M30-209-Y.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (29.9990 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 2 (-)
1) 1.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0

Target 3 (+) (59.9980 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 3 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 4 (+) (89.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 4 (-)
1) 1.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0

Target 5 (+) (119.9960 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 5 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 6 (+) (149.9950 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 6 (-)
1) 1.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0

Target 7 (+) (179.9940 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 7 (-)
1) 1.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0

Target 8 (+) (209.9930 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 9 (+) (239.9920 millimetres)
1) 0.0 2) -1.0 3) 0.0 4) -1.0
5) 0.0
Target 9 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 10 (+) (269.9910 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 10 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 11 (+) (299.9900 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) 0.0
Target 11 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 12 (+) (329.9890 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 12 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 13 (+) (359.9880 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 13 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 14 (+) (389.9870 millimetres)
1) 0.0 2) 1.0 3) 0.0 4) 0.0
5) 0.0
arget 14 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 15 (+) (419.9860 millimetres)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0
Target 15 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 16 (+) (449.9850 millimetres)
1) 1.0 2) -1.0 3) -1.0 4) 0.0
5) 0.0
Target 16 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M30-209-Z.RTL

Machine:MCV-450.
Date:16:13 Dec 12 2009
Axis:Z.

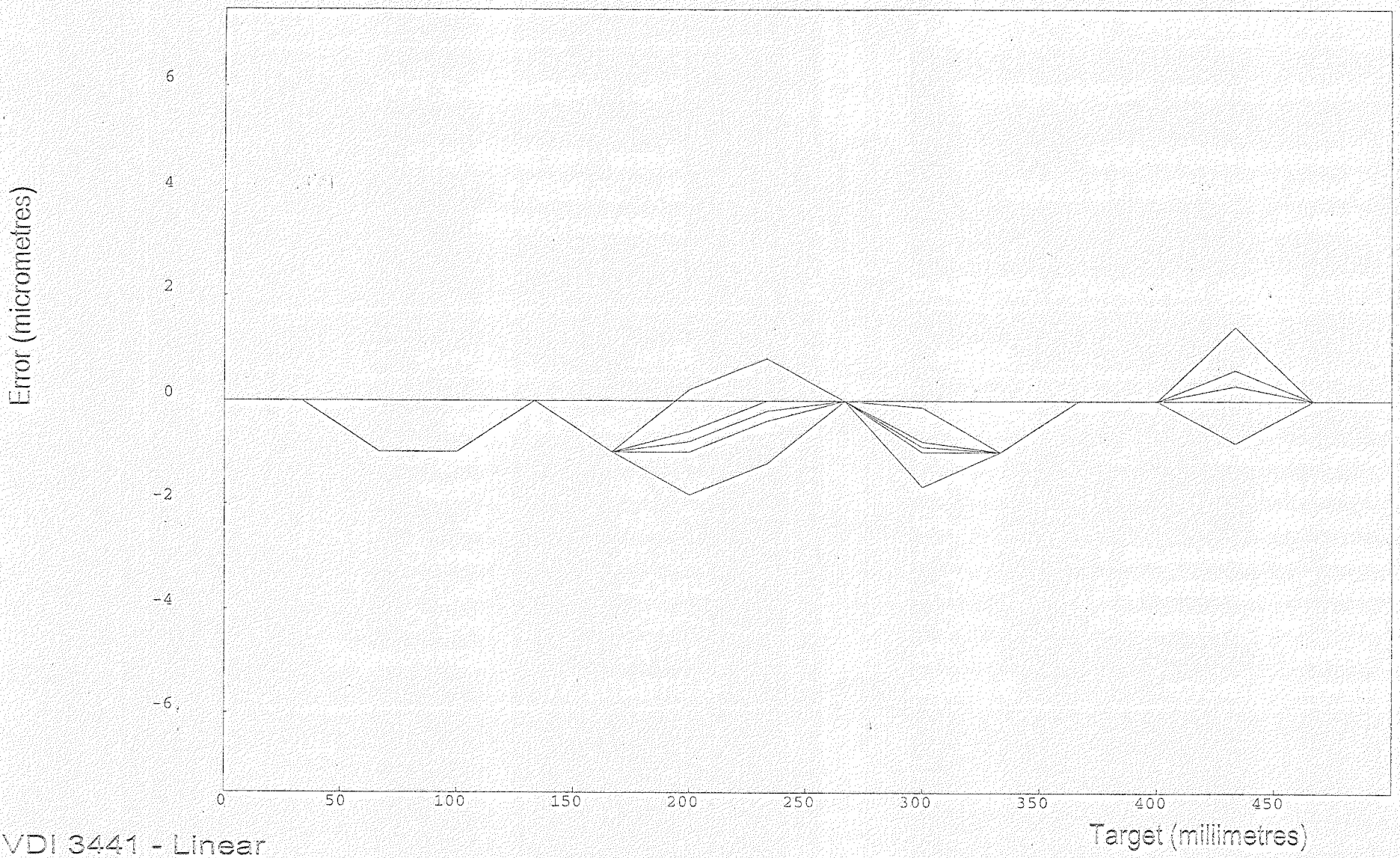
Serial No:M30-209.
By:QAS/SV.
Location:AMS.

TITLE:M/s.MAHAVIR DIE CASTINGS.

Data	Start	End	Units
Time :	16:13 Dec 12 2009	16:22 Dec 12 2009	
Air temperature :	24.15	24.15	deg C
Air pressure :	907.80	907.80	mbar
Rel. humidity :	71.49	71.64	%
Material Temperature 1 :	22.76	22.78	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31640940	0.31640931	

VDI 3441

M/s.MAHAVIR DIE CASTINGS.

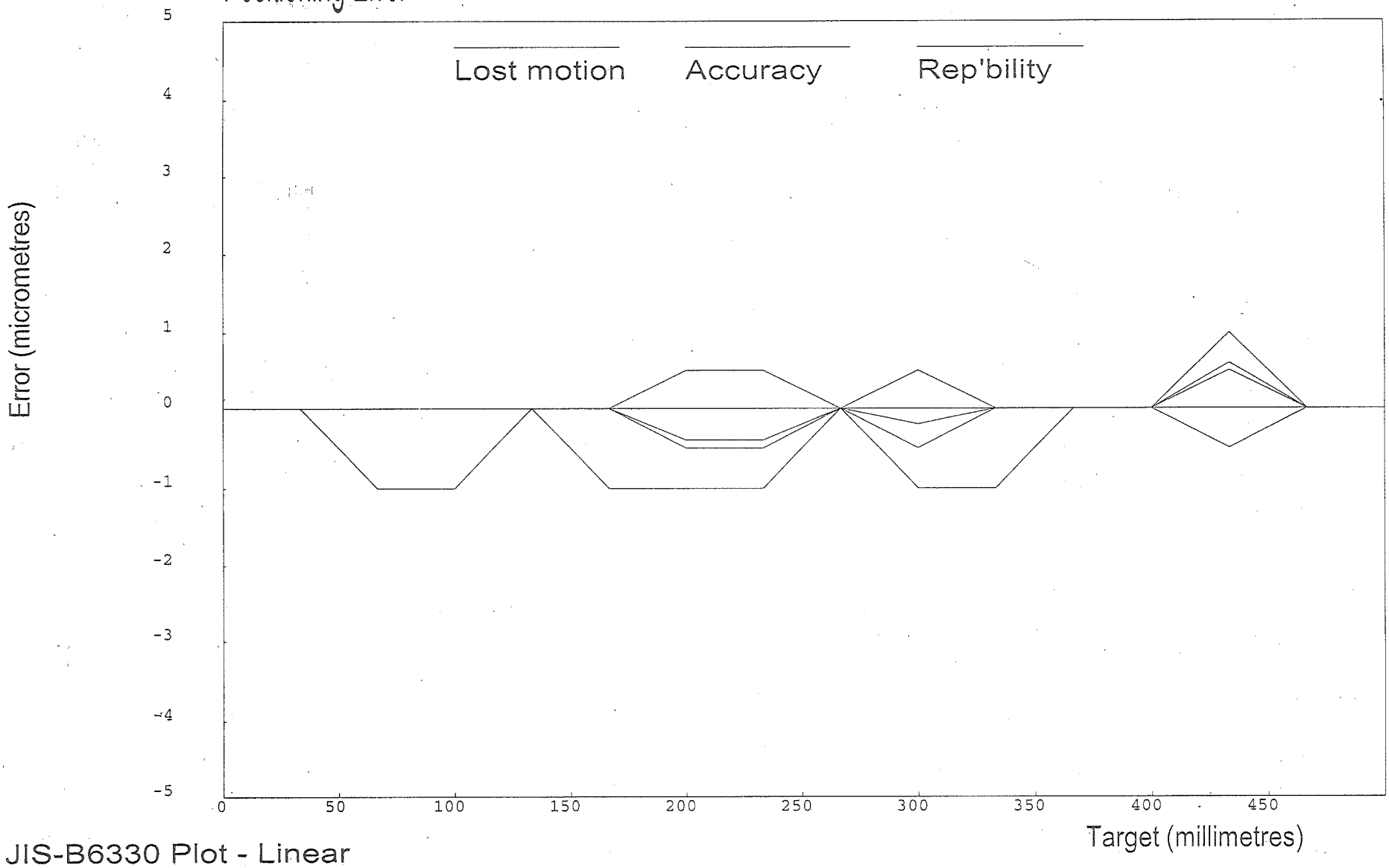


VDI 3441 - Linear

Machine:MCV-450.	Axis:Z.	U max : 0.600
Serial No:M30-209.	Location:AMS.	P : 3.243
Date:16:13 Dec 12 2009	U mean : 0.100	Ps max : 1.643
By:QAS/SV.	Ps mean : 0.392	Pa : 1.300

Positioning Error

M/s.MAHAVIR DIE CASTINGS.



JIS-B6330 Plot - Linear

Machine:MCV-450. Serial No:M30-209. Date:16:13 Dec 12 2009 By:QAS/SV.	Axis:Z. Location:AMS.	Lost motion: 0.600 Accuracy: 1.000 Rep'bility: 0.500

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-450. Serial No:M30-209.
Date:16:13 Dec 12 2009 By:QAS/SV.
Axis:Z. Location:AMS.
TITLE:M/s.MAHAVIR DIE CASTING Filename: M30-209-Z.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (33.3330 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 2 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 3 (+) (66.6660 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 3 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 4 (+) (99.9990 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 4 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 5 (+) (133.3320 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 5 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 6 (+) (166.6650 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 6 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 7 (+) (199.9980 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 7 (-)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0

Target 8 (+) (233.3310 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) -1.0

Target 8 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 9 (+) (266.6640 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 9 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 10 (+) (299.9970 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 10 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) 0.0

Target 11 (+) (333.3300 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 11 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 12 (+) (366.6630 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 12 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 13 (+) (399.9960 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 13 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 14 (+) (433.3290 millimetres)
1) 1.0 2) 1.0 3) 0.0 4) 0.0
5) 1.0
arget 14 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 15 (+) (466.6620 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 15 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 16 (+) (499.9950 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 16 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Ballbar diagnostics table
M30-209-XY-B

Operator: Shankar
Date: 2009-Dec-12 17:01:44

RENISHAW

Machine: Quick check
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash X	▸-0.4	◂0.9 µm	0.9 µm	(4%)	(6)
Backlash Y	▴0.7	▾-0.2 µm	0.7 µm	(2%)	(9)
Reversal spikes X	▸-2.7	◂-3.1 µm	3.1 µm	(12%)	(3)
Reversal spikes Y	▴-5.2	▾-5.4 µm	5.4 µm	(21%)	(2)
Lateral play X	▸-0.7	◂0.8 µm	0.4 µm	(2%)	(11)
Lateral play Y	▴1.0	▾0.1 µm	0.6 µm	(2%)	(10)
Cyclic error X	↑0.7	↓0.6 µm	0.7 µm	(3%)	(8)
Cyclic error Y	↑0.8	↓0.9 µm	0.9 µm	(3%)	(7)
Servo mismatch	0.01 ms		0.2 µm	(1%)	(13)
Squareness	42.2 µm/m		6.3 µm	(24%)	(1)
Straightness X	-0.4 µm		0.2 µm	(1%)	(12)
Straightness Y	2.8 µm		1.4 µm	(5%)	(4)
Scaling mismatch	-2.8 µm		1.4 µm	(5%)	(5)
Cyclic pitch X	19.0500 mm				
Cyclic pitch Y	25.4000 mm				
Scaling error X	-150.2 ppm				
Scaling error Y	-141.0 ppm				
Calculated feedrate	1000.2 mm/min				
Centre offset X	3.9 µm				
Centre offset Y	4.7 µm				
Positional tolerance	126.3 µm				
Best fit radius	149.9782 mm				
Circularity	10.7 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/0°/180°
Run sequence	CW CCW
Test position	
Sample rate	27.778Hz

ISO 230-4:1996(E) Circular Deviation
M30-209-XY-B

Operator: Shankar
Date: 2009-Dec-12 17:01:44

RENISHAW

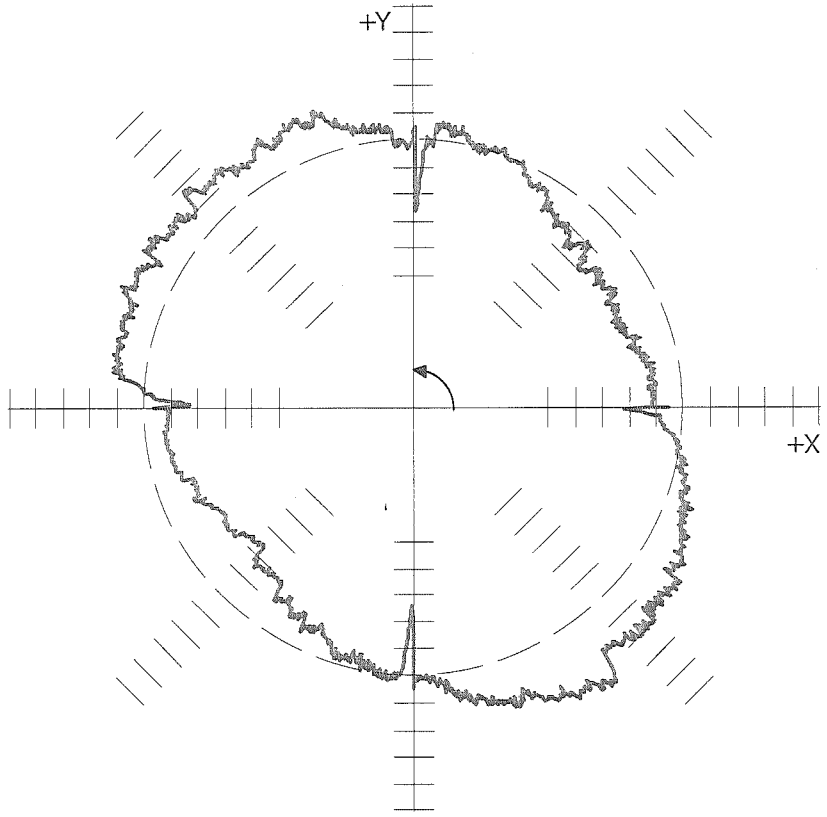
Machine: Quick check
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	10.7µm

Test Parameters

Radius	150.0000mm
Sample rate	27.778Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	XY
Test position	
Start angle	0°
End angle	0°
Overshoot angle	180°



Run 1

2.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M30-209-XY-B

Operator: Shankar
Date: 2009-Dec-12 17:01:44

RENISHAW

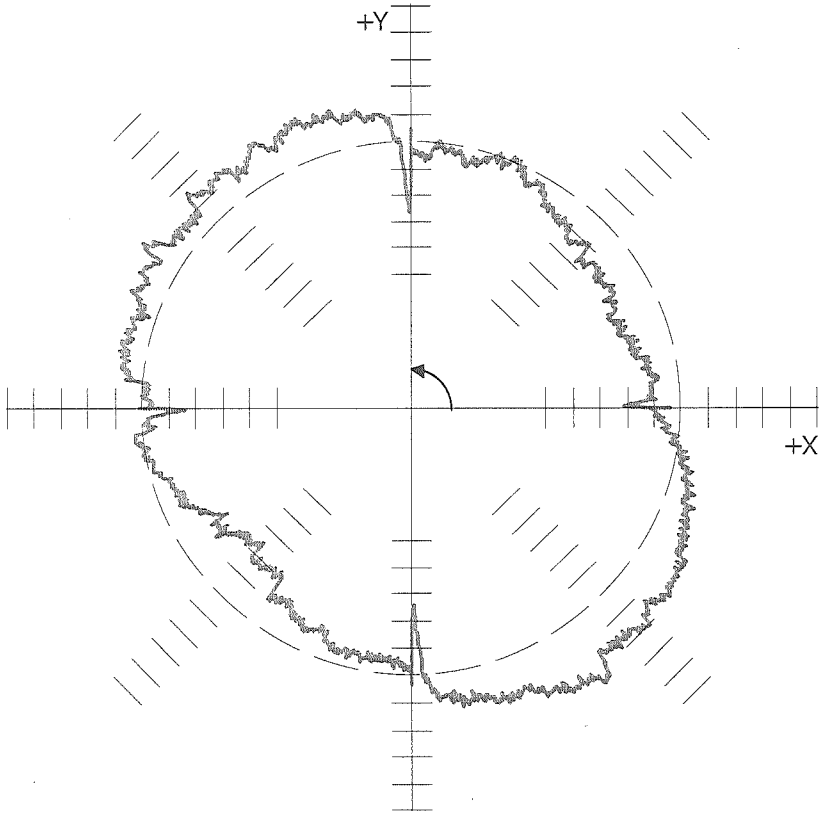
Machine: Quick check
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	10.3µm

Test Parameters

Radius	150.0000mm
Sample rate	27.778Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	XY
Test position	
Start angle	0°
End angle	0°
Overshoot angle	180°



Run 2

2.0 µm/div

Ballbar diagnostics table
M30-209-XZ-B

Operator: Shankar
Date: 2009-Dec-12 17:21:09

RENISHAW

Machine: Quick check
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash Z	▲-0.2	▼0.6 µm	0.6 µm	(3%)	(10)
Backlash X	▶0.0	◀-1.7 µm	1.7 µm	(7%)	(5)
Reversal spikes Z	▲-2.1	▼-2.1 µm	2.1 µm	(8%)	(4)
Reversal spikes X	▶-1.2	◀0.0 µm	1.2 µm	(5%)	(7)
Lateral play Z	▲-0.8	▼1.9 µm	1.1 µm	(4%)	(8)
Lateral play X	▶0.2	◀0.2 µm	0.2 µm	(1%)	(12)
Cyclic error Z	↑0.6	↓0.5 µm	0.6 µm	(2%)	(11)
Cyclic error X	↑1.6	↓1.3 µm	1.6 µm	(6%)	(6)
Servo mismatch	-0.01 ms		0.1 µm	(0%)	(13)
Squareness	23.5 µm/m		3.5 µm	(14%)	(1)
Straightness Z	-6.8 µm		3.4 µm	(13%)	(2)
Straightness X	5.8 µm		2.9 µm	(11%)	(3)
Scaling mismatch	-1.3 µm		0.7 µm	(3%)	(9)
Cyclic pitch Z	25.4000 mm				
Cyclic pitch X	20.0000 mm				
Scaling error Z	-139.7 ppm				
Scaling error X	-135.3 ppm				
Calculated feedrate	1000.2 mm/min				
Centre offset Z	-0.2 µm				
Centre offset X	-1.7 µm				
Positional tolerance	119.1 µm				
Best fit radius	149.9794 mm				
Circularity	10.9 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	90°/90°/180°
Run sequence	CW CCW
Test position	
Sample rate	27.778Hz

ISO 230-4:1996(E) Circular Deviation
M30-209-XZ-B

Operator: Shankar
Date: 2009-Dec-12 17:21:09

RENISHAW

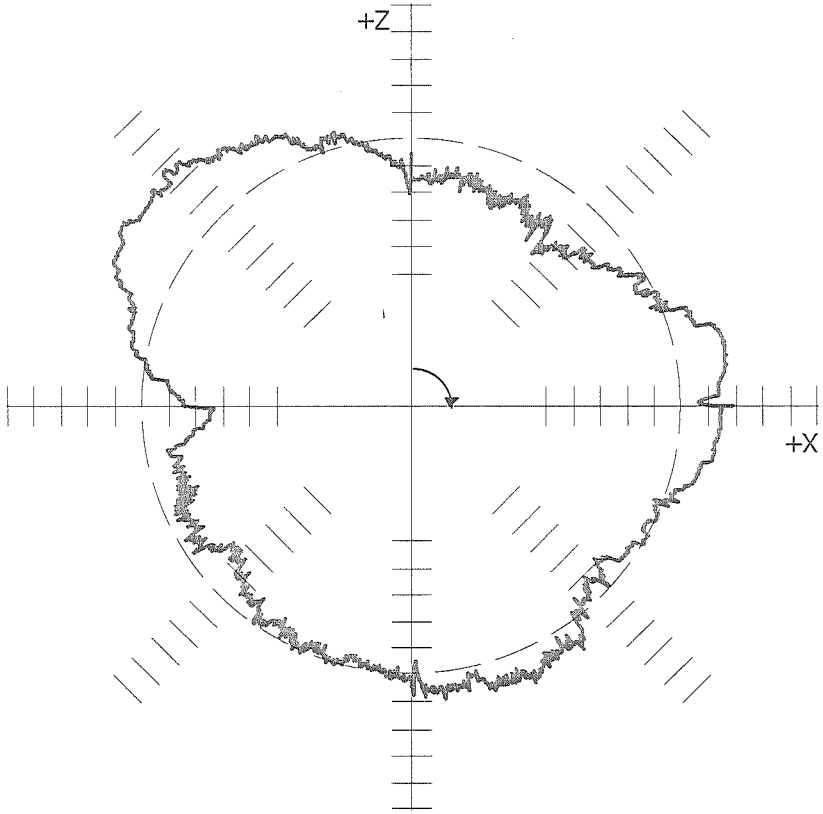
Machine: Quick check
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	10.8µm

Test Parameters

Radius	150.0000mm
Sample rate	27.778Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 1

2.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M30-209-XZ-B

Operator: Shankar
Date: 2009-Dec-12 17:21:09

RENISHAW

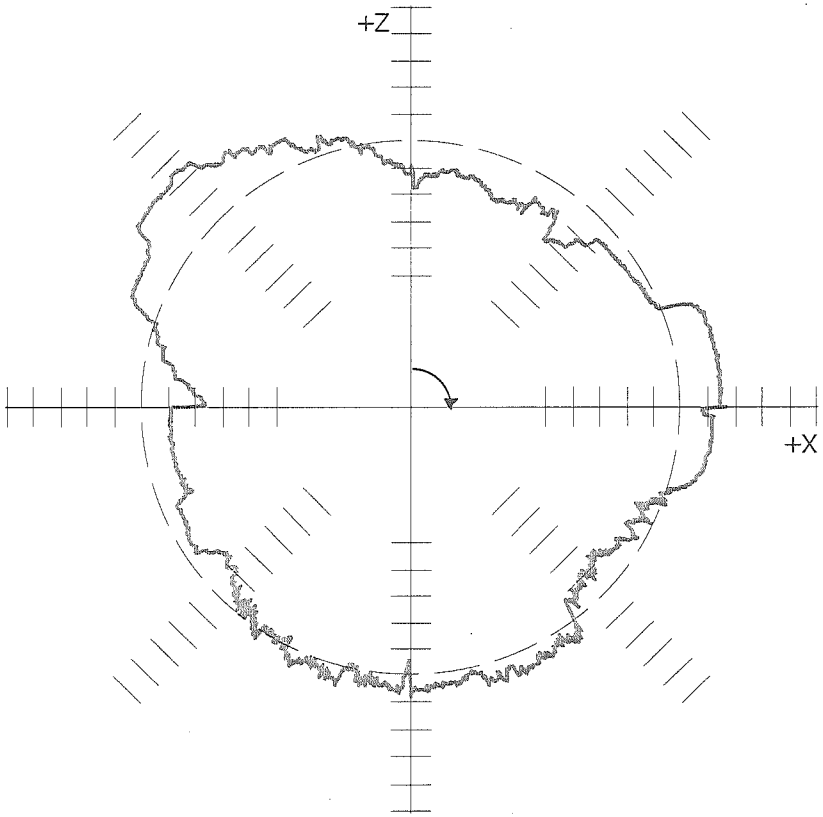
Machine: Quick check
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	9.6µm

Test Parameters

Radius	150.0000mm
Sample rate	27.778Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 2

2.0 µm/div

Ballbar diagnostics table
M30-209-YZ-A

RENISHAW

Operator: Shankar
Date: 2009-Dec-12 17:41:23

Machine: Quick check
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash Y	▸ -1.0	◂ -1.8 µm	1.8 µm	(6%)	(6)
Backlash Z	▴ -0.6	▾ 0.1 µm	0.6 µm	(2%)	(11)
Reversal spikes Y	▸ 0.9	◂ 0.0 µm	0.9 µm	(3%)	(9)
Reversal spikes Z	▴ -2.4	▾ -2.2 µm	2.4 µm	(8%)	(3)
Lateral play Y	▸ -3.7	◂ 1.5 µm	2.1 µm	(8%)	(5)
Lateral play Z	▴ 0.2	▾ -0.4 µm	0.2 µm	(1%)	(12)
Cyclic error Y	↑0.9	↓1.1 µm	1.1 µm	(4%)	(8)
Cyclic error Z	↑0.7	↓0.8 µm	0.8 µm	(3%)	(10)
Servo mismatch	0.01 ms		0.2 µm	(1%)	(13)
Squareness	-37.7 µm/m		5.6 µm	(20%)	(1)
Straightness Y	9.3 µm		4.6 µm	(16%)	(2)
Straightness Z	-3.2 µm		1.6 µm	(6%)	(7)
Scaling mismatch	4.7 µm		2.3 µm	(8%)	(4)
Cyclic pitch Y	20.0000 mm				
Cyclic pitch Z	25.4000 mm				
Scaling error Y	-114.0 ppm				
Scaling error Z	-129.6 ppm				
Calculated feedrate	1000.5 mm/min				
Centre offset Y	0.2 µm				
Centre offset Z	0.6 µm				
Positional tolerance	108.1 µm				
Best fit radius	149.9817 mm				
Circularity	15.7 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW
Test position	
Sample rate	27.778Hz

ISO 230-4:1996(E) Circular Deviation
M30-209-YZ-A

Operator: Shankar
Date: 2009-Dec-12 17:41:23

RENISHAW

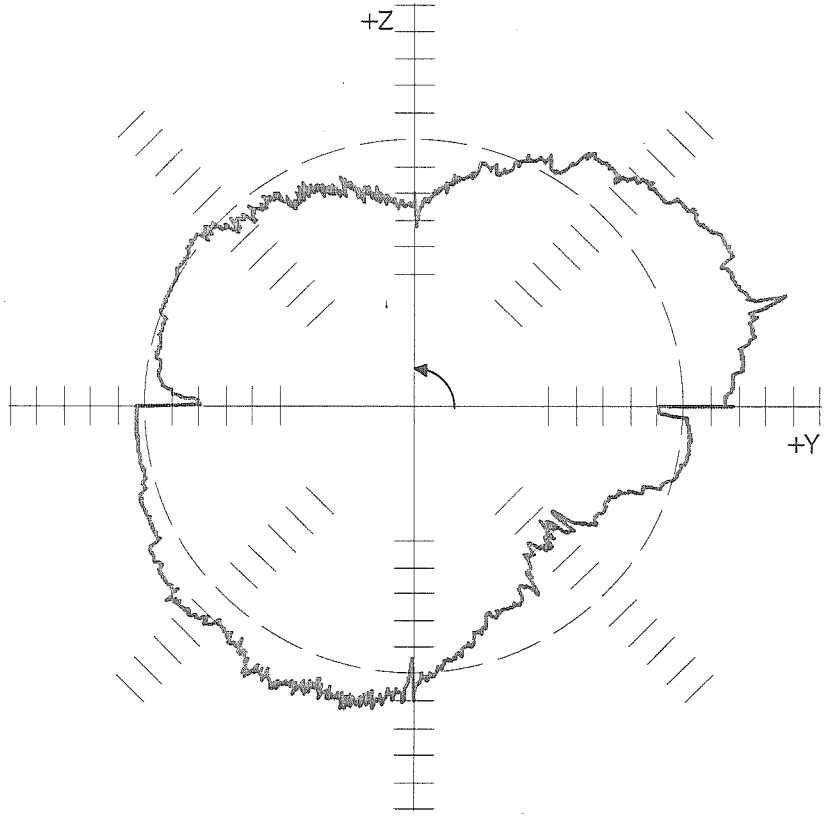
Machine: Quick check
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	16.1µm

Test Parameters

Radius	150.0000mm
Sample rate	27.778Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

2.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M30-209-YZ-A

Operator: Shankar
Date: 2009-Dec-12 17:41:23

RENISHAW

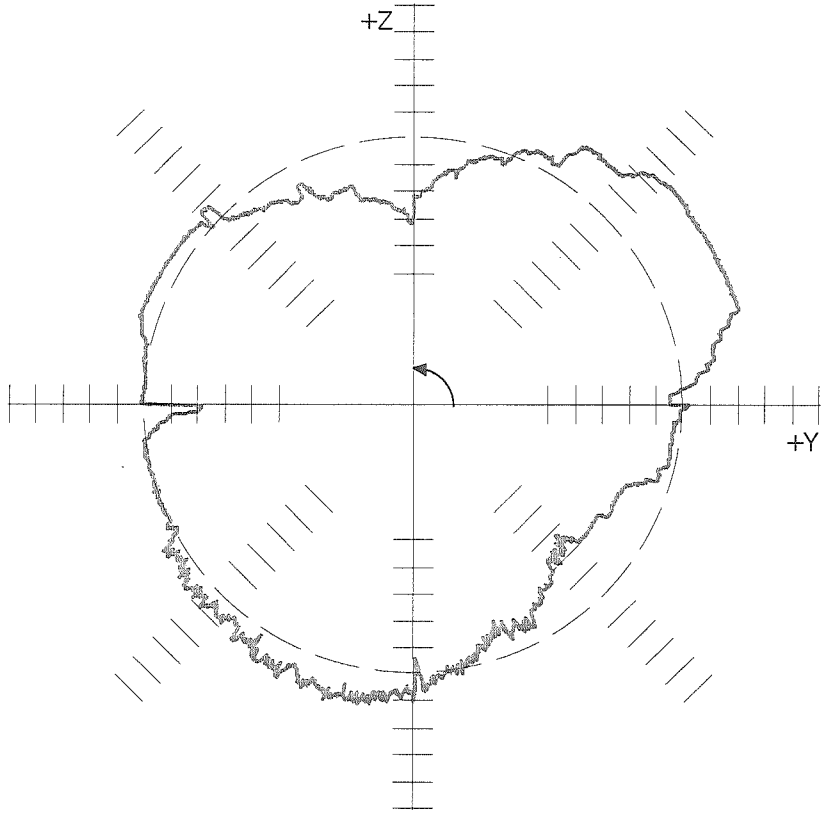
Machine: Quick check
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	11.4µm

Test Parameters

Radius	150.0000mm
Sample rate	27.778Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

2.0 µm/div

ACE MANUFACTURING SYSTEMS LTD

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FINAL ASSEMBLY - INSPECTION REPORT

CHECK POINTS M30-2Cg

I.AXES

SL.No	PARAMETERS	STATUS	REMARKS
1	Specifications of servo motors are as per work order	OK	
2	Axes strokes are as per work order. (P1320 onwards)	OK	
3	All servomotors are taking normal current while running	OK	
4	Motor ID & other parameters are matching with the motor (P2020)	OK	
5	Axis cables are routed properly with suitable clamps & ties	OK	
6	Cables are connected properly both at motor & drive end	OK	
7	Lubrication timers & signals are as required to the machine	OK	
8	Rapid rate & feed rate of axes as per specification (P1420/1428)	OK	

II.SPINDLE

SL.No	PARAMETERS	STATUS	REMARKS
1	Specifications of spindle motor /speed /power is as per work order	OK	
2	Speed locking parameters If any are perfectly set (P3735,3736,)	OK	
3	Power locking parameters If any are perfectly set (P4028,4029)	OK	
4	Rigid tap parameters are set (P4016#4,4044,4052,4085,5241,5261)	OK	
5	Spi. running is checked for direction, noise if any, low & high speed	OK	
6	Spindle motor fan rotation / direction is checked	OK	
7	Spindle orientation is perfectly set (Proximity signal, parameter)	OK	
8	Manual tool declamp in jog is checked in twin arm models	OK	
9	Switching of Contactors for required speed incase of star/delta mootr	–	
10	Cables are connected properly both at motor & drive end	OK	

III.ATC

SL.No	PARAMETERS	STATUS	REMARKS
1	Proper tool change program(**O9001**) is loaded to the system	OK	
2	Proper macro is loaded in twin arm machines	OK	
3	Check for Magazine indexing in both directions	OK	
4	Auto tool change is checked both in jog & auto mode	OK	
5	Position switch parameters related to tool change are set	OK	
6	Proximity sw cables are properly connected with tags & routed neatly	OK	
7	Tool display is matching with atc position	OK	
8	Power OFF conditions - Message and retrival are checked	OK	

IV.APC

SL.No	PARAMETERS	STATUS	REMARKS
1	Proper pallet change program(**O9003&100**) is loaded to the system		
2	APC is operating manually through mop & in auto through **M80**		
3	Proximity sw & limit sw signals are checked for operation during		
	different condition		
4	Working of pallet setup sequence		
6	Interrupt conditions checked & initialization procedure is displayed	NA	
7	Working of fixture interlock if any for pallet change is checked		
8	Cables are routed properly with suitable clamps & ties		

CHECKED BY:-	VERIFIED BY:-

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FINAL ASSEMBLY - INSPECTION REPORT

V.PNUEMATIC

SL.No	PARAMETERS	STATUS	REMARKS
1	All solenoids / pressure switches are working for specified condition	ok	
2	Messages related to pressure sw settingis appearing when disturbed	ok	
3	Suitable element tags & indicators are put	ok	
4	Cables / plugs are terminated & routed properly	ok	
5	FRL unit signals are functioning properly	—	

VI.HYDRAULIC

SL.No	PARAMETERS	STATUS	REMARKS
1	All solenoids /pressure switches are working for specified conditions		
2	Messages related to pressure sw settingis appearing when disturbed		
3	Hydraulic Motor / cooling fan direction is checked		
4	Oil level ,clog, temperature sensors if any, are working		
3	Suitable element tags & indicators are put	N/A	
4	Cables / plugs are terminated & routed properly		

VII.COOLANT

SL.No	PARAMETERS	STATUS	REMARKS
1	All coolant motors are running in cw direction	ok	
2	Timers for running all coolant / conveyor motors is checked	ok	
3	Float sw signals are functioning properly	ok	
4	Flow sw,clog sw & air blast is functioning properly in case of cts m/c	ok	
5	Door close condition for coolant 'on' is checked if it is a option	—	
6	Chip/ screw conveyors are working properly in auto & manual mode	—	
7	Limit sw / proximity sw signals are functioning properly	—	
8	MPCB used and settings done is as per the requirement	ok	

VIII.ROTARY/ INDEX TABLES

SL.No	PARAMETERS	STATUS	REMARKS
1	Table is checked for commanded positions & shortest path	ok	
2	Related logic is checked for proper running at safe position	ok	
3	Feedback signals are functioning properly	ok	
4	Related displays are marching with table position	ok	
5	Cable/ hoses are routed properly at safe position	ok	
	Rotary table primen only. checked		
	But all function checked.		

IX.PANEL

SL.No	PARAMETERS	STATUS	REMARKS
1	MCB's are proper rated & connected as per circuit diagram	ok	
2	Proper rated MPCB's are set & connected as per circuit diagram	ok	
3	Contactors' of proper rating & connected as per circuit diagram	ok	
4	Transformers of proper rating & connected as per circuit diagram	ok	
5	PCB's of proper rating & connected as per circuit diagram	ok	

CHECKED BY:-	VERIFIED BY:-

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FINAL ASSEMBLY - INSPECTION REPORT

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6	Relay boards of proper rating & connected as per circuit diagram	OK	
7	Cables of proper rating & properly crimped at connection end	OK	
8	Drive cooling fans are working normally	OK	
9	Panel cooler setting & working is checked	OK	
10	Proper identification stickers / tags are fixed for all elements	OK	
11	Cabinet is fully checked for loose connections if any	OK	
12	PCB , drive, I/O module connectors are firmly fixed	OK	
13	Machine operating panel connections are checked and routed	OK	
14	Moisture free beadings for cabinet are properly fixed	OK	
15	Moisture free beadings for operating panel are properly fixed	OK	

X.GENERAL

SL.No	PARAMETERS	STATUS	REMARKS
1	Illuminations of work lamps & indication lamps are satisfactory	OK	
2	Emergency circuit & monitor is functioning properly	OK	
3	Cables Emergency circuit & monitor is functioning properly	OK	
4	Protective covers for the elements are given if required	OK	
5	All channel covers are fixed with proper fasteners	OK	
6	Protective earth is connected for elements which require earthing	OK	
7	Safety instructions & other standard stickers are fixed on cabinet	OK	
8	PMC timer values are checked as per data sheet	OK	
9	New design changes are incorporated in the machine	OK	
	a)		
	b)		
	c)		

XI.KEEP RELAYS & PMC DATA

Instruction: Please note all keep relays which is enabled at the time of shipping .
Also give the details if it is used for special function

									REMARKS
K0	7	6	5	4	3	2	(1)	0	
K1	7	(6)	(5)	4	3	2	1	(0)	
K2	7	(6)	5	4	3	(2)	(1)	(0)	
K3	7	6	5	4	3	2	1	0	
K4	7	6	5	4	3	2	1	0	
K5	(7)	6	5	4	3	2	1	0	
K6	7	6	5	4	3	2	1	0	
K7	7	6	5	4	3	2	1	(0)	
K8	7	6	5	4	3	2	1	0	
K9	7	6	5	4	3	2	1	0	
K10	7	6	5	4	3	2	1	0	
K16	7	6	(5)	(4)	3	2	1	0	
			D5·3=1						
K			D6·1=1						
K									

CHECKED BY:-	VERIFIED BY:-

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FINAL ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

D100	00000010	0	1	✓
D110	00000000	1	1	✓
D491	00000010	0	409	✓
D1010	00000010	2	4	✓
D1124	00000010	2	2	✓
D0000	00000010	0	100	✓
D				
D				

XII.CNC PARAMETERS

Instruction: Please Set the following parameters at the time of shipping .
Also give the details if any special parameter has set

✓	1301#7 = 1 Over travel before move	OK
✓	3103#2 = 1 System alarm history display.	OK
✓	3106#4 = 1.Operation history enable.	OK
✓	3108#6 = 1 Spindle load meter	OK
✓	3111#7 = 0 (NPA) - Message display	OK
✓	3112#0 = 0 Graphics display enable	OK
✓	3112#2 = 1 Operator message history display	OK
✓	3112#3 = 1 Alarm description display in alarm history screen for external alarms	OK
✓	3112#5 = 0 Operator history enable	OK
✓	3117#1 = 1. Spindle position feedback display in dgn 442	OK
✓	3205#4 = 1 In the offset screen offset value CLEAR by soft key to be disabled.	OK
✓	3402#6 =1 (CLR)Set the parameter	OK
✓	1601#4 = 1 (RTO)Set the parameter	OK
✓	1722 = 10/10/10 Set for RTO percentage	OK
✓	1826 =(500,500,500) , 1827=(200,200,200)-check imposition width	OK
✓	6910 - 6965 -check position switches values for ATC and APC	OK
✓	8132.0=1, - TOOL LIFE Parameters	OK
✓	6800#0=1, 6801#3=1, 6801#6=1, 6810=100, 6811=6, 6844=100, 6845=5	OK
✓	1620 & 1621 Rapid time constant	OK

XIII.SHIPPING

Instruction: Please perform following activities before shipping .
Also give the details if any special activity is done

SL.No	PARAMETERS	REMARKS
✓1	Take the servo guide back up	OK
2	Check the various communication like RS232C,PCMCIA	
✓3	Edit title,& set time, date other original factory set values	OK
✓4	Take all programs back up &Lock 8000 & 9000 series programs (P3202)	OK
✓6	Take the back up of following :	OK
✓	a)CNC parameters b) Pitch error compensation c)PMC parameters	OK
✓	d) ID- Info f)M- info e) Ladder g) macro i)S-RAM data	OK
✓7	Make I/O channel '0' for RS232C communication	OK
✓8	Keep spares inside the cabinet if any	OK
✓9	Check relevant circuit diagram of latest edition	OK
✓10	Ensure all the elements are in good condition while shipping	OK

CHECKED BY:- VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

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MACHINE HISTORY - ELECTRICAL

M/C NUMBER: M30-209 CUSTOMER: M/S Mahavir die castera (P) Ltd

M/C MODEL: MCV-450 ROUTE CARD No.: RC-PDN-ElecCab-73-09

M/C BACKUP (PATH) : \\arya2\E-Backup-II\ELECTRICAL MACHINE DATA\AMS DISPATCH MACHINE BACKUPS (2009-10)\FANUC

CNC SYSTEM

0i-MC ☑ 0i-MATE-MC ☐ 0i-MD ☐ 0i-MATE-MD ☐

PART No.: A02B-0309-B520

SERIAL No.: C087B2807

DISPLAY:COLOUR ☐ MONOCHROME ☑ PENTIUM CARD ☐ ETHERNET ☐

β SERVO DRIVE SVPM H34371

5.5i (20/20/40) ☐ 15i (40/40/40) ☐ 15i (40/40/80) ☑

A06B-6134-H580-A

SERIAL No.: V09911019

α SERVO DRIVE PSM

11i ☐ 15i ☐ 26i ☐ 30i ☐ 37i ☐

PART No.: A06B-61□□-H0□□

SERIAL No.: NA

α SERVO DRIVE SPM

11i ☐ 15i ☐ 26i ☐ 30i ☐ 37i ☐

PART No.: A06B-61□□-H0□□ #550

SERIAL No.:

α SERVO DRIVE SVM1(1)

40i ☐ 160i ☐

PART No.: A06B-61□□-H10□

SERIAL No.:

α SERVO DRIVE SVM1(2)

40i ☐ 160i ☐

PART No.: A06B-61□□-H10□

SERIAL No.:

α SERVO DRIVE SVM2

40i/40I ☐ 160i/160i ☐

PART No.: A06B-61□□-H2□□

SERIAL No.:

FANUC I/O MODULE/ CARD

I/O MODULE WITH MPG ☑

PARTNo.: A02B-0309-C001

Sl No. E22817

I/O BOARD WITH MPG ☐

PART No.: A20B-2002-0520

I/O BOARD WITHOUT MPG ☐

PART No.: A20B-2002-0521

A.C REACTOR1:

0.13MH/46A ☐ 0.11MH/62A ☑ 0.07MH/110A ☐ ☐

PART No.: A81L-0001-0156

SERIAL No.: F09090675

CHECKED BY:- 26/11/09

VERIFIED BY:-

F-QA-FA-14-R0

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

X AXIS MOTOR

β4/4K ☐ β8/3K ☐ β12/3K ☑ β22/3K ☐ ☐

PART No.: A06B-0078-B103 #100

SERIAL No.: C099X4799

Y AXIS MOTOR

β4/4K ☐ β8/3K ☐ β12/3K ☑ β22/3K ☐ ☐

PART No.: A06B-0078-B103

SERIAL No.: C099M6708

Z AXIS MOTOR

β8/3K ☐ β12/3K ☐ β22/3K ☑ ☐ ☐

PART No.: A06B-0082-B403

SERIAL No.: C099B1770

B AXIS DETAILS

ROTARY TABLE ☐ INDEX TABLE ☐ Interway only

DRIVE: βSiV-20 ☐ βSiV-40 ☐ ☐

PART No.: A06B-____-H002

SERIAL No.:

MOTOR: β4/4K ☐ β8/3K ☐ ☐

PART No.: A06B-0___-B___03

SERIAL No.:

MOTOR: β4/4K ☐ β8/3K ☐ ☐

PART No.: A81L-0001-____ #3C

SERIAL No.:

DISCHARGE RESISTOR: 16 Ohms ☐ 30 Ohms ☐

PART No.: A06B-0___-H402

SERIAL No.:

PMM AXIS DETAILS

ROTARY TABLE ☐ TILTING TABLE ☐ NA SERVOMAG/ARM ☐

DRIVE: βSiV-20 ☐ βSiV-40 ☐ ☐

PART No.: A06B-____-H002

SERIAL No.:

MOTOR: β4/4K ☐ β8/3K ☐ ☐

PART No.: A06B-0___-B___03

SERIAL No.:

MOTOR: β4/4K ☐ β8/3K ☐

PART No.: A81L-0001-____ #3C

SERIAL No.:

DISCHARGE RESISTOR: 16 Ohms ☐ 30 Ohms ☐

PART No.: A06B-0___-H402

SERIAL No.:

SECIAL INFORMATION IF ANY:

—

CHECKED BY:- [signature]

VERIFIED BY:- [signature]

ACE MANUFACTURING SYSTEMS LTD

AMS — The Pursuit of Excellence

MACHINE HISTORY - ELECTRICAL

MAIN POWER CIRCUIT

ISOLATION SWITCH: (SW1) — AC 3 Ph 415V ☑

40A ☐ 63A ☑

MAKE: SALZER ☑ ☐

SERIAL/TYPE No.:

TRANSFORMER: (TFR 1) — PRIMARY: AC 3 Ph 415V ☑ — SECONDARY: AC 3 Ph 200V ☑

10KVA ☐ 12KVA ☐ 15.5KVA ☑ 17KVA ☐

MAKE:T.S INTERNATIONAL ☑ ☐

SERIAL No.: 7192-11-09

PRIMARY MCB (F15) — AC 3 Ph 415V / 3POLE ☑

25A ☐ 32A ☑ 40A ☐

MAKE: MERLIN GERIN ☑ SCHNEIDER ☐

SECONDARY MCB (F19) — AC 3 Ph 415V / 3POLE ☑

40A ☐ 50A ☐ 63A ☑

MAKE: MERLIN GERIN ☒ SCHNEIDER ☑

SPINDLE FAN MPCB (F56) — AC 3 Ph ~~415V~~ 200V / 3POLE ☑

0.1-0.16A ☑ 0.24-0.4A ☐

MAKE:SPRECHER &SCHU ☑ ☐

PSM CONTACTOR: (K19) — 24V DC ☐ — 230V AC ☑

CA3-30C ☐ CA3-43N-L11 ☐ CA3-72N-L11 ☑ ☐

MAKE: SPRECHER &SCHU ☑ ☐

REGULATED POWER SUPPLY CIRCUIT

REGULATED POWER SUPPLY: PS1 ☑ PS2 ☑ — INPUT: AC 1 Ph200V ☑ — OUTPUT:DC 24V 10A ☑

MAKE: TELEMECANIQUE(ABL 1RPM24100) ☐ OMRON(S8VS 24024) ☑ KB ELECTORNICS ☐

Sl.No: PS1: PS2:

TRANSFORMER: (TFR4) — PRIMARY: AC1 Ph 415V ☑ — SECONDARY: AC 1 Ph 200V ☑

700VA ☐ 900VA ☑ 1.5KVA ☐ ☐

MAKE:T.S INTERNATIONAL ☑ ☐

SERIAL No.: 1305-11-09

PRIMARY MCB (F16) — AC 1 Ph 415V / 2POLE ☑

1A ☐ 3A ☑ ☐

MAKE: MERLIN GERIN ☐ SCHNEIDER ☑

CONTROL CIRCUIT

SEALED KEYBOARD / MCP:

TYPE.: YSL ELECTRONICS (YSL-AMS-SKB-2-REV0-05/08) ☑ ☐

Sl.No:

ATC BRAKE RELAY (PCB 1) :

TYPE .: 1 SSR RELAY - YSL-SSRP-BR-2-REV4-08/09 ☐ — 2 SSR RELAY - YSL-SSRP-BR-4-REV4-08/09 ☑

Sl.No: YSL-S26-009-77-CN-REV0-07/09

I/O DISTRIBUTION PCB (PCB 2) :

TYPE: YSL ELECTRONICS (YSL-S26-009-056-REV0-04/08) ☑ ☐

Sl.No:

CHECKED BY:- VSh

VERIFIED BY:- Hashim

Shortage

① 2 SSR Relay Board - 1
16 Pin Connector Blade - 2

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

CNC ON/OFF PCB (PCB 3) :

TYPE: YSL ELECTRONICS (~~RP-002-EC-CP-1-REV4-08/09~~ YSL-SSR-CNC ON/OFF 08/09) ☑ ☐

Sl.No:

Z AXIS BRAKE RELAY (PCB 5) :

MAKE: YSL ELECTRONICS (YSL-SSRP-BR-2-REV4-08/09) ☑ ☑

Sl.No: YSL-SSRP-BR-2-REV4-08/09

24 V THROUGH ADD ON BLOCK K19-33/34 (CS4-P) ☐ 24 V THROUGH CONTACTOR K19-13/14 ☑

AC RELAY: Z AXIS BRAKE CIRCUIT (ACR 1) ☑ THERMALSW CIRCUIT (ACR 2) ☑

MAKE: OMRON (MK2P-I) ☑ ☐

Sl.No:

Z AXIS BRAKE - CURRENT SENSING RELAY (CSR1):

TYPE: YSL ELECTRONICS (CSR-1B) ☑ ☐

Sl.No:

RELAY BOARD-STD (X20) : 1RELAY(ELECTRO MAGNETIC-DC) ☑ ☐

TYPE: YSL ELECTRONICS (YSL-UMK-1RM DC-CN-FI-2) ☑ ☐

Sl.No:

RELAY BOARD-STD (X31) : 1RELAY(ELECTRO MAGNETIC-AC) ☑ ☐

TYPE: YSL ELECTRONICS (YSL-S26-009-062) ☑ ☐

Sl.No:

RELAY BOARD- APC (X25) : 8 RELAY(SSR-DC) ☐ N/A ☐

TYPE: YSL ELECTRONICS (YSL-S26-009-064-REV3-08/08) ☐ ☐

Sl.No:

RELAY BOARD- FIXTURE (X26) : 8 RELAY(SSR-DC) ☐ N/A 4 RELAY(SSR-DC) ☐ 2 RELAY(SSR-DC) ☐

TYPE: YSL ELECTRONICS (YSL-S26-009-064-REV3-08/08) ☐ YSL-S26-009-063-REV3-08/08 ☐ YSL-S26-009-062-REV3-08/08 ☐

Sl.No:

RELAY BOARD- : N/A ☐ ☐

MAKE: YSL ELECTRONICS ( ) ☐ ☐

Sl.No:

EMERGENCY CONTACTOR: (K22) 24V DC ☑

CS3C-40E ☑ ☐ MAKE:SPRECHER &SCHU ☑ ☐

POWER FAILURE DETECTION MODULE (PFDM) :

MAKE: YSL ELECTRONICS - ☐ NA ☐

Sl.No:

CAPACITOR BANK (DP1) :

MAKE: YSL ELECTRONICS - PF-CHU 1 ☑ ☐

Sl.No:

SECIAL INFORMATION IF ANY:

CHECKED BY:- YSL VERIFIED BY:- Ananthan

ACE MANUFACTURING SYSTEMS LTD

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MACHINE HISTORY - ELECTRICAL

SINGLE PHASE POWER SUPPLY CIRCUIT

SINGLE PHASE MCB (F18) AC 1 Ph 230V / 1POLE ☑

10A ☑ ☐

MAKE: MERLIN GERIN ☐ SCHNEIDER ☑

GANG BOX (230V 1Ph.)

MAKE; ANCHOR ☑

2 WAY SINGLE POLE SWITCH ☑ QTY: 02 Nos.

5 PIN DOMESTIC SOCKET ☑

CABINET LAMP (230V 1Ph.) :

MAKE: PEARL ☑ ☐

Sl.No:

PANEL COOLER (230V 1Ph.): AC 1 Ph200V/500W ☐ AC 1 Ph200V/1000W ☑

MAKE: ADVANCE (TURBO1000R33) ☐ ADVANCE (TURBO500R50) ☐ SUNBEAM (CNC2000EG) ☐ ACE (IGLOO) ☑

Sl.No: 8494 R-134a

SPINDLE STAR / DELTA CIRCUIT

STAR / DELTA CONTACTOR: (K20 & K21) 24V DC ☐ 230V AC ☐

CA3-72N-L11 ☐ CS7 - 85 ☐ ☐

ELE. INTER LOCK BLOCK (CS7) ☐

MAKE: SPRECHER &SCHU ☐ ☐

RELAY BOARD- (X27) : 2 RELAYS(ELECTRO MAGNETIC) ☐ ☐

MAKE:YSL ELECTRONICS - ☐ NA ☐

Sl.No:

SECIAL INFORMATION IF ANY:

CHECKED BY:-

VERIFIED BY:-

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MACHINE HISTORY - ELECTRICAL

MAGAZINE MOTOR

MPCB(F1):	1-1.6A	☑	☐	CONTACTOR (K1&K2)):	CA4-9C	☑	☐
MOTOR POWER:	1.6KVA/2.3A	☑	1.9KVA/2.6A ☐	MPCB SETTING:	1A	☑	☐
SOLID STATE RELAY (SSR1):	MAKE: ERI ELECTRONICS-				SJDA - 482524(DSPST)		☑

ARM MOTOR

MPCB(F2):	1-1.6A	☑	☐	CONTACTOR (K3):	CA4-9C	☑	☐
MOTOR POWER:	2.3KVA/3.2A	☐	1.9KVA/2.6A ☑	MPCB SETTING:	1A	☑	1.4A ☐
SOLID STATE RELAY (SSR3):	MAKE: ERI ELECTRONICS-				SJDA - 482524(DSPST)		☑

LUBRICATION MOTOR

MCB(F3):	1A/3POLE	☑	☐	CONTACTOR (K3):	CA4-9C	☑	☐
MOTOR POWER:	0.09KW / 0.5A	☐	☐	MPCB SETTING:		☐	

POWER CIRCUIT (STANDARD COOLANT)

FLOOD COOLANT MOTOR

MPCB(F4):	1-1.6A	☑	2.5-4A ☐	CONTACTOR (K5):	CA4-9C	☑	☐
MOTOR POWER:	0.37KW / 0.9A	☑	1.5KW / 3.4A ☐	MPCB SETTING:	1A	☑	3.4A ☐
SURGE SUPRESSOR(RC4)	MAKE:	YSL-NIR-RC3-1/022-575VAC	☑				

CHIP PUSH COOLANT MOTOR

MPCB(F5):	1-1.6A	☐	2.5-4A ☑	CONTACTOR (K6):	CA4-9C	☑	☐
MOTOR POWER:	0.37KW / 0.9A	☐	1.5KW / 3.4A ☑	MPCB SETTING:	1A	☐	3.4A ☑
SURGE SUPRESSOR(RC5)	MAKE:	YSL-NIR-RC3-1/022-575VAC	☑				

CHIP FLUSH COOLANT MOTOR

MPCB(F6):	1-1.6A	☐	2.5-4A ☐	CONTACTOR (K7):	CA4-9C	☑	☐
MOTOR POWER:	0.37KW / 0.9A	☐	1.5KW / 3.4A ☐	MPCB SETTING:	1A	☐	3.4A ☐
SURGE SUPRESSOR(RC6)	MAKE:	YSL-NIR-RC3-1/022-575VAC	☑		COMMON MPCB WITH CHIP PUSH (F5)		☑

COOLANT GUN MOTOR

MPCB(F27):	1-1.6A	☑ N/A	2.5-4A ☐	CONTACTOR (K25):	CA4-9C	☑	☐
MOTOR POWER:	0.37KW / 0.9A	☐	1.5KW / 3.4A ☐	MPCB SETTING:	1A	☐	3.4A ☐
SURGE SUPRESSOR(RC27)	MAKE:	YSL-NIR-RC3-1/022-575VAC	☑				

HIGH PRESSURE COOLANT MOTOR

MPCB(F9):	4-6.3A	☐	☐	CONTACTOR (K10):	CA4-12C	☐ N/A	☐
MOTOR POWER:	2.2KW / 4.9A	☐	☐	MPCB SETTING:	4.9A	☐	☐
SURGE SUPRESSOR(RC9)	MAKE:	YSL-NIR-RC3-1/022-575VAC	☐				

HYDRAUALIC MOTOR N/A

MPCB(F8):	2.5-4A	☐	☐	CONTACTOR (K9):	CA4-9C	☐	☐
MOTOR POWER:	1.1KW / 2.7A	☐	☐	MPCB SETTING:	2.7A	☐	☐
SURGE SUPRESSOR(RC8)	MAKE:	YSL-NIR-RC3-1/022-575VAC	☐				
OIL COOLING FAN CIRCUIT(230V 1Ph):		CA4-P	☐				

CHECKED BY:-	VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

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MACHINE HISTORY - ELECTRICAL

INDEX TABLE MOTOR POWER CIRCUIT

MPCB(F12): 1-1.6A ☐	☐	CONTACTOR (K15/K16):CA4-9C ☐	☐
MOTOR POWER: 1.6KVA/2.3A ☐	1.9KVA/2.6A ☐	MPCB SETTING: 2.3A ☐ NA	☐
SOLID STATE RELAY (SSR5):MAKE: ERI ELECTRONICS-		SJDA - 482524(DSPST)	☐

POWER CIRCUIT (CHIP CONVEYOR)

CHIP CONVEYOR 1 MOTOR

MPCB(F10): 0.63-1A ☐	☐	CONTACTOR (K11/K12):CA4-9C ☐	☐
MOTOR POWER: 0.18KW / 0.68A ☐	☐	MPCB SETTING: 0.74A ☐ NA	☐
SURGE SUPRESSOR(RC10) MAKE: YSL-NIR-RC3-1/022-575VAC ☐			

CHIP CONVEYOR 2 MOTOR

MPCB(F11): 0.63-1A ☐	☐	CONTACTOR (K13/K14):CA4-9C ☐	☐
MOTOR POWER: 0.18KW / 0.68A ☐	☐	MPCB SETTING: 0.74A ☐ NA	☐
SURGE SUPRESSOR(RC11) MAKE: YSL-NIR-RC3-1/022-575VAC ☐			

CONVEYOR DRUM WASH MOTOR

MPCB(F7): 1-1.6A ☐	2.5-4A ☐	CONTACTOR (K8): CA4-9C ☐	☐
MOTOR POWER:0.37KW / 0.9A ☐	1.5KW / 3.4A ☐	MPCB SETTING: 1A ☐ NA	3.4A ☐
SURGE SUPRESSOR(RC7) MAKE: YSL-NIR-RC3-1/022-575VAC ☐			

SCREW CONVEYOR 1 MOTOR POWER CIRCUIT

MPCB(F13): 1-1.6A ☐	☐	CONTACTOR (K17/K18):CA4-9C ☐	☐
MOTOR POWER: 0.37KW /1.1A ☐	☐	MPCB SETTING: 1.1A ☐ NA	☐
SURGE SUPRESSOR(RC13) MAKE: YSL-NIR-RC3-1/022-575VAC ☐			

OPTIONAL MOTOR 1 - F56 Spindle MPCB

☐ 0.10-0.16A ☐		CONTACTOR ( ): CA4-9C ☐	☐
MOTOR POWER: ☐	☐	MPCB SETTING: ☐ KA	☐
SURGE SUPRESSOR(RC) MAKE: YSL-NIR-RC3-1/022-575VAC ☐		Spare Mech.	

OPTIONAL MOTOR 2 -

MPCB(F7): ☐	☐	CONTACTOR ( ):CA4-9C ☐	☐
MOTOR POWER: ☐	☐	MPCB SETTING: ☐ NA	☐
SURGE SUPRESSOR(RC7) MAKE: YSL-NIR-RC3-1/022-575VAC ☐			

SECIAL INFORMATION IF ANY:

CHECKED BY:- Akshay	VERIFIED BY:- Raj

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MACHINE HISTORY - ELECTRICAL

ATC & SPINDLE DETAILS

SPINDLE MOTOR:	MODEL:	β3 ☐	β6 ☐	β8 ☐	β12 ☑	☐
	SPEED(IN RPM):	6000 ☐	7000 ☐	8000 ☑	10000 ☐	☐
	MOTOR POWER:	15	11 KW			
	PART No:	A06B-1447-B103#				
	Sl. No:	C09XJA1C1 102				
	GEAR RATIO:	1:1 ☑	1:1.25 ☐	1:1.5 ☐	1:2 ☐	DIRECT DRIVE ☐
	POWER LIMIT:	11 / 7.5 KW				
	SPEED LIMIT:	80 / 8000 RPM				

PROXIMITIES- SPINDLE

SPINNDLE ORIENTATION PROXIMITY(PXS1) ☑	M12 X73 MM CONN TYPE ☑	M12 X 40 MM NON CONN TYPE ☐	☐
MAKE; TEKNIC TYPE: 24V DC - PNP			
TOOL CLAMP PROXIMITY(PXS3) ☑	M12 X73 MM CONN TYPE ☑	M12 X 52 MM CONN TYPE ☐	☐
MAKE; TEKNIC TYPE: 24V DC - PNP			
TOOL DECLAMP PROXIMITY(PXS4) ☑	M12 X73 MM CONN TYPE ☑	M12 X 40MM NON CONN TYPE ☐	☐
MAKE; TEKNIC TYPE: 24V DC - PNP			
NO TOOL IN SPINDLE PROXIMITY(PXS32) ☑	M8 X50 MM NON CONN ☑	FG2X05APO24-5000LFLAT ☐	M8 X50 MM CONN ☐
MAKE; TEKNIC TYPE: 24V DC - PNP			
ONE ROTATION PROXIMITY(PXS39) ☐	M12 X73 MM CONN TYPE ☐	M12 X 40 MM NON CONN TYPE ☐	☐
MAKE; TEKNIC TYPE: 24V DC - PNP			

PROXIMITIES- ATC(DISC TYPE-AMS MAKE)

MAGAZINE IN POSITION PROXIMITY(PXS2) ☐	M12 X73 MM CONN ☐	M12 X68 MM NON CONN ☐	N/A	M12 X 40 MM NON CONN TYPE ☐
MAKE; TEKNIC TYPE: 24V DC - PNP				
MAG REFERENCE PROXIMITY(PXS46) ☐	M12 X73 MM CONN TYPE ☐	FG2X05APO24-5000LFLAT ☐		☐
MAKE; TEKNIC TYPE: 24V DC - PNP				
TOOL FEEDBACK ENCODER(ENC1) ☐	12 TOOL ☐	16 TOOL ☐		☐
MAKE; PRAGATHI TYPE:				

PROXIMITIES- ATC (TWIN ARM-PRAGATHI MAKE)

PROXIMITIES ☑ MAKE; TEKNIC		TYPE: EGT 12 X 02 AP - 024 SEM4L(24V DC - PNP) ☑	
MAGAZINE IN POSITION PROXIMITY (PXS2) ☑		MAG REFERENCE PROXIMITY (PXS46) ☑	
POCKET UP PROXIMITY (PXS5) ☑		POCKET DOWN PROXIMITY (PXS6) ☑	
ARM AT HOME PROXIMITY (PXS7) ☑			☐
TOOL CLAMP START PROXIMITY (PXS58) ☑		TOOL DECLAMP START PROXIMITY (PXS59) ☑	

CHECKED BY:- Harish

VERIFIED BY:- BM

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MACHINE HISTORY - ELECTRICAL

PNUEMATIC ELEMENTS (SPINDLE + ATC)

SOLENOIDS MAKE; FESTO	☑ ☐	TYPE: (24V DC -COIL)	☑
SPINDLE AIR JET SOLENOID (Y1)	☑	SPI AIR PURGE SOLENOID (Y115)	☐
POCKET UP SOLENOID (Y2)	☑	POCKET DOWN SOLENOID (Y3)	☑
TOOL CLAMP SOLENOID (Y51)	☑	TOOL DECLAMP SOLENOID (Y4)	☑
AIR THROUGH NOZZLE SOLENOID (Y99)	☐		☐
PRESSURE SWITCHES MAKE; SUCO	☑ ☐	TYPE: (24V DC - NO CONTACT)	☑
MAIN AIR PRESSURE SWITCH (PRS 2)	☑	SPINDLE AIR PRESSURE SWITCH (PRS 101)	☑
AIR PURGE PRESSURE SWITCH (PRS 102)	☐		☐

HYDRAULIC ELEMENTS (SPINDLE + ATC)

SOLENOIDS MAKE; YUKEN ☐ REXROTH	☐	TYPE: (24V DC -COIL) NA	☐
TOOL CLAMP SOLENOID (Y51)	☐	TOOL DECLAMP SOLENOID (Y4)	☐
TOOL CHANGE POSITION LIMIT SWITCH- (LS3) : 24V DC NO CONTACT	☑		☐
MAKE; TEKNIC SG-01-R-522	☑		☐

SECIAL INFORMATION IF ANY:

AUTO DOOR & CYCLE START DETAILS

SOLENOIDS MAKE; FESTO	☐ ☐	TYPE: (24V DC -COIL) NA	☐
AUTO DOOR OPEN SOLENOID (Y6)	☐	AUTO DOOR CLOSE SOLENOID (Y7)	☐
AUTO DOOR SAFETY SOLENOID (Y8)	☐		☐
DOOR SWITCHES MAKE; TECNIK ☐ FESTO	☐	TYPE: LS5 - NG1 RS 510 (24V DC -NO CONTACT)	☐ ☐
CYCLE START SWITCH MAKE;TELEMECHANIC	☐	TYPE: LS45 - XCE-154 (24V DC -NO CONTACT)	☐ ☐
DOUBLE HAND CYCLE START BOX MAKE; TEKNIC	☐	CYCLE START PB (MUSHROOM HEAD GREEN) WITH N.O CONTACT (02 Nos, EMERGENCY PB (MUSHROOM HEAD SPRING RETURN) WITH N.C CONTA	☐ ☐
AUTO DOOR OPERATION MAKE; TEKNIC	☐	AUTO DOOR OPEN/CLOSE PB (GREEN) WITH N.O CONTAC AUTO DOOR INTERLOCK KEY SW WITH N.O CONTACT	☐ ☐

SECIAL INFORMATION IF ANY:

CHECKED BY:- Harish M	VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

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MACHINE HISTORY - ELECTRICAL

COOLANT TANK / CONVEYORS DETAIL

COOLANT LEVEL FLOAT SWITCH (PXS 8) MAKE; FILPRO ☑	TYPE: LIM 18PF SC(24V DC PNP) ✓
PROXIMITY SWITCHES MAKE; TEKNIC	TYPE: M12 X 40MM CONNECTOR TYPE(24V DC-PNP) M12 X 72MM CONNECTOR TYPE(24V DC-PNP)
SCREW CONVEYOR 1 O/L PROXIMITY (PXS20)	SCREW CONVEYOR2 O/L PROXIMITY (PXS104)
LIMIT SWITCHES MAKE; TEKNIC	TYPE: NA NG1 RS 510 (24V DC-NO CONTACAT)
CHIP CONVEYOR 1 O/L LIMIT SW (LS12)	CHIP CONVEYOR 2 O/L LIMIT SW (LS13)
M/C DOOR CLOSE LIMIT SW (LS5)	
H.P.COOLANT FLOW SWITCH (FLW1) MAKE; IFM	TYPE: SH0 500(24V DC PNP)
H.P.COOLANT CLOG SWITCH (FCLG1) MAKE; HIRLEKAR SALVIN	TYPE: NA (24V DC-NO CONTACT)
HP AIR BLAST SOLENOID (Y5) MAKE; FESTO	TYPE: (24V DC -COIL)
SELECTOR SWITCHES MAKE; TEKNIC	COOLANT GUN ON/OFF (2 POSITION) WITH N.O CONTACT SCREW/CHIP CONYR FWD/REV (3 POSI SPRING RETURN) WITH 2 N.O CONT.
COOLANT PLUG (HC1) MAKE; INDO ELECTRIC ☑	TYPE: 16 PIN ☑ 24 PIN ☐ LOCATION(FROM M/C REAR) : LHS ☐ RHS ✓
CHIP CONVEYOR PLUG (HC3) MAKE; INDO ELECTRIC	TYPE: 10 PIN ☐ 16 PIN ☐
SPECIAL INFORMATION:	

HYDRAULIC TANK DETAIL

OIL LEVEL FLOAT SWITCH (FL4) MAKE; VIDYUT	TYPE: NA (24V DC PNP)
HYDRAULIC PLUG (HC2) MAKE; INDO ELECTRIC	TYPE: 16 PIN ☐ 24 PIN ☐ LOCATION(FROM M/C REAR) : LHS ☐ RHS ☐
SPECIAL INFORMATION:	

CHECKED BY:- Harinder

VERIFIED BY:- RHS

ACE MANUFACTURING SYSTEMS LTD

AMS — An Instrument of Excellence

MACHINE HISTORY - ELECTRICAL

LUBRICATION TANK / FRL DETAIL

OIL LEVEL FLOAT SWITCH (FL1)		TYPE:	(24V DC - NO CONTACT) ☑
MAKE; CENLUB ☑	☐		☐
LUB PRESSURE SWITCH (PRS1)		TYPE:	(24V DC - NO CONTACT) ☑
MAKE; SUCO ☑	☐		☐
AUTO DRAIN SOLENOID (Y44)		TYPE:	(24V DC - COIL) ☑
MAKE; ROTEX JANATICS ☑	☐		☐
MACHINE LAMPS: ML1 ☑ ML2	☑	230V 1 Ph. (11W) ☐	230V 1 Ph. (18W) ☐
MAKE; PATANKAR ☑	☐	230V 1 Ph. (36W) ☑	☐
MACHINE INDICATION LAMP(PL5):		1 - TIER ☐	3 - TIER ☑
MAKE; PATANKAR ☑	☐	3 TIER WITH HOOTER ☐	☐
TPM TRACK:		NA VER 1.077	☐
MAKE; PIONEER ☐	☐		☐
SERVO VOLTAGE STABILIZER:		POWER : 15 KVA ☐ 25KVA ☐	☐
MAKE; SERVOMAX ☐	☐	INPUT :	NA
Sl. No:		OUTPUT:	
EMERGENCY MONITOR (PCB 12)		YSL-EMG-PCB-4-REV1-07/06 ☑	
HAND HELD MPG:		NA	☐
MAKE; FANUC ☐	☐		☐

SPECIAL INFORMATION: —

FIXTURE DETAIL

HYDRAULIC ☐	PNEUMATIC ☐	No. OF STATIONS: NA	
SOLENOIDS		TYPE: (24V DC -COIL)	☐
MAKE; REXROTH ☐	YUKEN ☐		
STATION 1 CLAMP SOLENOID (Y70)	☐	STATION 1 DECLAMP SOLENOID (Y71)	☐
STATION 2 CLAMP SOLENOID (Y72)	☐	STATION 2DECLAMP SOLENOID (Y73)	☐
STATION 3 CLAMP SOLENOID (Y84)	☐	STATION 3 DECLAMP SOLENOID (Y85)	☐
STATION 4 CLAMP SOLENOID (Y86)	☐	STATION 4 DECLAMP SOLENOID (Y87)	☐
PRESSURE SWITCHES		TYPE: (24V DC -NO -CONTACT)	☐
MAKE; POLYHYDRON ☐	☐		
STATION 1 CLAMP CLAMP PRSW (PRS20)	☐	STATION 2 CLAMP CLAMP PRSW (PRS22)	☐
STATION 1 CLAMP CLAMP PRSW (PRS36)	☐	STATION 2 CLAMP CLAMP PRSW (PRS37)	☐

SPECIAL INFORMATION: —

CHECKED BY:- Haresh.M	VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

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MACHINE HISTORY - ELECTRICAL

APC (LINEAR TYPE) DETAIL

NH

MANUAL OPERATION PANEL		YSL-AMS-APC-PCB-REV0-07/06 ☐
PUSH BUTTONS ☐ MAKE; TEKNIC		PALLET SET UP PB (I LUMINOUS GREEN) WITH N.O CONTACT ☐ EMERGENCY PB (MUSHROOM HEAD RED) WITH N.C CONTACT ☐
PALLET RECEIVED PROXIMITY(PXS11) MAKE; TEKNIC TYPE: PNP		M12 X73 MM CONN TYPE ☐ M12 X 40 MM NON CONN TYPE ☐
APC STATION PROXIMITIES MAKE; TEKNIC ☐	TYPE:	EGT 12 X 02 AP - 024 SEM4L(24V DC - PNP) ☐
ARM 1 IN PROXIMITY (PXS13) ☐		ARM 1 OUT PROXIMITY (PXS14) ☐
ARM 2 IN PROXIMITY (PXS15) ☐		ARM 2 OUT PROXIMITY (PXS16) ☐
LIMIT SWITCHES MAKE; JAI BALAJI ☐ ☐	TYPE:	JLSSPR (24V DC-NO CONTACAT) ☐ ☐
PALLET 1 ON STATION LIMIT SW (LS10) ☐		PALLET 2 ON STATION LIMIT SW (LS11) ☐
APC DOOR OPEN LIMIT SW (LS6) ☐		APC DOOR CLOSE LIMIT SW (LS7) ☐
PNEUMATIC SOLENOIDS MAKE; FESTO ☐ ☐	TYPE:	(24V DC -COIL) ☐
APC DOOR OPEN SOLENOID (Y12) ☐		APC DOOR CLOSE SOLENOID (Y13) ☐
PALLET PAD CLEAN SOLENOID (Y18) ☐		☐
HYDRAULIC SOLENOIDS MAKE; YUKEN ☐ REXROTH ☐	TYPE:	(24V DC -COIL) ☐
PALLET CLAMP SOLENOID (Y10) ☐		PALLET DECLAMP SOLENOID (Y11) ☐
HYDRAULIC PRESSURE SWITCHES MAKE;POLYHYDRON ☐ ☐	TYPE:	(24V DC -NO -CONTACT) ☐
PALLET CLAMP PRSW (PRS4) ☐		PALLET DECLAMP PRSW (PRS60) ☐

SPECIAL INFORMATION:

CHECKED BY:-	VERIFIED BY:-

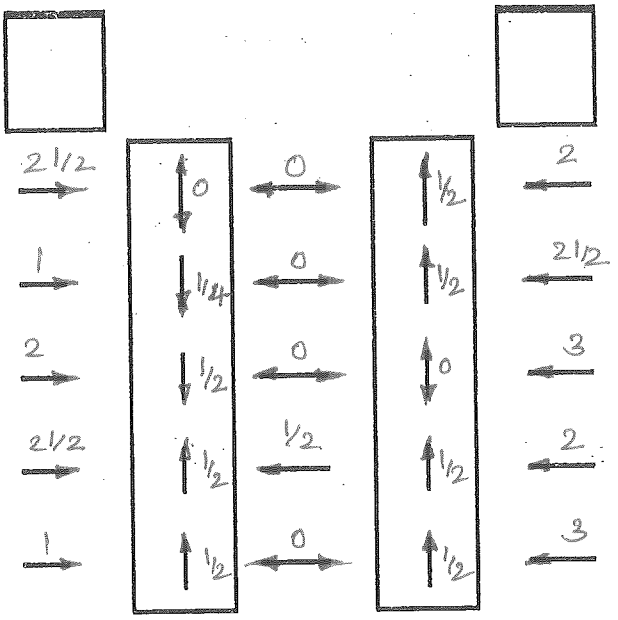
ACE MANUFACTURING SYSTEMS LIMITED

INSPECTION REPORT – BASE BUILDING

AMS *In Pursuit of Excellence*

MACHINE NUMBER: 30-209 **DATE:** 17-11-09

I. MACHINE BASE LEVELING: Least count of the Instrument: 1 div = 0.020mm / meter



SL NO	DESCRIPTION	PERMISSIBLE (mm)	ACTUAL(mm)		
1	Leveling - Length wise	1 div	1/2 div		
2	Leveling - Individual	3 div	3 div		
3	Leveling - Height variation	1 div	1/2 div		
4	Straightness of locating guide way	0.005	0.002 mm		
5	Parallelism of Non locating guide way	0.005	0.002		
6	Ball screw Stretching	0.025 /met (±0.002 /met)	0.025		
7	Ball screw Run out at Motor end	0.010	0.01		
8	Ball screw Run out at Brake end	0.010	0.01		
9	Ball screw alignment at Top	0.020	0.01		
10	Ball screw alignment at Side	0.010	0.01		
			M.E	CEN	B.E
11	Ball screw Torque	0.9 to 1.8 N-m	1.25	1.25	1.25 N-m
12	Slide & Jib Torque (Slide Torque for LM guide way)	Ball Screw Torque + 0.5 to 1.0	1.5	1.5	1.5 N-m
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0 + 0.1 to 0.15	—	—	—
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0 + 0.2 to 0.3	—	—	—
15	Final slide Torque	2.5 to 3 N-m	1.5	1.5	1.5 N-m

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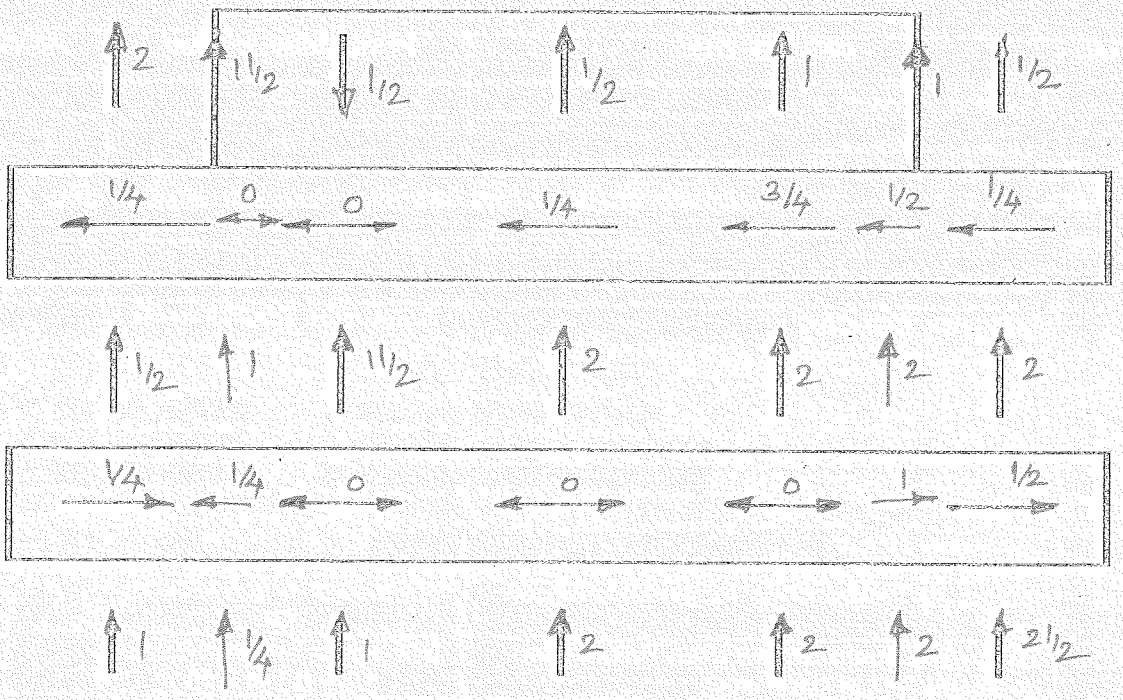
INSPECTION REPORT – BASE BUILDING

AMS — In Pursuit of Excellence

MACHINE NUMBER: M30-209 DATE: 18-11-09

SL NO	Check Points	Remark
1	Ensure all leveling screws & locknuts are fixed to the base.	✓
2	Ensure guide way butting face blue matching is minimum 80%	✓
3	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	✓
4	Ensure position of LM block ground surface butted to slide butting face.	✓
5	Ensure required height blocks are fitted to the base as per Work Order / Planning chart.	✓

II. SADDLE FINAL READING: — Least count of the Instrument: 1 div = 0.020mm / met



NAME & SIG					
	PREPARED	CHECKED	APPROVED	PAGE NO	FORMAT NO
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INSPECTION REPORT – BASE BUILDING

AMS *In Pursuit of Excellence*

MACHINE NUMBER: **DATE:** 18-11-09

SL NO	DESCRIPTION	PERMISSIBLE (mm)	ACTUAL(mm)		
1	Leveling - Length wise	1 div	1 div		
2	Leveling - Individual	3 div	2 div		
3	Leveling - Height variation	1 div	2 div		
4	Straightness of locating guide way	0.005	0.005		
5	Parallelism of Non locating guide way	0.005	0.004		
6	Ensure Squareness of Base & Saddle	0.010	[illegible]		
7	Ball screw alignment at Top	0.020	0.02		
8	Ball screw alignment at Side	0.010	0.01		
9	Ball screw Stretching	0.025 /met (± 0.002 /met)	0.025		
10	Ball screw Run out at Motor end	0.010	0.01		
11	Ball screw Run out at Brake end	0.010	0.01		
			M.E	CEN	B.E
12	Ball screw Torque	0.9 to 1.8 N-m	1.5	1.5	1.5 N-m
	Slide & Jib Torque (Slide Torque for LM guide way)	Ball Screw Torque + 0.5 to 1.0	1.75	1.75	1.75 N-m
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0 + 0.1 to 0.15	—	—	—
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0 + 0.2 to 0.3	—	—	—
15	Final slide Torque	2.5 to 3 N-m	1.75	1.75	1.75 N-m

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene.	✓
2	Ensure guide way butting face & saddle bottom blue matching is minimum 80%	✓
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	NA
4	Ensure proper matching of jib taper. ((H G m/c only)	NA
5	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	✓
6	Ensure position of LM block ground surface butted to slide butting face. (L M m/c only)	✓
7	Ensure Equal or minimum 0.5 mm gap between ball nut & nut housing bore.	✓
8	Ensure Nut housing alignment is with in 0.010 mm	✓
9	Ensure White purcian matching between nut housing & ball nut butting face is 80 %	✓
10	Ensure bearing housing Dowelling and grub locking.	✓
11	Ensure Ball screw lock nut grub locking with locktite.	✓
12	Ensure guide way wipers are fixed. (H G m/c only)	NA

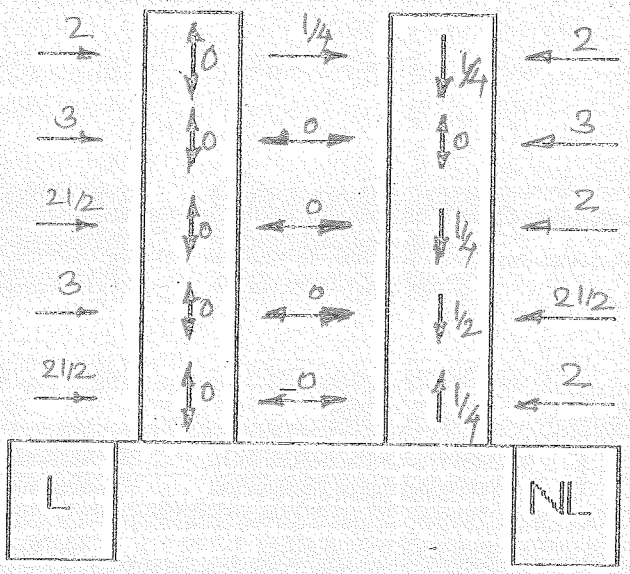
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INSPECTION REPORT – BASE BUILDING

AMS — In Pursuit of Excellence

13	Ensure Nylon stoppers are fixed as per specification.	✓

MACHINE NUMBER: M30-209 DATE: 21-11-09

III. COLUMN READING: Least count of the Instrument: 1 div = 0.020mm / met



SL NO	DESCRIPTION	PERMISSIBLE (mm)	ACTUAL(mm)		
1	Leveling - Length wise	1 div	1/4 Div		
2	Leveling - Individual	3 div	3 Div		
3	Leveling - Height variation	1 div	1/4 Div		
4	Straightness of locating guide way	0.005	0.003		
5	Parallelism of Non locating guide way	0.005	0.004		
6	Ball screw alignment at Top	0.020	0.010		
7	Ball screw alignment at Side	0.010	0.010		
8	Ball screw Stretching	0.025 /met (± 0.002 /met)	0.025		
9	Ball screw Run out at Motor end	0.010	0.01		
10	Ball screw Run out at Brake end	0.010	0.01		
11	Mandrel run out	0.010			
12	Mandrel reading at Top	0.010	0.007		
13	Mandrel reading at side	0.010	0.005		
			M.E	CEN	B.E
14	Ball screw Torque	0.9 to 1.8 N-m	1.25	1.25	1.25 N-m
15	Slide & Jib Torque (Slide Torque for LM guide way)	Ball Screw Torque + 0.5 to 1.0	1.5	1.5	1.5 N-m
16	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0 + 0.2 to 0.25	—	—	—
17	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0 + 0.4 to 0.5	1.5	1.5	1.5 N-m

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INSPECTION REPORT – BASE BUILDING

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18	Final slide Torque	2.5 to 3.5			

MACHINE NUMBER: **DATE:** 21-11-09

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene.	✓
2	Ensure guide way butting face, head stock bottom face,& column bottom face blue matching is minimum 80%	✓
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	NA
4	Ensure proper matching of jib taper. ((H G m/c only)	NA
5	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	✓
6	Ensure position of LM block ground surface butted to slide butting face. (L M m/c only)	✓
7	Ensure Equal or minimum 0.5 mm gap between ball nut & nut housing bore.	✓
8	Ensure nut housing alignment is with in 0.010 mm	✓
9	Ensure White purcien matching between nut housing & ball nut butting face is 80 %	✓
10	Ensure spindle cartridge butting face blue matching is 80%	✓
11	Ensure head stock is fixed as per requirement (ID, DD, and DD CTS).	✓
12	Ensure bearing housing Dowelling and grub locking.	✓
13	Ensure Ball screw lock nut grub locking with locktite.	✓
14	Ensure guide way wipers are fixed. (H G m/c only)	NA
15	Ensure Nylon stoppers are fixed as per specification.	✓
16	Ensure machine lifting tackles are fitted.	✓

ACE MANUFACTURING SYSTEMS LIMITED

INSPECTION REPORT – BASE BUILDING

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MACHINE NUMBER: DATE:

IV. SIDE ARM ALINGMENT READING: (Turret Type ATC only)

SL NO	DESCRIPTION	PERMISSIBLE (mm)	ACTUAL (mm)
1	Mandrel reading at Top	0.020	
2	Mandrel reading at side	0.020	
3	Square ness of locating side	0.010	

SL NO	MODEL	DISTANCE BETWEEN GUIDEWAY TOP TO SIDE ARM BORE CENTER (mm)		DISTANCE BETWEEN SIDE ARM BUTTING FACE TO SIDE ARM BORE CENTER (mm)	
		PERMISSIBLE READING	ACTUAL READING	PERMISSIBLE READING	ACTUAL READING
1	SPARK (M14-6TOOL)	596		740	
2	DTC 300 (M13 -12 TOOL)	596		740	
3	DTC 300 (M13 -16 TOOL)	575.5		722.5	
4	DART (M05-12 TOOL)	596		740	
5	DART (M05-16 TOOL)	575.5		722.5	
6	DART PLUS (M15-12 TOOL)	695		828	
7	DART PLUS (M15-16 TOOL)	677		810.5	
8	MCV 300 (M10-12 TOOL)	626		773	
9	WINNER (M26-12 TOOL)	688		838	
10	WINNER XL (M25-12 TOOL)	695		828	
11	CHALLENGER	728.5		861	

SL NO	Check Points	Remark
1	Ensure machine lifting tackles are fitted.	
2	Ensure side arm Dowelling.	

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INSPECTION REPORT – BASE BUILDING

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MACHINE NUMBER: M30-209 **DATE:** 26-11-09

PERMISSIBLE TOLERANCE 50 % OF TEST CHART VALUE

V. LEVELLING OF MACHINE:

L.C of Instrument: 1 div = 0.020mm / m

Along X-axis			Along Y-axis		
Motor End	Middle	Brake End	Motor End	Middle	Brake End
0	0	0	1	1/2	1/2
0	1/2	1/2	1/2	1/4	0
1/2	0	0	1/4	0	0

VI. TABLE TOP READING ALONG X- AXIS:

0	-8	-3	
0	-10	-5	
0	-10	-3	

VII. TABLE TOP READING ALONG Y- AXIS:

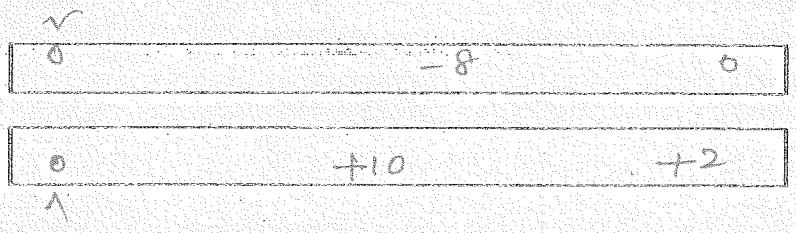
0	0	0	
-6	-5	-8	
-4	-4	-5	

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INSPECTION REPORT – BASE BUILDING

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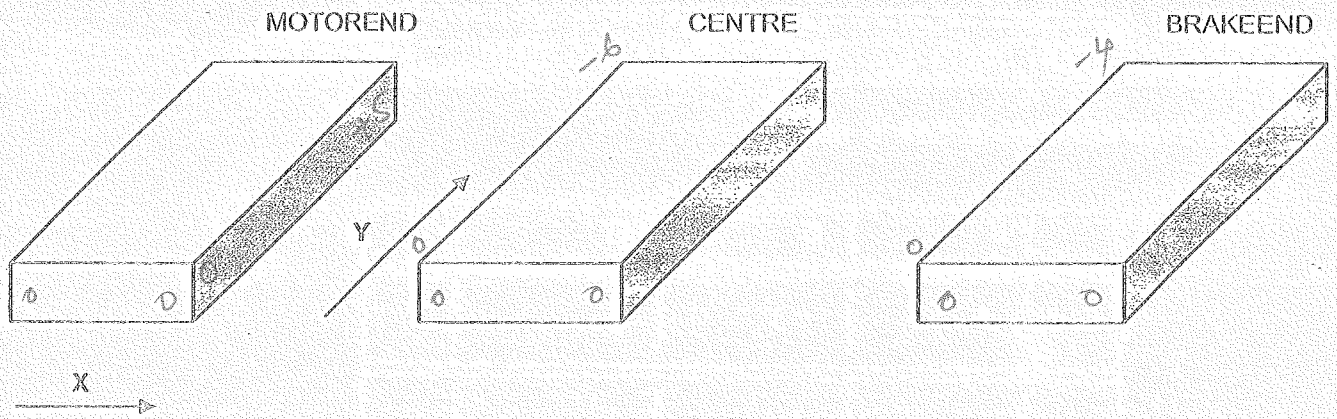
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VIII. TABLE TENON - SLOT READING AT 4 POSITIONS:



SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene.	✓
2	Ensure table bottom face blue matching is minimum 80%	✓
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	NA
4	Ensure proper matching of jib taper. ((H G m/c only)	NA
5	Ensure position of LM block ground surface butted to slide butting face.	✓
6	Ensure Equal or minimum 0.5 mm gap between ball nut & nut housing bore.	✓
7	Ensure nut housing parallelism is with in 0.010 mm	✓
8	Ensure Blue matching between nut housing & ball nut butting face is 80 %	✓
9	Ensure Table Tenon slot size as per specification (14 / 18 mm)	✓

IX. SQUARENESS OF Y –AXIS MOVEMENT WITH RESPECT TO X – AXIS MOVEMENT:



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MACHINE NUMBER: —

DATE: 28-11-09

X. COLUMN READING (INITIAL):

XZ PLANE(SIDE)

500	
450	+13
400	
350	
300	
250	
200	
150	
100	
50	
0	0

YZ PLANE (FRONT)

500	
450	-35
400	
350	
300	
250	
200	
150	
100	
50	
0	0

COLUMN READING - FINAL:

XZ PLANE(SIDE)

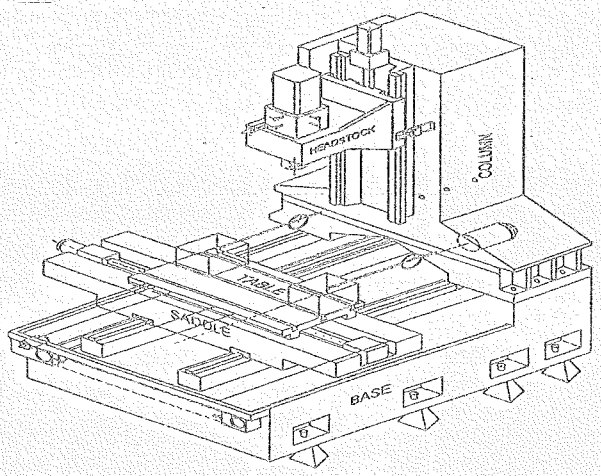
500	
450	
400	
350	
300	
250	
200	
150	
100	
50	
0	

YZ PLANE (FRONT)

500	
450	
400	
350	
300	
250	
200	
150	
100	
50	
0	

MACHINE NUMBER: DATE:

XI. COLUMN ALINGMENT READING WITH RESPECT TO X-AXIS MOVEMENT:



0	100	200	300	400	500	600	700	800	900	1000

ACE MANUFACTURING SYSTEMS LTD

LUBRICATION ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

MACHINE NUMBER : M30 -209.

I. Lubrication metering cartridge specification in CC:

SL NO	Axis	Ball Screw	Guideway Top / LM Block	Guideway Side	Jib	Keeper plate	
1	X -Axis	0.25 cc	0.1 cc	-NA-	-NA-	-NA-	
2	Y -Axis	0.25 cc	0.1 cc	-NA-	-NA-	-NA-	
3	Z-Axis	0.25 cc	0.1 cc	-NA-	-NA-	-NA-	

Sl NO	Check points:	Remark
1	Ensure X,Y & Z guideways and ball screw are lubricating properly	✓
2	Ensure lubrication pressure set to 15 Bar.	✓
3	Ensure there is no leakage of lubrication oil from joints, cartridges & manifolds etc.	✓
4	Ensure that the lubrication hoses do not rub or foul with the stationary parts during axis movement.	✓
5	Ensure that the lubrication hoses of appropriate length are used and properly clamped and routed.	
6	Ensure metering cartridge are fixed as per circuit diagram.	

II Telescopic cover Assembly:

	Sl No	Axis	Extension block Parallelism		Length Variation -	
			Top -Permissible 0.020 mm	Side-Permissible 0.030 mm	Permissible 0.050 mm	
	1	X - Motor End	0.010 MM -11-	0.010 MM -11-	0.010MM	
	2	X - Brake End	0.010 mm -11-	0.010MM 0.020MM.	0.010MM	
	3	Y - Motor End	0.020MM 0.010 MM	0.030MM 0.010MM	-NA-	
	4	Y - Brake End	0.010MM -11-	0.010MM -11-	0.010MM.	
	5	Z - Brake End	0.020MM -11-	0.010MM -11-	0.010MM.	

III. Saddle cover alignment (If required) :

	Sl NO	Top - Permissible 0.050 "mm"	Side - Permissible 0.050 "mm"	
	1	-NA-	-NA-	

Sl NO	Check points:	Remark
1	Check telescopic cover sliding smoothly	
2	Ensure Top & Side surface of the extension blocks are matching with respect to guideways.	
3	Ensure telescopic cover screws are tightened in box close condition	

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LUBRICATION ASSEMBLY - INSPECTION REPORT

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MACHINE NUMBER : M30 - 209

IV. Motor Assembly:

Motor mounting bracket Concentricity & Face out:

Sl No	Axis	Face out -Permissible 0.030 mm	Concentricity-Permissible 0.060 mm
1	X- Axis	0.030mm	0.030MM
2	Y- Axis	0.020mm	0.040MM
3	Z- Axis	0.030 MM	0.020 MM

SI NO	Check points:	Remark
1	Ensure motor shaft and coupling bore are free from oil / grease before fitting of coupling	✓
2	Ensure Axis Motor & Motor mounting bracket are fitted as per specification.	✓
3	Ensure coupling 3 slots (Steel Bellows, Clamping ring & Bush) are in same line.	✓
4	Check the tightness of coupling and set to 17 N-m (M6 screw), 35 N-m (M8 screw) Torque.	✓

V. Counter balance Assembly (If Required)

SI NO	Check points:	Remark
1	Ensure Smooth movement of Z- Axis with out fouling of dead weight (MCV 500 / MCV 600 only)	NA
2	Ensure the parallelism of counter balance cylinder piston is with in 0.050 mm both XZ & YZ plane with respect to Z Axis movement.	NA
3	Ensure Slide is balanced properlly in both upward and down ward movement	NA
4	Ensure the counter balance bracket Dowelling.	NA

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